Ramsey Alignment and the Postwar Tax System

Jackson Mejia*

February 2026

Was the postwar shift from corporate to personal taxation welfare-improving? I show that it was: a small optimal reform in the 1950s would have increased welfare by 0.7% of GDP, but by the 1980s almost no gains remained, and little remains today. That conclusion rests on a geometric framework I develop that measures alignment with the Ramsey optimum via the angle between welfare and revenue gradients. I use narrative tax shocks to identify the gradients in general equilibrium, revealing temporal and cross-base spillovers that static partial equilibrium approaches understate by an order of magnitude.

JEL-Classification: H21, H25, E62, C32

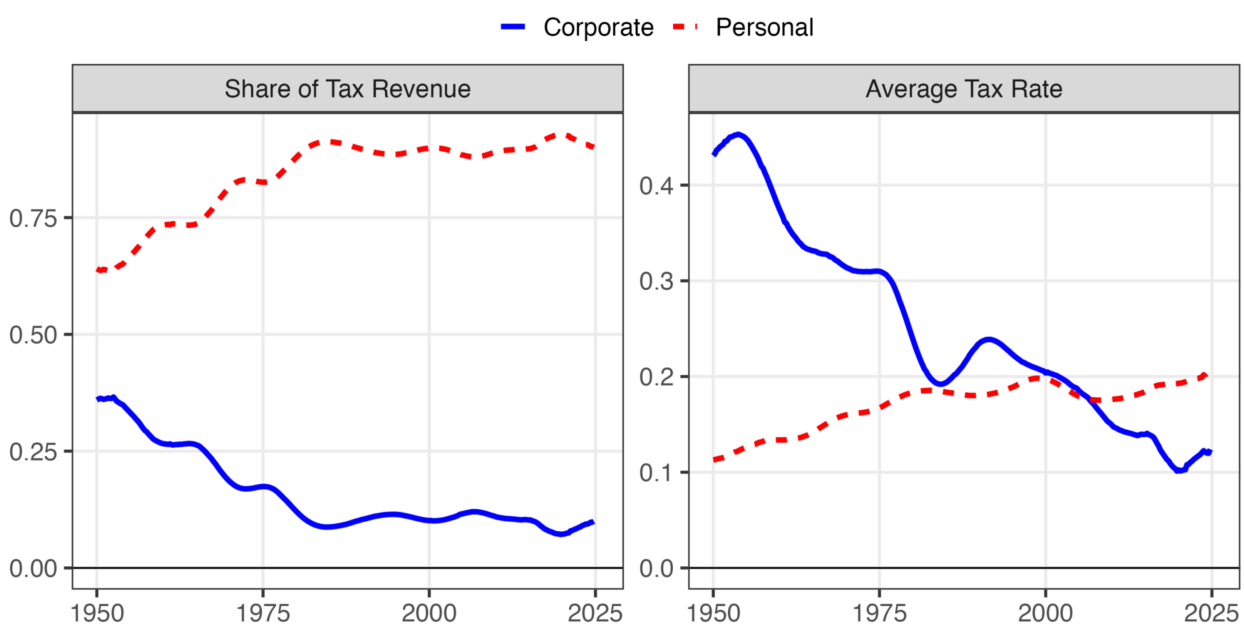
Keywords: Optimal taxation, tax reform, sufficient statistics, marginal cost of public funds, corporate taxation, narrative identification, fiscal policy

*Massachusetts Institute of Technology, jpmejia@mit.edu. First Draft: January 2026. *The views and conclusions contained in this document are mine and should not be interpreted as representing the official policies, either expressed or implied, of any employer or affiliated organization.*

1 Introduction

At the end of World War II, corporate income taxes accounted for nearly half of federal tax revenue. Since then, the share of receipts from personal taxes has climbed to over 85 percent, while the corporate share has dwindled to below 15 percent. This was driven by a secular decline in the corporate tax rate and a corresponding increase in the personal tax rate. Was this sea change in the tax system welfare-improving?

Figure 1: Changes in Postwar Sources of Revenue and Tax Rates



Note: The right panel assumes that personal and corporate taxes make up 100% of tax revenue. In reality, they sum to just over 90%. All series are trend components from the two-sided Christiano and Fitzgerald (2003). I strip out all components with a frequency less than forty quarters. See the appendix for details on data construction.

Within Ramsey models, the shift from corporate to personal taxation was a good idea if it meant the government better balanced revenue and welfare objectives. Whether that balance improved depends on two objects: a welfare gradient defining the marginal welfare cost of each tax, and a revenue gradient defining the efficiency cost of each tax in dynamic general equilibrium. The welfare gradient captures how we value the claimants to personal and corporate income; if policymakers place greater weight on personal income claimants, shifting the burden toward personal taxation may not be desirable. The revenue gradient captures the fiscal externalities associated with taxing each margin—the elasticity of *total government revenue* with respect to each tax rate, including effects on other tax bases statically and dynamically. When corporate taxes fall, the effects extend beyond the corporate base: higher investment could lead to higher wages and expand the personal tax base. That channel constitutes the primary supply-side motivation for corporate tax cuts (Lucas 1990).

This paper, building on a classical literature on marginal tax reform (Dixit 1979; Tirole and Guesnerie 1981), shows how to characterize the alignment of the tax system with the Ramsey optimum using a single scalar metric. Those papers show that, given the welfare and revenue gradients, the optimal revenue-neutral reform is the projection of the welfare gradient onto the hyperplane of reforms that hold revenue fixed—but they do not measure how far the system is from optimal or quantify the welfare gains available from reform. I show that the angle between the two gradients answers both questions. At the Ramsey optimum, welfare and revenue gradients are proportional and no revenue-neutral reform can improve welfare. Away from the optimum, they diverge, and the angle θ between them characterizes distance from optimality. $\cos\theta$ measures proximity to the Ramsey optimum—a direct answer to "how far is the system from optimal?" that comparing pairwise marginal values of public funds struggles to provide.[1] $\sin\theta$ translates that misalignment into the welfare gain available from revenue-neutral reform.

I operationalize the marginal value of public funds framework (Finkelstein and Hendren 2020) to measure these gradients in dynamic general equilibrium. The welfare gradient captures how marginal changes in each tax affect social welfare. By the envelope theorem, the first-order welfare cost of raising a tax is the mechanical burden on taxpayers. Under utilitarian weights this is simply the vector of tax bases, which are directly observable from national accounts data. The revenue gradient is more challenging: it requires estimating how total government revenue responds to tax changes, accounting for behavioral responses across all bases and over time. The relevant behavioral response is what Hendren (2016) terms the policy elasticity—the causal effect of the policy, requiring no decomposition into income and substitution effects. I estimate the revenue gradient using narrative tax shocks to the personal and corporate income tax bases. Mertens and Ravn (2013a) and Cloyne et al. (2022), extending Romer and Romer (2010), construct exogenous shocks for both corporate and personal income taxes, which are sufficient to identify the elasticities of each tax base with respect to a tax shock in dynamic general equilibrium. To get each own- and cross-base elasticity, I cumulate discounted impulse responses with respect to corporate and personal tax shocks, which are then sufficient to construct present-value revenue gradients. The approach bridges structural and sufficient statistics methods: it captures cross-base spillovers that partial equilibrium misses and dynamic accumulation that static approaches ignore, while avoiding the parametric disagreements that divide structural models.

1. The marginal value of public funds as defined in this paper follows Mayshar (1990) and Hendren and Sprung-Keyser (2020). It differs from the traditional MVPF of Atkinson and Stern (1974) and Stiglitz and Dasgupta (1971), which bundles the cost of taxation with the value of expenditure. See Bastani (2025) for a discussion of the distinction.

The elasticities this approach delivers differ from taxable income elasticities in the public finance literature, which measure individual responses to marginal rates holding prices fixed. Standard estimates range from 0.1 to 0.4 (Saez, Slemrod, and Giertz 2012). Badel, Huggett, and Luo (2020) and Kleven et al. (2025) argue that even these understate long-run responses because they miss human capital accumulation. Mertens and Montiel Olea (2018) estimate larger aggregate elasticities around 1.2 using narrative shocks, closer to my approach. My estimates go further: they capture how aggregate bases respond to aggregate average rate changes—the combined effect of marginal rates, deductions, credits, and base definitions that move together in actual legislation—allowing wages, interest rates, and organizational form to adjust in general equilibrium over a five-year horizon. The own-base elasticities are comparable to existing estimates—around 1 for personal income, 4 for corporate profits—but the cross-elasticity from personal tax shocks to the corporate base is large, around 9. The magnitude is a predictable consequence of operating leverage: corporate profits are the residual after firms pay wages and cover costs that are slow to adjust, so a modest output expansion translates into a proportionally larger increase in profits. Because personal tax cuts generate roughly twice the output response of corporate tax cuts, the amplification falls disproportionately on the cross-elasticity.

Applying this framework to postwar U.S. data reveals a dramatic improvement in tax system efficiency. The alignment metric $\cos\theta$ rose from around 0.3 in the early 1950s to nearly one by the 2010s, indicating the tax system moved from substantial misalignment to near-optimality on the personal-corporate margin. The welfare stakes were large: in the early 1950s, a 1 percentage point reform in the optimal revenue-neutral direction would have improved welfare by approximately 0.7% of GDP. By the 2020s, the same reform would yield only 0.05% of GDP. Every major reform since 1980—including the Economic Recovery Tax Act of 1981, the Tax Reform Act of 1986, and the Tax Cuts and Jobs Act of 2017—has a welfare contribution on this margin indistinguishable from zero. Partial equilibrium estimates, which capture own-base responses but ignore cross-base spillovers, would have understated these early gains by an order of magnitude, concluding the system was roughly efficient when substantial welfare improvements were available.

What drove the improvement? A decomposition attributes changes in alignment to two channels: tax wedges, which policy directly controls, and revenue composition, which reflects both policy and structural shifts in the economy. Tax policy changes account for essentially all of the convergence. Reductions in effective corporate tax rates—through investment credits, accelerated depreciation, and statutory rate cuts—lowered the fiscal externality on corporate taxation until it equalized with the personal tax margin. Shifts in the composition of the tax base, including the rise of pass-through entities, contributed little. These results are robust to alternative discount rates, estimation methods, welfare weights, and the treatment of S-corporations.The practical implication: there is little left to gain from reforming the income tax mix. The framework's own decomposition shows where the remaining welfare stake lies: as composition converges to optimality, the entire welfare gradient rotates into the revenue direction, so that any remaining gains require changing the level of taxation rather than its mix.

It is important to note the scope of these findings. The framework evaluates the composition of revenue between corporate and personal income taxes, not the structure within either base. Efficiency gains from flattening rate schedules, broadening bases, or reforming the treatment of capital gains operate on margins this analysis does not capture. The framework also abstracts from payroll taxes, consumption taxes, and state and local taxation, other margins where cross-base spillovers may create misalignment. Finally, the welfare analysis adopts a utilitarian benchmark. The results are robust to varying the relative weights on corporate versus personal income taxpayers. However, the framework treats all personal income taxpayers as a single group, abstracting from the distribution of tax burdens within the personal tax base.

Related Literature Existing approaches to evaluating tax composition face a fundamental trade-off between capturing general equilibrium forces and avoiding parametric disagreement. Structural general equilibrium models capture both cross-base spillovers and dynamic accumulation, but require parametric commitment that leads to fundamentally different conclusions. For example, a subclass of neoclassical models would conclude the postwar shift toward personal taxation was optimal (Chari, Nicolini, and Teles 2020), while overlapping generations models reach the opposite conclusion (Conesa, Kitao, and Krueger 2009). Even within the neoclassical framework, the optimal tax on capital—and hence the optimal direction of reform—depends sensitively on preference specifications and heterogeneity (Straub and Werning 2020). Sufficient statistics approaches avoid this model dependence by working directly with causal elasticities (Chetty 2009; Hendren 2016). However, existing implementations typically rely on static own-base elasticities estimated from micro data (Saez 2001; Hendren and Sprung-Keyser 2020), missing both the cross-base spillovers and dynamic accumulation that are first-order for evaluating tax composition.[2] Both approaches typically characterize global optima, which implicitly define optimal reform directions from any given baseline.[3] Their parametric disagreements over where the optimum lies therefore translate into disagreements over which direction to reform.

I address this by directly characterizing optimal reform directions without computing global

2. Other limitations arise even within the Diamond and Mirrlees (1971) model from which the sufficient statistics approach stems. Bhandari, Borovička, and Yao (2025) point to robustness concerns, echoing Mankiw, Weinzierl, and Yagan (2009) that the Diamond and Saez (2011) result depends critically on the unknown distribution of types. Saez, Slemrod, and Giertz (2012) argue, as this paper does, that it is critical to account for spillovers between instruments—in their case, between capital gains and labor income.

3. Vergara and Swonder (2025) incorporate cross-base fiscal externalities from corporate taxes on the labor base in an optimal corporate tax formula, but their framework is static and characterizes the global optimum rather than evaluating marginal reforms from the status quo.

optima, combining tools from the classical marginal reform literature with modern empirical methods. Following Feldstein (1976), I focus on marginal reforms rather than globally optimal taxes. A classical literature characterizes the directions along which small tax changes improve welfare: Dixit (1975) derives conditions for welfare-improving pairwise swaps, Guesnerie (1977) characterizes the full cone of feasible improvement directions, and Tirole and Guesnerie (1981) construct a process of reform based on gradient projections, showing that the optimal revenue-neutral direction is the projection of the welfare gradient onto the revenue-neutral hyperplane.[4] I build on their approaches to marginal reform to give a first-order characterization of how far the tax system is from optimal, without computing the optimum itself. I extend Ahmad and Stern (1984)'s application to actual tax data in two ways: first, by estimating dynamic general equilibrium elasticities that capture cross-base spillovers rather than requiring complete elasticity matrices from cross-sectional data; second, by extracting scalar alignment metrics from the projection that measure distance from optimality. By estimating those metrics with the correct elasticities, I can draw robust conclusions about the efficiency of the tax mix.

The empirical implementation builds on the marginal value of public funds literature's insight that fiscal externalities can be estimated from causal evidence (Hendren 2016; Finkelstein and Hendren 2020). I estimate these fiscal externalities using narrative identification, constructing present-value revenue gradients from impulse responses to tax shocks. This builds on McKay and Wolf (2023) and Caravello, McKay, and Wolf (2024), who show that impulse responses are sufficient to construct Lucas-critique-robust counterfactuals under alternative policy rules; my application is narrower, evaluating whether actual tax changes moved toward or away from the locally optimal direction rather than simulating counterfactual policies. The geometric approach—testing whether welfare and revenue gradients are aligned—also parallels Barnichon and Mesters (2023), who evaluate monetary policy by testing whether the gradient of the loss function with respect to policy shocks is zero; both frameworks detect suboptimal policies from sufficient statistics without requiring a fully specified structural model.

Roadmap. The remainder of the paper proceeds as follows. Section 2 develops the projection framework for static revenue-neutral reform, showing that aggregate gradients are sufficient statistics and deriving the geometric solution. Sections 3 and 4 implement the framework empirically, estimating present-value revenue and welfare gradients from narrative shocks and evaluating the complete history of postwar federal reforms. Section 5 concludes.

4. Diewert (1978) and Dixit (1979) extend the marginal reform framework to many-consumer economies. Dixit (1979, Section 5) characterizes the best local welfare improvement subject to a feasibility constraint, independently yielding a projection result similar to Tirole and Guesnerie (1981).

2 The Geometry of Reform

This section formalizes the projection framework for evaluating multi-instrument tax reforms. The framework shows that the optimal direction of revenue-neutral reform depends on two sufficient statistics—the welfare gradient and the revenue gradient—which can be estimated from reduced-form evidence without specifying a full structural model. Despite the generality of the setup, the framework yields a simple geometric characterization: the optimal reform is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane, and system efficiency is measured by a single scalar: the alignment between welfare and revenue objectives. This alignment metric is simply the cosine of the angle between the two gradients.

Section 2.1 introduces notation and derives the welfare and revenue gradients. Section 2.2 establishes the projection formula and shows the revenue gradient is a sufficient statistic. Section 2.3 develops scalar metrics that measure proximity to the Ramsey optimum and the scope for revenue-neutral reform. Section 2.4 specializes to the two-instrument case used in the empirical application and demonstrates how partial equilibrium approaches can prescribe reforms in the opposite direction from the optimum.

2.1 Setup: Taxes, Bases, and Gradients

Policy environment. Let the government choose tax rates $\{\tau_{i,t}\}_{i=1,\dots,n}^{t=0,\dots,T}$ across n instruments and $T+1$ periods. Stack these into a policy vector:

$$\boldsymbol{\tau} := (\tau_{1,0}, \dots, \tau_{n,0}, \tau_{1,1}, \dots, \tau_{n,T})^\top \in \mathbb{R}^{n(T+1)}.$$

Tax bases $B_{i,t}(\boldsymbol{\tau})$ depend on the entire policy sequence, and they respond to tax rates for standard economic reasons: behavioral responses grow with rates, and revenue may eventually decline. Importantly, there are spillovers across bases and across time. A change in $\tau_{i,t}$ affects not only base i at date t but also other bases at other dates. For example, a corporate tax cut today raises investment, growing the capital stock and expanding future labor tax bases through production complementarity. The government discounts its stream of revenue at β, so present-value revenue is:

$$R(\boldsymbol{\tau}) := \sum_{t=0}^{T} \beta^t \sum_{i=1}^{n} \tau_{i,t} B_{i,t}(\boldsymbol{\tau}). \tag{1}$$

I assume that aggregate tax base responses are well-behaved.

Assumption 1. *$R(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

This is an assumption about aggregate equilibrium responses, not individual behavior. It requires the equilibrium to be unique and to vary smoothly with policy. Standard neoclassical, heterogeneous agent, and overlapping generations models all satisfy this condition. The key exclusion is economies with discontinuous equilibrium responses, such as models with multiple equilibria where small tax changes trigger discrete jumps. Given differentiability, define the present-value elasticity

$$\varepsilon_{ik} := \sum_{s=0}^{T} \beta^s \frac{\partial \ln B_{i,s}}{\partial \ln(1-\tau_{k,0})} \tag{2}$$

measuring the cumulative discounted response of base i to a reform of tax k. Entries with $i = k$ capture own-base responses; entries with $i \neq k$ capture cross-base spillovers. Both include effects that accumulate over time—through capital deepening, human capital investment, or anticipatory adjustments—as well as contemporaneous responses. The object is well-defined for any reform persistence; Section 3 describes how to normalize estimates from transitory shocks into permanent-equivalent elasticities, bridging a gap between the public finance literature, which typically defines elasticities with respect to permanent reforms, and the macroeconomics literature, which estimates impulse responses to shocks that decay over time.

The revenue gradient. The revenue gradient $r := \nabla_{\boldsymbol{\tau}} R \in \mathbb{R}^{n(T+1)}$ measures how present-value revenue responds to marginal tax changes. Consider raising $\tau_{i,t}$ by a small amount. There are three effects on present-value revenue: a mechanical revenue gain from taxing the existing base, an own-base behavioral loss as base i shrinks, and spillovers to other bases. Formally, define discounted bases $\tilde{B}_{i,t} := \beta^t B_{i,t}$. Then:

$$r_{i,t} = \tilde{B}_{i,t} \left[1 - \frac{\tau_{i,t}}{1-\tau_{i,t}} \varepsilon_{ii} - \Lambda_{i,t} \right] \tag{3}$$

where $\Lambda_{i,t}$ aggregates all cross-base spillovers:

$$\Lambda_{i,t} = \frac{1}{1-\tau_{i,t}} \sum_{k \neq i} \tau_{k,t} \frac{\tilde{B}_{k,t}}{\tilde{B}_{i,t}} \varepsilon_{ki}. \tag{4}$$

The three components have clear economic interpretations. The mechanical effect (+1) is how much revenue would rise absent any behavioral response. The own-base effect (ε_{ii}) captures how base i contracts when its tax rises—including both the contemporaneous response and accumulated future losses from reduced investment or human capital. The spillover term ($\Lambda_{i,t}$) captures how other bases respond—for instance, a corporate tax increase that erodes the labor base through production complementarity. Because the present-value elasticities already aggregate dynamic responses, the revenue gradient formula has the same structure whether spillovers operate within a period or across decades. These spillovers can be decomposed by when revenue effects occur:

$$r_{i,t} = \underbrace{\beta^t \frac{\partial R_t}{\partial \tau_{i,t}}}_{\text{Contemporaneous}} + \underbrace{\sum_{s>t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Persistence}} + \underbrace{\sum_{s<t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Anticipation}}. \tag{5}$$

The contemporaneous term captures revenue effects at date t itself. The persistence and anticipation terms capture intertemporal spillovers: how tax changes at t affect revenue at other dates. Both are embedded in $\Lambda_{i,t}$. Two examples illustrate the economic channels underlying these spillovers.

Example 1: Human capital accumulation (persistence, own-base). Consider an increase in the labor tax at $t = 0$. Workers respond by reducing human capital investment (education, training), as the after-tax return to skill accumulation has fallen. This shrinks the labor tax base $B_{L,t}$ in all future periods:

$$\frac{\partial B_{L,t}}{\partial \tau_{L,0}} < 0 \quad \text{for all } t > 0. \tag{6}$$

The persistence terms in equation (5) capture these effects. Because the revenue losses accumulate over time, the present-value revenue gradient $r_{L,0}$ is smaller than the contemporaneous effect alone would suggest. Badel, Huggett, and Luo (2020) and Kleven et al. (2025) show that accounting for human capital accumulation substantially raises the long-run own-base elasticity of labor income relative to static micro estimates, underscoring why present-value elasticities are the appropriate object for evaluating tax reform.

Example 2: Capital accumulation (persistence, cross-base). A corporate tax increase at $t = 0$ reduces investment, shrinking the capital stock over time. If capital and labor are complements in production ($F_{KL} > 0$), the reduced capital stock lowers the marginal product of labor (wages), eroding the labor tax base in future periods:

$$\frac{\partial B_{L,t}}{\partial \tau_{K,0}} < 0 \quad \text{for all } t > 0.$$

The present-value revenue gradient for the capital tax includes this cumulative cross-base effect:

$$r_{K,0} = \beta^0 \frac{\partial R_0}{\partial \tau_{K,0}} + \sum_{t=1}^{T} \beta^t \left(\frac{\partial R_{K,t}}{\partial \tau_{K,0}} + \frac{\partial R_{L,t}}{\partial \tau_{K,0}} \right),$$

where the second term in the sum captures the cross-base spillover. This channel—production complementarity linking corporate taxes to labor revenue—is the primary supply-side motivation for corporate tax cuts (Lucas 1990).[5]

Welfare. Let $W(\boldsymbol{\tau})$ denote the money-metric social welfare cost of the tax system, measured in present-value dollars. Higher W is worse. We measure welfare by the present-value compensating variation aggregated across agents using social welfare weights $\{\omega_g\}$.

Assumption 2 (Welfare regularity)**.** *$W(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

The welfare gradient $g := \nabla_{\boldsymbol{\tau}} W \in \mathbb{R}^{n(T+1)}$ has a simple characterization. By the envelope theorem applied to individual optimization, the first-order welfare cost of raising $\tau_{i,t}$ is the mechanical burden on taxpayers, discounted to date zero:

$$g_{i,t}(\bar{\boldsymbol{\tau}}) = \beta^t \tilde{\gamma}_{i,t} B_{i,t}(\bar{\boldsymbol{\tau}}). \tag{7}$$

Each piece has an economic reason. The discount factor β^t appears because the burden lands at date t. The base $B_{i,t}$ appears because that is what the tax is levied on. The incidence share $\tilde{\gamma}_{i,t} := \sum_g \omega_g s_{g,i,t}$ appears because that is who bears it, where $s_{g,i,t}$ is group g's share of the incidence from instrument i at date t.

The key to this result is that it applies the envelope theorem to individual optimization, not social optimization. Each agent takes the tax sequence as given and optimizes over their lifetime—labor supply, savings, human capital, organizational form. At that private optimum, the agent's behavioral response to a tax perturbation has zero first-order welfare effect. When there are multiple distorted margins, behavioral responses to $\tau_{i,t}$ do change tax payments on other bases and at other dates. But these are transfers between the agent and the government: they affect revenue (entering through $\Lambda_{i,t}$ in the revenue gradient) without creating additional welfare costs beyond the mechanical burden. The welfare gradient captures who bears the tax; the revenue gradient captures what the behavioral responses cost the fisc. The separation is exact to first order.

This separation requires that the taxes in the framework are the only wedge between private prices and social resource costs (Hendren 2016). When behavioral responses to $\tau_{i,t}$ interact with other distortions, they generate welfare effects beyond the mechanical burden that equation (7) does not capture. Three categories matter in principle. First, *taxes outside the system.* If agents face payroll taxes, state and local income taxes, or consumption taxes that the framework does

5. A third channel is anticipation: forward-looking agents adjust behavior today in response to announced future reforms, so that $\partial B_{K,t}/\partial \tau_{K,5} \neq 0$ for $t < 5$. These effects mean the present-value revenue cost of a reform exceeds the contemporaneous calculation. Mertens and Ravn (2012) exploit announcement timing to estimate these effects directly. The empirical implementation in Section 3 sidesteps this channel by excluding reforms implemented more than 90 days after announcement.

not model, behavioral responses to federal income tax changes create fiscal externalities on those governments' budgets—welfare-relevant costs absent from the revenue gradient r, which measures only federal income tax revenue. Second, *non-tax distortions*. Market power, externalities, or informational frictions create additional wedges between private and social costs. If a corporate tax cut stimulates production in a polluting industry, the behavioral response has a welfare cost beyond the mechanical tax burden. Third, *binding constraints*. If agents face binding borrowing constraints, behavioral responses to a tax change can have first-order welfare effects because the agent is not at an interior optimum on the constrained margin.[6] In each case, the welfare gradient would require additional terms beyond the tax bases. The framework as implemented treats federal personal and corporate income taxes as the relevant distortions, maintaining the same assumption as the MVPF literature when applied to federal tax policy (Hendren 2016; Hendren and Sprung-Keyser 2020). This assumption is most defensible for the early postwar period, when state and local taxes were smaller and the economy was relatively closed, and becomes more tenuous as other tax margins grow in importance.

Under utilitarian weights, $\tilde{\gamma}_{i,t} = 1$ for domestic taxes and the welfare gradient simplifies to discounted tax bases:

$$g_{i,t} = \beta^t B_{i,t}. \tag{8}$$

This is directly observable from national accounts data. It also cleanly separates the empirical challenge: the welfare gradient g can be constructed from baseline data, while the revenue gradient r must be estimated.

The marginal value of public funds. Pairing the welfare and revenue gradients yields the marginal value of public funds for each instrument-date pair:[7]

$$\text{MVPF}_{i,t} = \frac{g_{i,t}}{r_{i,t}} = \frac{\tilde{\gamma}_{i,t}}{1 - \frac{\tau_{i,t}}{1-\tau_{i,t}}\varepsilon_{ii} - \Lambda_{i,t}}.$$

At a global optimum, MVPFs are equalized across instruments and dates (Appendix I.1 derives this formally). When MVPFs differ, which is the typical case empirically, there exists scope for welfare-improving revenue-neutral reform. Section 2.2 characterizes the optimal direction of such reforms.

6. The differentiability assumption on W implicitly requires that such constraints do not bind at the margin.

7. See Mayshar (1990), Hendren and Sprung-Keyser (2020), and Bastani (2025) for a more detailed analysis of the MVPF.

2.2 Policy as Projection

The problem of reform arises when MVPFs differ across instruments. When this occurs, there exist directions of tax adjustment that improve welfare while holding total revenue fixed. The planner's local problem is to identify the reform direction $d\boldsymbol{\tau}$ that maximizes the first-order welfare gain subject to revenue neutrality:

$$\max_{d\boldsymbol{\tau}} \quad -g(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} \quad \text{subject to} \quad r(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} = 0. \tag{9}$$

The welfare gradient g measures how marginal tax changes affect social welfare; the revenue gradient r measures how they affect government revenue, accounting for all behavioral responses including cross-base spillovers. This constrained optimization problem has a simple geometric solution: the optimal reform direction is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane.

Lemma 1 (Gradient projection)**.** *Let $\bar{\boldsymbol{\tau}}$ be a baseline with welfare gradient $g(\bar{\boldsymbol{\tau}})$ and revenue gradient $r(\bar{\boldsymbol{\tau}}) \neq 0$. The optimal revenue-neutral reform direction is*

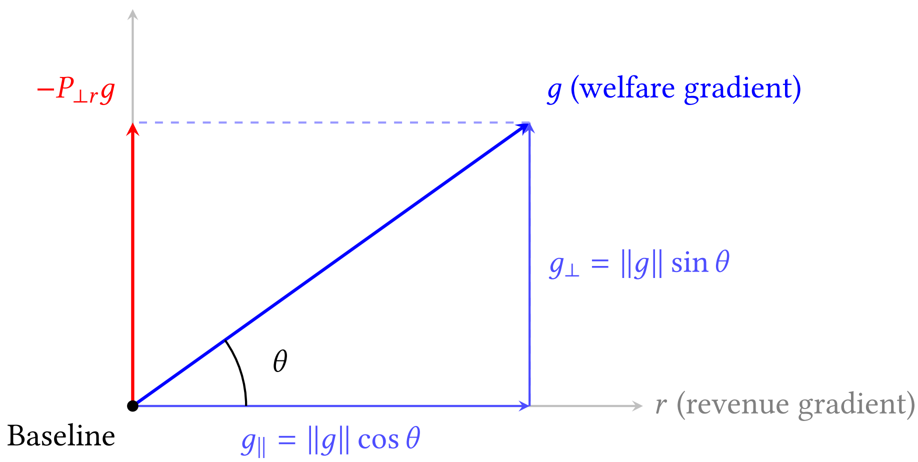
$$d\boldsymbol{\tau}^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}, \tag{10}$$

where $\alpha > 0$ is the reform magnitude in percentage points and $P_{\perp r} := I - r(r^\top r)^{-1} r^\top$ is the projection operator onto the hyperplane orthogonal to r.

The derivation is in Appendix I.2. Tirole and Guesnerie (1981) construct a process of revenue-neutral reform based on gradient projections for linear commodity taxes. Dixit (1979) independently derives the same geometric structure—the optimal local welfare improvement is the projection of the welfare gradient onto the feasible set—in many-consumer economies. The contribution here is twofold. First, the revenue gradient incorporates intertemporal spillovers and cross-base fiscal externalities, so the projection aggregates dynamic general equilibrium responses that those static frameworks do not capture. Second, Section 2.3 extracts scalar alignment metrics from the projection that measure how far the tax system is from optimal—an object neither framework provides.[8] Tirole and Guesnerie (1981) establish the projection result in static economies; here the revenue gradient incorporates intertemporal spillovers, so the projection aggregates dynamic responses that their framework does not capture.

8. The projection framework nests several classical results as special cases: Chamley's (1986) zero capital tax theorem, Barro's (1979) tax smoothing result, and Ramsey's (1927) optimality condition emerge when present-value revenue gradients take particular limiting forms. Appendix K develops these connections. Appendix J extends the projection formula to deficit-financed reforms, where the welfare and revenue gradients are discounted at different rates.

Figure 2: Geometric Decomposition of Optimal Policy



Note: The welfare gradient g (blue) decomposes into a parallel component $g_{||} = ||g|| \cos\theta$ along the revenue gradient r and an orthogonal component $g_{\perp} = ||g|| \sin\theta$ perpendicular to r. The optimal revenue-neutral reform direction is $-P_{\perp r}g$ (red), which points opposite to the perpendicular component of welfare cost.

Figure 2 illustrates this decomposition. The welfare gradient g (blue) splits into a parallel component $g_{||} = ||g|| \cos\theta$ along the revenue gradient r and an orthogonal component $g_{\perp} = ||g|| \sin\theta$ perpendicular to r. If we could move freely along g—ignoring the revenue constraint—we would achieve the steepest welfare improvement. However, moving along g would change total revenue, violating revenue-neutrality. The projection operator resolves this tension by extracting the perpendicular component, yielding $P_{\perp r}g$ (red) as the optimal revenue-neutral reform direction. Because the revenue gradient is the only behavioral object in the projection formula, welfare conclusions follow from the estimated gradients regardless of the specific channels through which spillovers operate.

2.3 Alignment Metrics

The geometry in Figure 2 depends entirely on the angle θ between the welfare and revenue gradients. Existing characterizations of tax optimality focus on MVPF equalization, which answers a binary question—equal or not—without measuring how costly the deviation is. The angle θ fills this gap: $\Delta W = \alpha ||g|| \sin\theta$ converts the MVPF gap into the welfare gain available from a reform of size α in the optimal direction, an object that comparing MVPFs across instruments cannot deliver. Define system alignment as:

$$\cos\theta = \frac{g^{\top} r}{||g|| \, ||r||} \in [-1, 1]. \tag{11}$$

This is the standard inner product formula for the angle between vectors.

Proposition 1 (System alignment and revenue-neutral potential)**.** *The efficiency of the tax system is characterized by two scalar metrics:*

1. ***System alignment****:* $\cos\theta$ *measures proximity to the Ramsey optimum. When* $\cos\theta = 1$*, welfare and revenue objectives are perfectly aligned and the system is optimal. When* $\cos\theta = 0$*, the gradients are orthogonal and inefficiency stems purely from tax composition. When* $\cos\theta < 0$*, raising taxes increases welfare cost but reduces revenue.*

2. ***Revenue-neutral potential****:* $\sin\theta$ *measures the share of potential welfare gains achievable through revenue-neutral reform. The first-order welfare gain from a reform of size* α *(in percentage points) in the optimal direction is*

$$\Delta W \approx -\alpha \|g\| \sin\theta. \tag{12}$$

Proof. The optimal reform direction is $d\tau^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}$, where α is the reform magnitude in percentage points. The first-order welfare change is $\Delta W = g^\top d\tau^* = -\alpha \|P_{\perp r} g\| = -\alpha \|g\| \sin\theta$. □

For intermediate values, the metrics have quantitative bite. When $\cos\theta = 0.5$, for example, $\sin\theta \approx 0.87$—nearly nine-tenths of potential welfare gains are achievable revenue-neutrally. Negative values ($\cos\theta < 0$) indicate that raising taxes increases welfare cost but reduces total revenue; this occurs when cross-base spillovers dominate own-revenue effects. Appendix Table I.1 summarizes. While $\cos\theta$ measures how far the system is from optimal, $\sin\theta$ measures what can be done about it. The revenue-neutral potential $\sin\theta$ reflects a fundamental decomposition: the welfare gradient g splits into a component parallel to the revenue gradient (which revenue-neutral reform cannot address) and a perpendicular component (which it can). Because $\sin\theta$ depends only on the perpendicular component, it is maximized when the gradients are orthogonal ($\theta = 90°$) and declines symmetrically toward zero as θ approaches either 0° or 180°. The limiting case $\sin\theta = 0$ is the geometric counterpart of the optimality criterion in Saez and Stantcheva (2016): at the optimum, no budget-neutral perturbation increases the weighted sum of gains and losses, precisely because the welfare gradient has no component perpendicular to the revenue gradient. Whether the system is moderately misaligned on the right side ($\theta = 60°$) or the wrong side ($\theta = 120°$) of the Laffer curve, the scope for revenue-neutral improvement is the same.

Together, $\cos\theta$ and $\sin\theta$ fully characterize the efficiency of tax composition—but they also reveal what composition cannot address. The perpendicular component $\|g\| \sin\theta$ measures the welfare gains available from revenue-neutral reform. The parallel component $\|g\| \cos\theta$ measures the welfare stake in the level of taxation—gains that require changing total revenue, not just its mix. As the system approaches alignment ($\cos\theta \to 1$), the perpendicular component vanishes and the entire welfare gradient lies along the revenue direction. The framework cannot say whether taxes should be higher or lower—that requires comparing the cost of revenue to the value of spending—but it can quantify how large that question is. It also requires only local information. Assumption 1 asks that the revenue function be differentiable at the baseline $\bar{\boldsymbol{\tau}}$; it does not require that the elasticities embedded in the gradients remain stable under counterfactual policies. If a reform is large enough to change the elasticities, the welfare gain formula (12) ceases to be exact—but the alignment diagnostic remains valid as a characterization of the current system. The framework answers "is the current tax mix efficient?" and "which direction should the next marginal reform go?," not "what would welfare be at the global optimum." This locality allows the analysis to remain agnostic about whether elasticities are policy-invariant, sidestepping the parametric disagreements that divide structural models computing global optima.

Beyond measuring system efficiency, the framework evaluates whether historical reforms moved in the right direction. For an observed reform $d\boldsymbol{\tau}^{\text{actual}}$, we compute its directional alignment with the optimal reform $d\boldsymbol{\tau}^*$:

$$\cos\phi = \frac{(d\boldsymbol{\tau}^{\text{actual}})^\top d\boldsymbol{\tau}^*}{\|d\boldsymbol{\tau}^{\text{actual}}\| \, \|d\boldsymbol{\tau}^*\|}. \tag{13}$$

This statistic inherits the same interpretation as $\cos\theta$. When $\phi \approx 0^\circ$ ($\cos\phi \approx 1$), the actual reform moved nearly parallel to the optimal direction. It was well-designed given the economy's position. When $\phi \approx 90^\circ$ ($\cos\phi \approx 0$), the reform was orthogonal, neither helping nor hurting. When $\phi > 90^\circ$ ($\cos\phi < 0$), the reform moved opposite to the welfare-improving direction, making the system less efficient. Together, $\cos\theta$ (system position) and $\cos\phi$ (reform direction) provide a complete characterization of tax policy.[9]

The projection formula also enforces the classic intuition of MVPF equalization. Rewriting equation (10) component-wise:

$$d\tau_i^* \propto -g_i + \left(\frac{r^\top g}{r^\top r}\right) r_i = -r_i \left(\frac{g_i}{r_i} - \frac{r^\top g}{r^\top r}\right).$$

Define $\text{MVPF}_i = g_i/r_i$ and the revenue-weighted average $\overline{\text{MVPF}} = (r^\top g)/(r^\top r)$:

Corollary 1 (MVPF equalization)**.** *The optimal reform direction satisfies*

$$d\tau_i^* \propto -r_i(MVPF_i - \overline{MVPF}). \tag{14}$$

9. The reform evaluation statistic $\cos\phi$ is related to the welfare effectiveness index of Raimondos-Møller and Woodland (2014), who use the cosine between a proposed tariff reform and the welfare gradient to rank reforms in a small open economy. The system-level diagnostic $\cos\theta$, measuring alignment between welfare and revenue gradients as separate objects, is new.

The projection automatically lowers taxes on instruments with above-average MVPFs and raises taxes on instruments with below-average MVPFs. That MVPFs are equalized at the optimum is a standard implication of the Ramsey first-order conditions (Jacobs 2018; Bastani 2025). The geometric derivation adds a quantitative dimension: when MVPFs are *not* equalized, the angle θ measures how costly the deviation is, converting the MVPF gap into the welfare gain $\Delta W = \alpha \|g\| \sin\theta$ available from reform in the optimal direction—an object that comparing MVPFs across instruments cannot deliver.

2.4 Specializing to Two Instruments

The empirical application focuses on personal (L) and corporate (C) income taxes. In this case, the alignment condition reduces to a simple requirement: the marginal fiscal externality from each tax must be equal. This externality depends on three observable quantities—tax wedges, revenue composition, and behavioral elasticities including cross-base spillovers—which jointly determine how far the system is from optimal and what drives changes in alignment over time.

Fiscal externalities and alignment decomposition. Corollary 1 shows that the optimal reform equalizes MVPFs across instruments. Specializing to personal (L) and corporate (C) taxes, and adopting utilitarian welfare ($g_{i,t} = \tilde{B}_{i,t}$), MVPF equalization requires $r_L/\tilde{B}_{L,t} = r_C/\tilde{B}_{C,t}$. From equation (3), this ratio equals

$$\frac{r_i}{\tilde{B}_{i,t}} = 1 - \eta_i \varepsilon_{ii} - \Lambda_{i,t}$$

where $\eta_i = \tau_i/(1-\tau_i)$ is the tax wedge on instrument i. The spillover term $\Lambda_{i,t}$ from equation (4) involves base ratios, which can be converted to revenue ratios using $\tilde{B}_{j,t}/\tilde{B}_{i,t} = (R_j/R_i)(\tau_i/\tau_j)$. Substituting:

$$\Lambda_{L,t} = \eta_L \cdot \frac{R_C}{R_L} \cdot \varepsilon_{CL}, \qquad \Lambda_{C,t} = \eta_C \cdot \frac{R_L}{R_C} \cdot \varepsilon_{LC}.$$

Collecting terms, define the *effective elasticity* for each instrument:

$$\tilde{\varepsilon}_L = \varepsilon_{LL} + \frac{R_C}{R_L}\varepsilon_{CL}, \qquad \tilde{\varepsilon}_C = \varepsilon_{CC} + \frac{R_L}{R_C}\varepsilon_{LC}.$$

The effective elasticity captures both the own-base response and the revenue-weighted spillover from the other base. Then

$$\frac{r_i}{\tilde{B}_{i,t}} = 1 - \eta_i \tilde{\varepsilon}_i \equiv 1 - A_i$$

where $A_i = \eta_i \tilde{\varepsilon}_i$ is the *marginal fiscal externality*—the total revenue leakage (as a share of the mechanical revenue gain) from raising tax i. MVPF equalization reduces to $A_L = A_C$: alignment requires equalizing marginal fiscal externalities across instruments.

This representation connects to the classical Ramsey intuition. The inverse elasticity rule states that optimal tax rates should be inversely proportional to elasticities. With cross-base spillovers, the relevant object is not the own-elasticity ε_{ii} but the effective elasticity $\tilde{\varepsilon}_i$, which incorporates the fiscal externality imposed on other bases. A tax instrument with large positive spillovers (high ε_{ji}) has a higher effective elasticity than its own-base response would suggest, calling for a lower tax wedge.

Corollary 2 (Decomposition of alignment)**.** *In the two-instrument case, changes in system alignment can be attributed to three policy levers: (i) tax wedges η_i, (ii) revenue composition R_C/R_L, and (iii) behavioral elasticities ε_{ij}. Holding elasticities fixed, alignment improves when tax wedges and revenue shares adjust to equalize fiscal externalities $A_i = \eta_i\tilde{\varepsilon}_i$ across bases.*

The corollary decomposes alignment changes into observable components. Consider an economy that moves from misalignment ($A_L \neq A_C$) toward alignment ($A_L = A_C$). This convergence can occur through three channels. First, *tax wedges*: if one instrument has an excessive fiscal externality ($A_L > A_C$), reducing the wedge η_L directly lowers A_L. Second, *revenue composition*: the revenue ratio R_C/R_L enters the effective elasticities, so shifts in the tax base mix change both $\tilde{\varepsilon}_L$ and $\tilde{\varepsilon}_C$. Third, *behavioral elasticities*: if the underlying elasticities ε_{ij} change over time—due to globalization, financial innovation, or changing industrial structure—the alignment condition shifts even with fixed policy. Section 3 quantifies the contribution of each channel to postwar alignment improvements.

Why general equilibrium matters. The effective elasticity $\tilde{\varepsilon}_i$ incorporates cross-base spillovers through the terms ε_{CL} and ε_{LC}. What happens if a planner ignores these and uses only the own-base elasticity? Define the PE revenue gradient by setting $\Lambda_{i,t} = 0$:

$$r_{i,t}^{\mathrm{PE}} = \tilde{B}_{i,t}\left[1 - \frac{\tau_{i,t}}{1-\tau_{i,t}}\varepsilon_{ii}\right]. \tag{15}$$

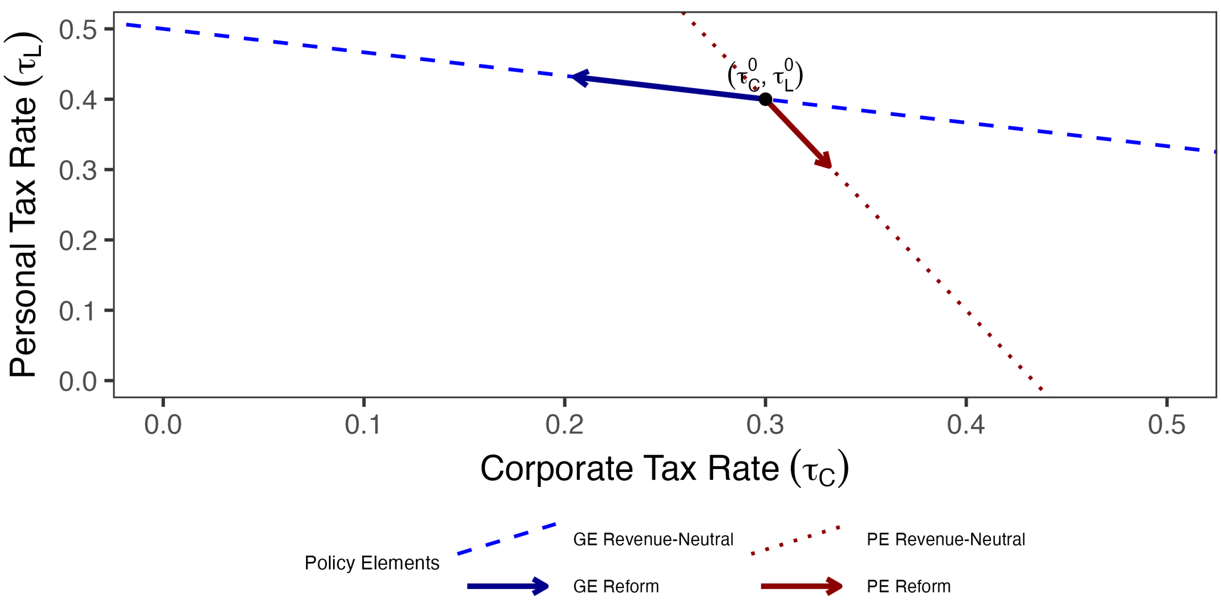
This corresponds to using only the own-base elasticity ε_{ii}, ignoring how each instrument affects the other's base. If capital and labor are complements in production, a higher corporate tax reduces investment, shrinking the capital stock and lowering wages, which erodes the personal tax base. Vergara and Swonder (2025) formalize this wage-incidence channel in an optimal corporate tax formula, showing it creates a strategic complementarity between corporate taxes and in-work transfers. The present-value elasticities here capture this channel, among others. This spillover means the true revenue gradient for corporate taxes is smaller than the PE estimate:

$$r_C^{\mathrm{GE}} = \tilde{B}_{C,t}\left(1 - \eta_C\varepsilon_{CC} - \Lambda_{C,t}\right) < r_C^{\mathrm{PE}} = \tilde{B}_{C,t}\left(1 - \eta_C\varepsilon_{CC}\right),$$

where $\Lambda_{C,t} > 0$ captures the cross-base spillover. The PE planner thinks corporate taxes raise more revenue than they actually do.

Figure 3 illustrates the consequences. The GE revenue-neutral line (blue) and PE revenue-neutral line (red) have different slopes because PE ignores spillovers. The projection formula tells the planner to project the welfare gradient onto the revenue-neutral line—but the PE planner projects onto the wrong line. The result: GE prescribes lowering corporate taxes and raising personal taxes, while PE prescribes the opposite. When cross-base spillovers are large, partial equilibrium analysis can reverse the optimal reform direction entirely.

Figure 3: The Geometric Failure of Partial Equilibrium



Note: Starting from baseline policy (black dot), the partial equilibrium reform (red arrow) projects onto the PE revenue-neutral line (red dashed) and prescribes raising corporate taxes. The general equilibrium reform (blue arrow) projects onto the correct revenue-neutral line (blue dashed) and prescribes lowering corporate taxes—the opposite direction.

The example illustrates a general point: the alignment metric $\cos\theta$ is only as good as the revenue gradient used to compute it. The gap between partial and general equilibrium is one dimension along which the revenue gradient can go wrong. The other is dynamics. Most existing MVPF calculations use static partial equilibrium—only own-base contemporaneous responses from micro data. The framework requires the opposite extreme: dynamic general equilibrium elasticities that cumulate cross-base spillovers over time. Between these sit two intermediate cases: dynamic PE, which cumulates own-base responses over time but misses cross-base fiscal externalities, and static GE, which captures contemporaneous cross-base spillovers but misses their compounding through capital accumulation. The ranking between these intermediate cases is genuinely ambiguous—it depends on whether cross-base spillovers or persistence is the larger channel empirically. In the personal/corporate application, the cross-base channel dominates: ε_{CL} is an order of magnitude larger than the own-base terms, and it is this fiscal externality that drives most of the early misalignment. But this ranking need not hold in other settings. Section 3 estimates dynamic general equilibrium elasticities directly, capturing both channels simultaneously.

3 Estimation of General Equilibrium Dynamic Elasticities

As an application of the geometric framework, I examine whether the postwar decline in corporate tax burdens relative to personal tax burdens moved the U.S. tax system toward or away from efficiency. Figure 1 shows that the corporate share of federal revenue fell from nearly 50% in the late 1940s to around 10% by 2019, while the personal income tax share grew correspondingly. This shift raises a natural question: did the rebalancing improve welfare, or would revenue-neutral reforms in the opposite direction—higher corporate taxes, lower personal taxes—have been preferable?

The projection framework answers this question by characterizing the alignment between welfare and revenue objectives. The alignment metric $\cos\theta$ measures how close the tax system is to the Ramsey optimum, while $\sin\theta$ measures the share of potential welfare gains achievable through revenue-neutral rebalancing. If welfare and revenue gradients point in similar directions ($\cos\theta$ near one), the system is near-optimal and the postwar shift may have been efficient. If the gradients are misaligned ($\cos\theta$ near zero or negative), substantial welfare gains were available, and whether the observed reforms moved toward or away from the optimum depends on the sign of their directional alignment with the optimal reform. In this section, I assemble the ingredients for estimating these alignment statistics—the revenue and welfare gradients for personal and corporate income taxes—using narratively-identified VARs. Section 4 then applies these estimates to evaluate historical reforms.

3.1 From Theory to Estimation

The key empirical object is the revenue gradient r_j from equation (3), which captures how an exogenous change in tax rate j affects present-value government revenue, accounting for dynamic adjustments and general equilibrium spillovers across tax bases. Specializing to the two-instrument case from Section 2.4:

$$r_j = \tilde{B}_j \cdot \left[1 - \frac{\tau_j}{1-\tau_j}\varepsilon_{jj} - \frac{\tau_k}{1-\tau_j}\frac{\tilde{B}_k}{\tilde{B}_j}\varepsilon_{kj}\right], \tag{16}$$

where $\tilde{B}_j$ is the discounted base and each present-value elasticity ε_{ij} measures the cumulative discounted response of base i to a permanent change in tax j, as defined in equation (2). The welfare gradient under utilitarian weights is:

$$g_j = \tilde{B}_j, \tag{17}$$

since the common annuity factor cancels when computing alignment metrics. The welfare gradient is directly observable from national accounts data. The empirical challenge is estimating the four elasticities ε_{ij} that enter the revenue gradient.

I estimate these elasticities from impulse responses of log tax bases to shocks in log retention rates $\ln(1-\tau_j)$. Cumulating and discounting the impulse responses yields the present-value base response; dividing by the cumulative retention rate persistence $\kappa_j := \sum_{s=0}^{H} \beta^s \cdot \mathrm{IRF}_s(\ln(1-\tau_j))$ converts these into permanent-equivalent elasticities. This normalization is exact when base responses scale linearly with shock size, so that the impulse response to a shock of size a is a times the unit-shock response. The baseline uses $\beta = 0.9926$ (approximately 3% annual discount rate) and $H = 20$ quarters, following the five-year horizon in Mertens and Montiel Olea (2018).

3.2 Identification

The narrative approach provides the exogenous variation needed to estimate the revenue gradient. I follow the identification strategy developed by Romer and Romer (2010) and extended by Mertens and Ravn (2013a) to distinguish personal and corporate income taxes. Romer and Romer (2010) read the Congressional Record and Treasury documents to classify major federal tax legislation from 1945 to 2007 according to the motivations policymakers articulated at the time. They code a reform as exogenous when the legislative record indicates policymakers pursued objectives unrelated to the contemporaneous business cycle—such as long-run deficit reduction, ideological commitments, or tax simplification—rather than short-run stabilization. Mertens and Ravn (2013a) refine the series by separating personal and corporate tax changes and excluding reforms implemented more than 90 days after announcement to avoid anticipation effects. I extend their series through 2019 using the classification in Cloyne et al. (2022).

Each reform receives a quantitative measure equal to the projected revenue impact scaled by the lagged tax base, as announced at the time of enactment. This scaling delivers an approximation to the change in the average tax rate on that instrument. The resulting series are denoted m_t^{PI} for personal income taxes and m_t^{CI} for corporate income taxes.

The narrative measures serve as instruments for latent structural tax shocks rather than entering as regressors directly. As discussed in Mertens and Ravn (2013a), the two instruments are contemporaneously correlated because Congress often adjusts personal and corporate taxes in the same legislation. This correlation means the instruments identify only the two-dimensional subspace spanned by the structural shocks, not the individual shocks themselves. Additional restrictions are required to separate them.[10] Following Mertens and Ravn (2013a) and Cloyne et al. (2022), I use the narrative measures as external instruments to identify the contemporaneous impact matrix A_0. Let Z_t denote the vector of endogenous variables. A reduced-form VAR produces residuals u_t, which relate to structural shocks ε_t via:

$$u_t = A_0 \varepsilon_t$$

where ε_t are mutually orthogonal structural shocks with unit variance and A_0 is the contemporaneous impact matrix. The external instruments identify the columns of A_0 corresponding to tax shocks through the moment conditions:

$$E[m_t \varepsilon_{tax,t}'] = \Phi \neq 0, \quad E[m_t \varepsilon_{other,t}'] = 0$$

where $m_t = (m_t^{PI}, m_t^{CI})'$, $\varepsilon_{tax,t}$ denotes the two structural tax shocks, and $\varepsilon_{other,t}$ denotes all other structural disturbances. The first condition requires the instruments to correlate with the tax innovations; the second requires them to be uncorrelated with non-tax shocks.

The two narrative instruments are correlated, so the proxy-SVAR identifies a two-dimensional subspace of tax shocks but does not uniquely separate personal from corporate shocks within this subspace. To isolate each shock, I apply a Cholesky decomposition to the identified subspace, ordering the shock of interest last. When estimating responses to the personal tax shock, I order the personal tax rate last, imposing that the personal shock has no contemporaneous effect on the corporate rate; when estimating responses to the corporate shock, I reverse the ordering. This approach does not require a single structural model that simultaneously satisfies both restrictions. Each column of the elasticity matrix is estimated from a separate single-shock experiment, and the alignment metric requires only the individual revenue gradients—not a joint structural interpretation of both shocks.

The revenue gradient requires the total derivative of tax bases with respect to tax rates: the full response a policymaker would observe, including effects operating through output, invest-

10. Cloyne et al. (2022) shows that both VAR-identified instruments are strong.

ment, and wages. The four-variable specification is not a parsimony choice but a theoretical requirement. Adding variables like GDP or government spending would partial out these channels, estimating a conditional elasticity rather than the total elasticity the theory requires. Narrative identification does not require additional controls; exogeneity comes from the Romer-Romer selection criterion, not from the conditioning set. Appendix G develops this argument in detail and shows that a seven-variable specification following Mertens and Ravn (2013a) yields qualitatively similar but less precise estimates.

3.3 Estimation

I estimate impulse responses using the two-stage Bayesian local projection framework developed by Cloyne et al. (2022). The first stage estimates a Bayesian VAR with hierarchical priors following Giannone, Lenza, and Primiceri (2015). Hyperparameters governing overall shrinkage are treated as random and sampled via Metropolis-Hastings. For each draw from the BVAR posterior, I apply the identification procedure described above to recover a draw of A_0. This delivers the causal impact of each tax shock on all variables in the system at horizon zero.

The second stage estimates local projections at each horizon $h \geq 1$:

$$Z_{t+h} = c^{(h)} + B_1^{(h)} Z_{t-1} + \sum_{j=2}^{p} b_j^{(h)} Z_{t-j} + u_{t+h}, \quad \text{var}(u_{t+h}) = \Omega_h \tag{18}$$

where $p = 4$ lags. At horizon zero, the residuals u_t relate to the structural shocks via $u_t = A_0 \varepsilon_t$. The impulse response at horizon h is $IRF_h = B_1^{(h-1)} A_0$. I estimate equation (18) with Bayesian methods using a Minnesota prior and Student-t errors to accommodate fat tails. Appendix B provides complete details on prior specification and the MCMC algorithm.

Bayesian estimation offers three advantages: it propagates identification uncertainty from A_0 into the impulse response posterior, it improves precision at long horizons where LP variance is high (Li, Plagborg-Møller, and Wolf 2024), and it produces posterior draws for straightforward inference on the alignment metrics. I use local projections rather than propagating shocks through a VAR because the present-value gradient requires reliable estimates over extended horizons, where VARs can suffer from misspecification bias due to lag truncation. However, I also show that the main results are robust to a variety of alternative estimation methods including standard local projections, penalized local projections, smooth local projections, Bayesian VARs, and proxy SVARs.

The elasticities are estimated once from the full sample and held fixed across time. Time variation in the alignment metric $\cos\theta$ comes entirely from changes in tax rates (τ_L, τ_C) and the relative size of the two bases (B_C/B_L), which I measure using the long-run (Christiano and Fitzgerald

(2003) filtered) components of the data. The revenue gradient formula (16) combines these time-varying bases and rates with fixed elasticities to produce a quarterly series of alignment statistics. Section 4.2 addresses the plausibility of the fixed-elasticity assumption directly.

Data. The baseline specification includes four variables:

$$Z_t = \left[\log\left(1 - \mathrm{APITR}_t\right), \log\left(1 - \mathrm{ACITR}_t\right), \ln B_t^{PI}, \ln B_t^{CI} \right],$$

where APITR_t and ACITR_t are the average personal and corporate income tax rates, and B_t^{PI} and B_t^{CI} are the corresponding tax bases. The sample covers 1947Q1–2019Q4. Following standard practice in this literature, the corporate base is BEA corporate profits, which includes both C-corporations and S-corporations; the implications of this measurement choice for interpreting the cross-elasticity are discussed in Section 3.4. Appendix A provides detailed variable definitions and data sources. As a robustness check, I also estimate the model using adjusted bases that reallocate S-corporation income from the corporate to the personal base using IRS Statistics of Income data, details of which are in Appendix H.[11]

The impulse responses are estimated on the raw (unfiltered) data, which is standard in the VAR literature. For the alignment calculations in Section 4, I extract the long-run component of tax rates and bases using the Christiano and Fitzgerald (2003) band-pass filter, retaining only frequencies longer than 40 quarters (10 years). This filtering removes business-cycle variation that is orthogonal to the permanent reform experiments the framework evaluates and smooths measurement error in quarterly tax data, which can be noisy due to timing of payments and refunds.

3.4 Present-Value Elasticities

Table 1 reports the baseline elasticities. The diagonal tells a familiar story: both own-elasticities are positive, with the corporate base considerably more responsive ($\varepsilon_{CC} \approx 4.3$) than the personal base ($\varepsilon_{LL} \approx 1.0$). The striking feature is the off-diagonal asymmetry. A one percent reduction in personal taxes expands the corporate base by roughly nine percent ($\varepsilon_{CL} \approx 9$), while corporate tax

11. The empirical analysis uses average tax rates rather than statutory marginal rates for three reasons. First, the theoretical framework in Section 2.1 defines revenue as $R_i = \tau_i B_i$, where τ_i is the average rate. The revenue gradient $\partial R / \partial \tau$ therefore requires variation in average rates. Second, the U.S. tax code features graduated rate schedules with rates varying across income brackets, types of income, and categories of taxpayers, so no single statutory rate summarizes the policy stance. The average rate—total revenue divided by the base—provides the natural aggregate measure. Third, many historical reforms changed the effective tax burden without altering statutory rates at all. Base-broadening measures such as limiting deductions, accelerating depreciation schedules, or restricting credits raise the average rate even when the statutory schedule remains unchanged. The revenue-based narrative measure captures these reforms, which statutory rate series would miss.

cuts have essentially no effect on the personal base ($\varepsilon_{LC} \approx -0.1$). Personal tax policy generates enormous fiscal externalities on corporate revenue; the reverse channel is absent.[12]

Table 1: Present Value Elasticities of Tax Bases with Respect to Retention Rates

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	4.25	-0.11	9.03	0.98
	(3.10, 5.76)	(-0.38, 0.18)	(7.03, 11.28)	(0.51, 1.48)
LP	5.51	0.08	14.01	1.12
	(4.14, 7.54)	(-0.56, 0.71)	(10.33, 21.74)	(0.25, 2.23)
SLP	5.52	0.08	13.99	1.11
	(4.16, 7.56)	(-0.56, 0.71)	(10.25, 21.78)	(0.25, 2.24)
LP-BC	5.45	0.04	13.64	1.07
	(4.10, 7.51)	(-0.58, 0.66)	(10.01, 21.23)	(0.22, 2.15)
BVAR	5.13	-0.46	4.65	1.38
	(3.12, 8.20)	(-0.75, -0.22)	(1.20, 8.43)	(0.63, 2.28)
SVAR-IV	6.60	-0.09	16.58	1.64
	(2.82, 20.39)	(-2.14, 1.45)	(10.50, 33.55)	(-0.19, 6.30)

Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the four-variable system comprised solely of tax bases and rates.

The estimate of $\varepsilon_{CL} \approx 9$ is robust across estimation methods and specifications. Table 1 reports the cross-elasticity under Bayesian local projections, frequentist local projections, smooth local projections (Barnichon and Brownlees 2019), bias-corrected local projections (Herbst and Johannsen 2024), Bayesian VARs, and proxy SVARs. All methods produce estimates between 5 and 17 in the four-variable specification. The seven-variable specification, which adds output, government spending, and debt to the conditioning set (Appendix Table G.1), yields ε_{CL} estimates between 11 and 20—uniformly larger than the four-variable baseline. This rules out the

12. Appendix Figure E.6 plots the discounted cumulative impulse responses underlying these elasticities. Each panel shows the running sum $\sum_{s=0}^{h} \beta^s \cdot \mathrm{IRF}_s$ as the horizon extends from 0 to 20 quarters, with $\beta = 0.9926$ and $H = 20$ following Mertens and Montiel Olea (2018). The terminal values, divided by cumulative retention rate persistence κ_j, yield the elasticities in Table 1.

concern that the four-variable estimate is inflated by omitting controls; if anything, it is conservative. Later, I show that reallocating S-corporation income from the corporate to the personal base (Appendix H) leaves ε_{CL} essentially unchanged at 9.1, ruling out organizational form shifting as the primary explanation.

These elasticities differ conceptually from the taxable income elasticities in the public finance literature. Standard estimates—from bunching at kinks, difference-in-differences around state tax changes, or tax return panel data—identify individual responses to *marginal* rates, holding wages and prices fixed. The elasticities here capture aggregate base responses to *average* rates, incorporating general equilibrium adjustments in wages, interest rates, and organizational form, cumulated over a five-year horizon. The two objects answer different questions: the micro elasticity asks how one taxpayer responds to their own marginal rate; the macro elasticity asks how total revenue responds to an aggregate tax change. Saez, Slemrod, and Giertz (2012) survey the taxable income elasticity literature and conclude that the best available estimates range from 0.12 to 0.40. Mertens and Montiel Olea (2018), using narrative identification similar to the present study, find larger aggregate elasticities around 1.2—slightly above my $\varepsilon_{LL} \approx 1.0$, though well within the credible interval. Vergara and Swonder (2025) survey 25 estimates of the elasticity of corporate taxable profits with respect to the net-of-tax rate, finding a median of 0.58 and a range from 0.08 to 4.79; my $\varepsilon_{CC} \approx 4.3$ is toward the upper end of this range, as expected given that I cumulate general equilibrium responses over a five-year horizon rather than estimating short-run or partial equilibrium effects. The cross-elasticity $\varepsilon_{CL} \approx 9$ has no analog in the existing literature, which tends to focus on own-base responses.

Rationalizing the Cross-Elasticity. The cross-elasticity from personal taxes to the corporate base is the key empirical object in the paper, and it is large. The explanation is operating leverage. Corporate profits are a residual claim—revenue minus costs that are slow to adjust—so any expansion in output translates into a proportionally larger increase in profits. If the labor share γ is roughly constant at the margin, the elasticity of the corporate tax base with respect to output is $(1-\gamma)/s_\pi$, where s_π is the profit share of corporate value added. This ratio is directly observable from the National Income and Product Accounts and ranges from approximately 2 to 3.5 over the postwar period (Appendix C, Figure C.1). The cross-elasticity is then the product of this leverage and the output response to personal tax cuts: $\varepsilon_{CL} \approx [(1-\gamma)/s_\pi] \times \varepsilon_{YL}$.

When we add GDP to the VAR specification in Appendix G, the output elasticity with respect to the personal retention rate is $\varepsilon_{YL} \approx 2.1$. Even under the extreme assumption that labor adjusts frictionlessly to tax changes, Appendix C shows that combining a postwar average leverage of 2.6 with the output elasticity of 2.1 predicts $\varepsilon_{CL} \approx 5.5$. In other words, national accounting explains more than half the baseline estimate. This is a lower bound. The remaining gap reflects wage stickiness: when compensation adjusts slowly, a larger share of output gains accrues to profits, raising effective leverage above the frictionless benchmark. At a five-year horizon, partial wage adjustment is the natural benchmark, and the baseline estimate of 9 would be straightforward to generate with a model of sticky wages. Appendix C develops the full decomposition and shows it rationalizes the complete pattern of elasticities in Table 1.

Organizational Form and Elasticity Magnitudes A separate concern is whether the cross-elasticity reflects genuine economic responses or an artifact of how the corporate base is measured. As noted in Section 3.3, the corporate base is BEA corporate profits, which includes both C-corporations and S-corporations. C-corporations pay corporate income taxes, but S-corporations are pass-through entities: their profits flow to shareholders and are taxed at personal rates. When personal taxes fall, S-corporation activity expands, and this response appears in the measured corporate base even though S-corporation income does not generate corporate tax revenue. The cross-elasticity therefore captures both traditional general equilibrium channels—demand effects, labor-capital complementarity, investment responses—and the direct sensitivity of pass-through business activity to personal tax incentives. The latter channel is quantitatively important: S-corporations grew from a negligible share of business income in the 1970s to roughly half by the 2010s (Dyrda and Pugsley 2024; Smith et al. 2022). This growth accelerated sharply after the Tax Reform Act of 1986, which lowered the top personal rate below the corporate rate for the first time in the postwar era, making pass-through status newly attractive. If the measured cross-elasticity simply reflected firms relabeling C-corporation income as S-corporation income in response to personal tax changes—with no real effects on economic activity—the estimates would be difficult to interpret.

To address this concern, I construct adjusted tax bases that reallocate S-corporation income from the corporate to the personal base using IRS Statistics of Income data. The procedure, detailed in Appendix H, applies the annual S-corporation share of total corporate profits to NIPA data, yielding a corporate base that isolates C-corporation activity and a personal base that includes all pass-through income. Table H.1 reports the results. The own-elasticity of the corporate base falls from 4.25 to 3.20, confirming that the baseline estimate captures some organizational form arbitrage: part of what appears as corporate base elasticity reflects firms shifting between C-corp and S-corp status rather than changes in real corporate activity. More importantly, the cross-elasticity ε_{CL} remains large—9.11 under Bayesian LP estimation, virtually unchanged from the baseline 9.03. The large spillover from personal to corporate taxes is not an artifact of S-corporation dynamics; even when S-corporation income is stripped from the corporate base entirely, personal tax cuts still generate substantial expansions in C-corporation profits, suggesting the cross-elasticity reflects genuine general equilibrium channels rather than mere organizational form shifting. The remaining elasticities move in predictable directions: ε_{LC} becomes more negative (-0.43 versus -0.11) and ε_{LL} rises modestly (1.54 versus 0.98), both mechanical consequences of moving S-corporation income—which responds positively to personal tax cuts—from the corporate base to the personal base.

4 The Postwar Evolution of Tax System Efficiency

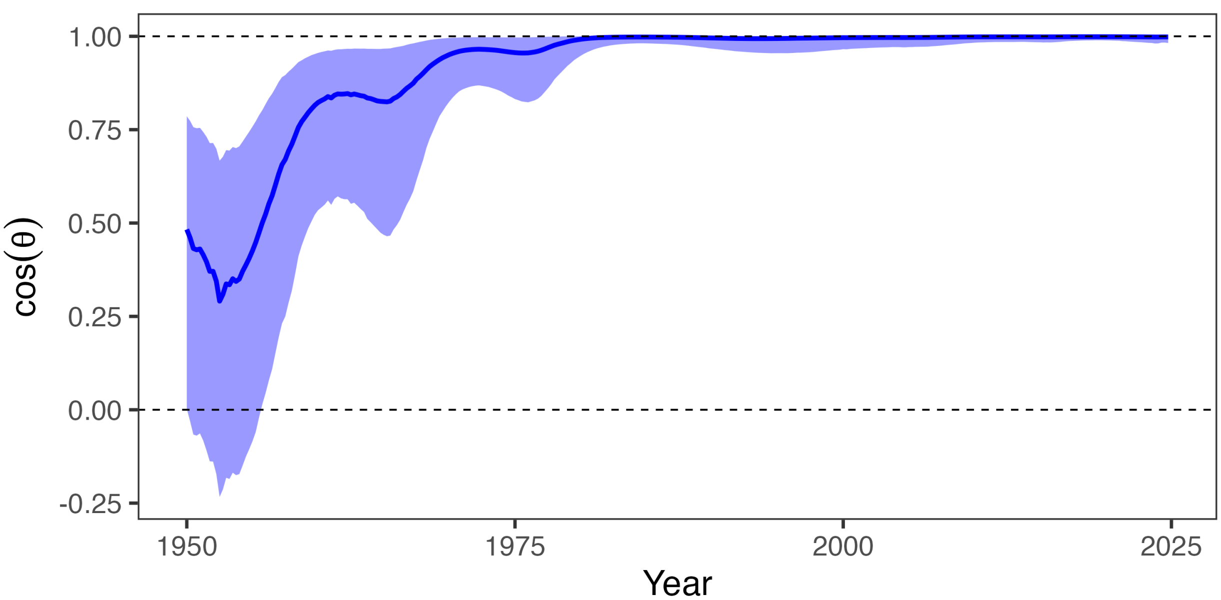
The previous section estimated the revenue gradient from narrative tax shocks; the welfare gradient is simply the vector of tax bases under utilitarian weights. This treats a marginal dollar of corporate profit identically to a marginal dollar of wage income—a strong assumption given that corporate claimants are wealthier on average. In the framework, discounting corporate income corresponds to setting $\omega_C < 1$; Section 4.2 shows results are nearly insensitive to this choice. With both gradients in hand, I now apply the projection framework to evaluate U.S. federal income tax policy from 1950 to 2025. For each quarter, I compute the alignment statistics developed in Section 2: $\cos\theta$ measuring proximity to the Ramsey optimum, the welfare gain from optimal reform, and $\cos\phi$ measuring whether actual reforms moved toward or away from the optimal direction.

4.1 Results

Figure 4 presents the central finding, plotting $\cos\theta$, the alignment between welfare and revenue gradients. Time variation in $\cos\theta$ comes entirely from changes in tax rates (τ_L, τ_C) and base composition (B_C/B_L), with elasticities held fixed at full-sample estimates. Starting in the 1950s, $\cos\theta$ was around 0.3—indicating that welfare and revenue gradients were close to orthogonal. Around 1980, alignment sharply improved as $\cos\theta$ approached 0.8. By around 2020, $\cos\theta$ exceeded 0.95. The tax system moved from misalignment to near-optimality. What drove this improvement? Figure 1 shows two trends moving together: both corporate tax rates and the corporate share of revenue declined. Both could explain the convergence with the Ramsey optimum. Lower corporate tax wedges directly reduce the fiscal externality $A_C = \eta_C\tilde{\varepsilon}_C$ on corporate taxes. But a shrinking corporate revenue share also matters: it enters the effective elasticity $\tilde{\varepsilon}_L = \varepsilon_{LL} + (R_C/R_L)\varepsilon_{CL}$, so as corporate revenue fell relative to personal revenue, the cross-base spillover that made personal taxes look cheap became less important. The question is which channel dominated.

The theoretical framework offers a natural decomposition. Corollary 2 shows that alignment requires equalizing marginal fiscal externalities $A_i = \eta_i\tilde{\varepsilon}_i$ across instruments, where $\eta_i = \tau_i/(1-\tau_i)$ is the tax wedge and $\tilde{\varepsilon}_i$ is the effective elasticity incorporating cross-base spillovers. Changes in alignment can therefore be attributed to two channels: tax wedges η_i, which policy directly controls, and revenue composition R_C/R_L, which enters the effective elasticities and reflects both policy choices and structural shifts in the economy.

Figure 4: The Evolution of Tax System Efficiency

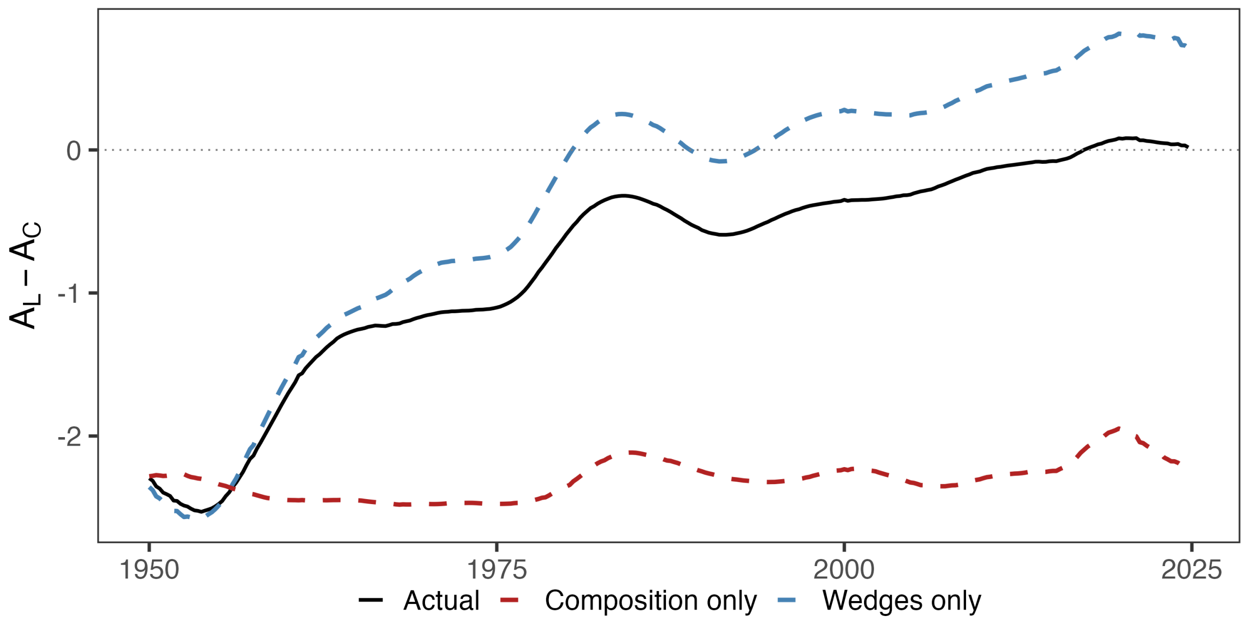


Note: $\cos\theta$ is the alignment between welfare and revenue gradients; $\cos\theta = 1$ indicates the Ramsey optimum. Shaded region shows 90% credible interval.

Figure 5 decomposes the convergence of fiscal externalities into these channels. The solid black line plots the actual gap $A_L - A_C$, which starts around −2.3 in 1950—indicating that corporate taxes had substantially higher fiscal externalities than personal taxes—and converges to zero by 1980. The dashed blue line holds revenue composition fixed at 1950 values and allows only tax wedges to vary; the dashed red line does the reverse. The wedge channel accounts for essentially all of the convergence: the blue line tracks the actual gap almost perfectly, while the red line remains flat near its 1950 value throughout the sample. Tax policy changes that reduced effective corporate tax rates—the investment tax credit in 1962, accelerated depreciation schedules, the growing use of foreign tax credits, and eventually statutory rate cuts—drove the efficiency gains. Shifts in the composition of the tax base, including deindustrialization and the rise of pass-through entities, contributed little.

This finding clarifies the mechanism behind improved alignment and resolves the ambiguity in Figure 1. The corporate share of revenue fell both because effective corporate tax rates declined (wedge channel) and because the economy shifted toward services and pass-through entities (composition channel). The decomposition establishes that the first mechanism dominated. The policy implication is that alignment gains are fragile to policy reversal: if Congress raised effective corporate rates sufficiently—by eliminating bonus depreciation, lengthening cost recovery, or raising tax statutory rates—misalignment could reemerge even without any structural change in the economy.

Figure 5: Decomposing the Convergence of Fiscal Externalities

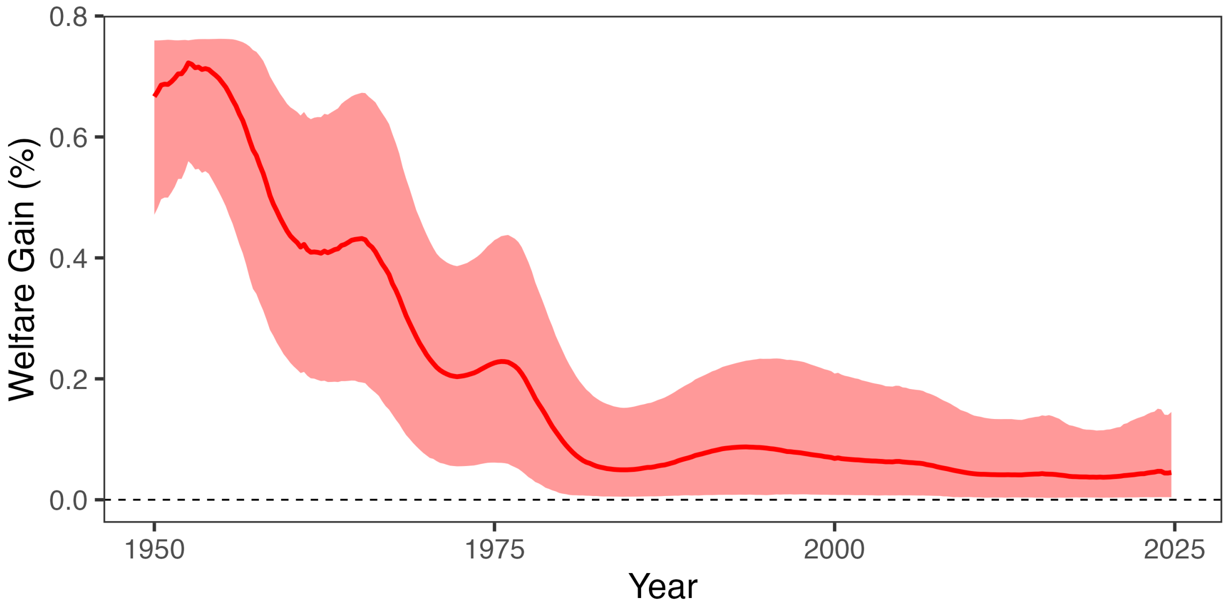


Note: The solid black line plots the gap in marginal fiscal externalities $A_L - A_C$. The dashed blue line holds revenue composition fixed at 1950 values; the dashed red line holds tax wedges fixed at 1950 values. Alignment requires $A_L = A_C$, i.e., the gap equals zero.

What did this misalignment cost? Figure 6 translates the alignment metric into dollars. In the early 1950s, a 1 percentage point reform in the optimal revenue-neutral direction would have improved welfare by approximately 0.7% of GDP per year—over $200 billion in 2024 dollars.[13] By 2020, the same reform would yield only 0.05% of GDP. The large efficiency gains have already been realized. Contemporary debates over corporate versus personal taxation, including the 2017 Tax Cuts and Jobs Act and the 2025 One Big Beautiful Bill Act, are not irrelevant, but the efficiency stakes are an order of magnitude smaller than they were two generations ago. If large welfare gains remain available, they lie in the level of taxation rather than its composition. As Section 2.3 shows, near-perfect alignment means the entire welfare gradient now points along the revenue direction: the remaining welfare stake is in how much revenue the system collects, not how it divides the burden.

13. Appendix Figure D.3 plots the evolution of $\sin\theta$ over the same time span.

Figure 6: Welfare Gains from Optimal Revenue-Neutral Reform

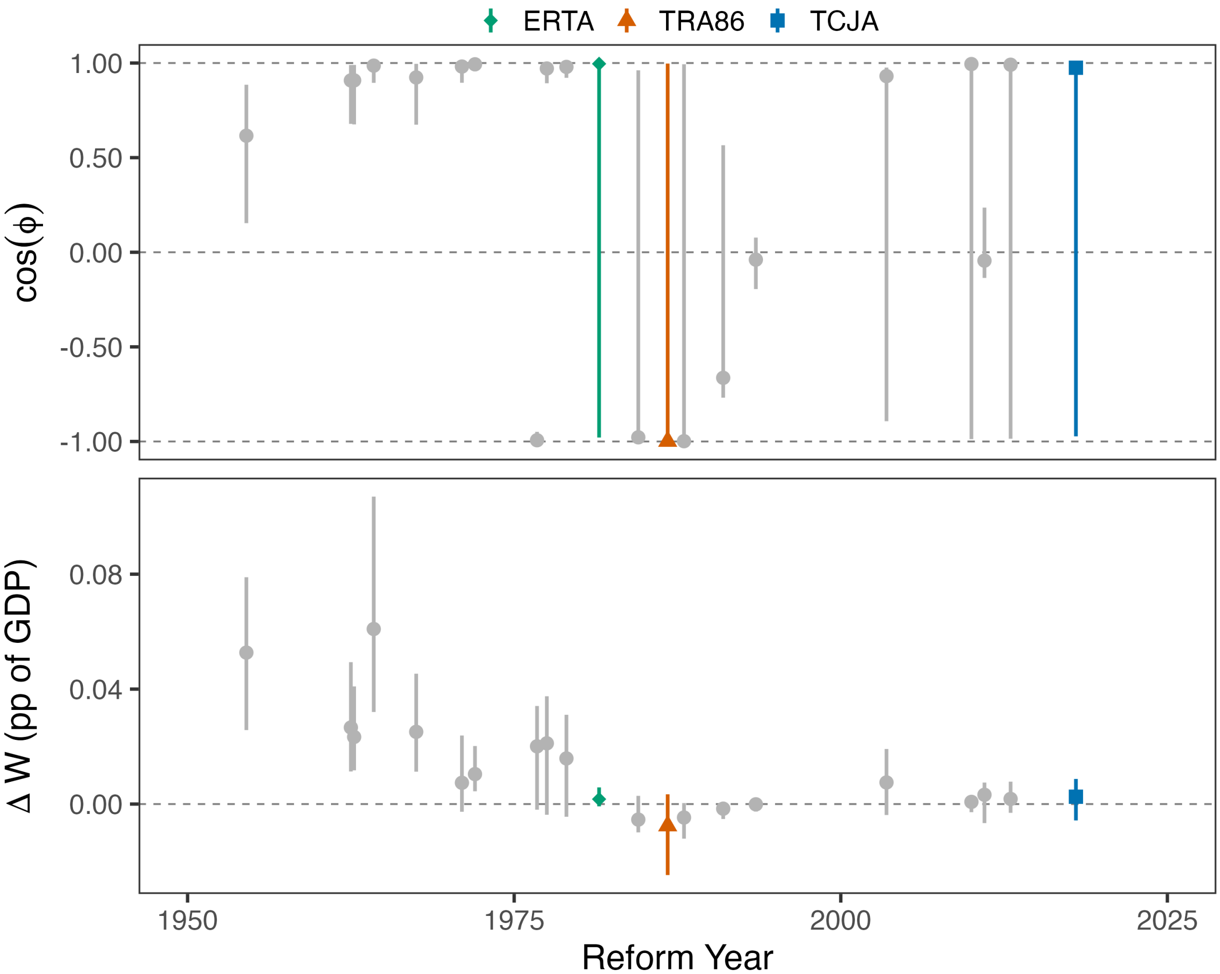


Note: Figure shows the welfare gain from a 1 percentage point reform in the optimal revenue-neutral direction, expressed as a percent of GDP. Welfare gains are computed as $\Delta W = \alpha \|g\| \sin\theta$, where $\alpha = 0.01$ is the reform size, $\|g\|$ is the norm of the welfare gradient (the tax bases), and $\sin\theta$ measures the share of welfare gains achievable under revenue neutrality. Shaded region shows 90% credible interval.

Which reforms contributed to this convergence? Figure 7 evaluates each major tax reform on two dimensions. The top panel plots $\cos\phi$, the directional alignment between the actual reform and the optimal revenue-neutral reform at that date. When $\cos\phi \approx 1$, the reform moved nearly parallel to the welfare-improving direction; when $\cos\phi < 0$, it moved in the opposite direction. The bottom panel plots the welfare gain attributable to each reform, constructed by comparing actual outcomes to a counterfactual where the reform did not occur. For each reform, we remove its estimated effects on tax rates and bases using the impulse responses from Section 3.4, recompute ΔW under the counterfactual, and plot the difference averaged over the five years following enactment. Positive values indicate reforms that captured efficiency gains by moving the system closer to optimal.

The two panels tell a consistent story. Before 1980, most reforms moved in the right direction ($\cos\phi$ near 1) and captured substantial welfare gains. The Revenue Act of 1964, which cut the top corporate rate from 52% to 48% and the top personal rate from 91% to 70%, improved welfare more than any other reform. After 1980, the picture changes: credible intervals on $\cos\phi$ widen dramatically, with point estimates spanning the full range from -1 to 1. But the bottom panel reveals that this apparent imprecision is inconsequential—every post-1980 reform has a welfare contribution indistinguishable from zero. The wide intervals on $\cos\phi$ reflect geometric degeneracy at near-optimality: when the welfare and revenue gradients are nearly collinear, the optimal reform direction is poorly identified, and small perturbations swing $\cos\phi$ across its entire range. Direction ceases to matter when there is nowhere left to go.

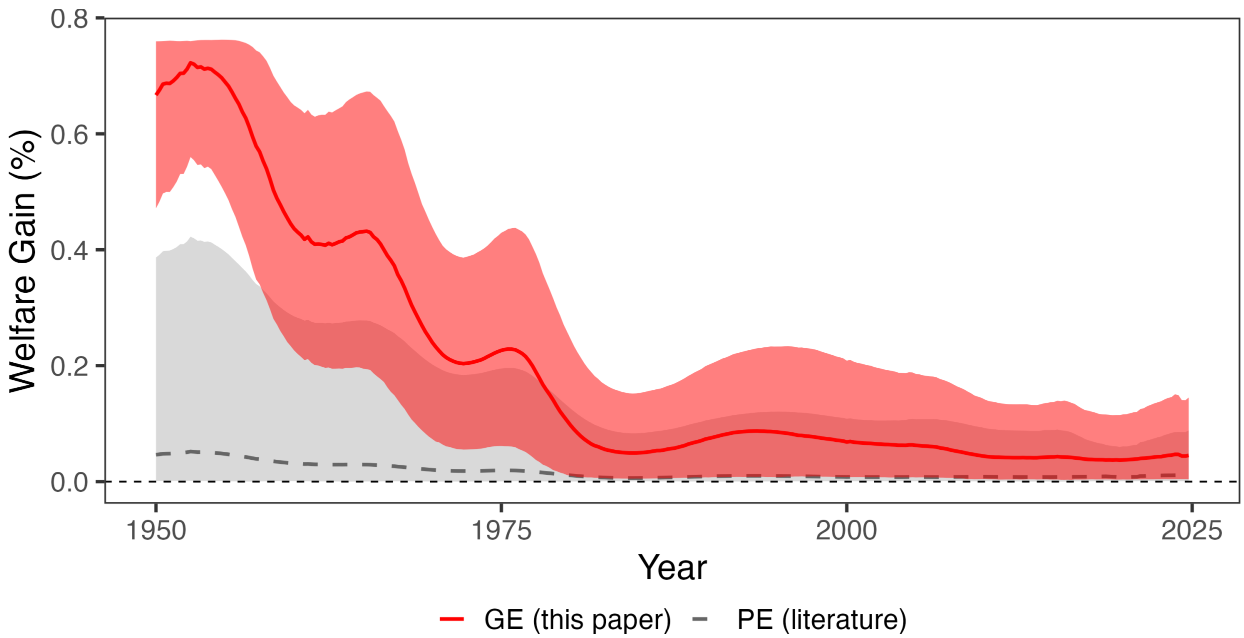
Figure 7: Reform Direction and Welfare Contribution



Note: Top panel: $\cos\phi$ measures the directional alignment between each actual reform and the optimal revenue-neutral reform; $\cos\phi = 1$ indicates the reform moved in the welfare-improving direction. Bottom panel: welfare gain attributable to each reform, computed as the difference in ΔW between a no-reform counterfactual and actual outcomes, averaged over 20 quarters following enactment. Counterfactuals remove each reform's effects on tax rates and bases using the impulse responses from Section 3.4. Bars show 90% credible intervals.

Among the reforms with welfare contributions indistinguishable from zero are the Economic Recovery Tax Act of 1981, the Tax Reform Act of 1986, and the 2017 Tax Cuts and Jobs Act. These later reforms were not necessarily poorly designed; they arrived after the efficiency problem had already been largely solved on the personal-corporate margin. Whatever case exists for these reforms must rest on considerations beyond compositional efficiency: the distribution of the tax burden, the timing of taxation through deficit finance, or margins outside the personal and corporate income tax system entirely. That carries a similar implication for the recent One Big Beautiful Bill Act: since the system is near-optimal on the personal-corporate mix by 2025, any evaluation of that reform on this margin would likewise find it difficult to distinguish from zero.

Figure 8: Welfare Gains: General vs. Partial Equilibrium



Note: Solid red line shows welfare gains from general equilibrium estimates with cross-base spillovers. Dashed gray line and shaded band show welfare gains using partial equilibrium own-elasticities from the literature with cross-elasticities set to zero. The PE band spans all combinations of $\varepsilon_{CC} \in [0.08, 4.79]$ and $\varepsilon_{LL} \in [0.12, 0.40]$.

The welfare gains documented in both the aggregate series (Figure 6) and the reform-level evaluation (Figure 7) depend on the general equilibrium elasticities estimated in Section 3.4. Figure 8 shows what partial equilibrium estimates would have concluded. Using own-base elasticities from the literature—$\varepsilon_{CC} \in [0.08, 4.79]$ from Vergara and Swonder (2025) and $\varepsilon_{LL} \in [0.12, 0.40]$ from Saez, Slemrod, and Giertz (2012)—with cross-base spillovers set to zero, welfare gains range from 0 to 0.4% of GDP in the early 1950s, with a median around 0.05%, compared to 0.7% in GE.[14] The upper bound on the PE estimates converge to zero decades before the GE estimates. The difference is the cross-elasticity $\varepsilon_{CL} \approx 9$: personal tax cuts substantially expand the corporate

14. The only documented large ε_{CC} in partial equilibrium are from studies of Costa Rica (Bachas and Soto 2021), Germany pre-unification (Buettner 2003), and Slovakia (Bukovina et al. 2025). Those studies are why the upper bound on the early welfare losses is not closer to zero for the PE comparison, but it is also unclear if they have external validity for the tax elasticity of the corporate base in the United States.

base, creating fiscal externalities invisible to partial equilibrium analysis. When the corporate tax share was large, these spillovers pushed the revenue gradient far from the welfare gradient. PE, capturing only own-base responses, understates early-period misalignment by an order of magnitude. The PE series also provides a bound on sensitivity to time-varying elasticities: if cross-base spillovers were smaller in the early postwar period, the true alignment path would lie between the GE and PE series, but even the limiting case of $\varepsilon_{CL} = 0$ preserves the qualitative trajectory. Section 4.2 discusses the plausibility of time-varying elasticities directly.

4.2 Robustness

The main results rely on present-value general equilibrium elasticities generated by a Bayesian local projection using utilitarian welfare and a particular discount rate. How sensitive are the findings to these choices?

S-corporation adjustment. Section 3.4 showed that reallocating S-corporation income from the corporate to the personal base reduces the own-elasticity ε_{CC} but leaves the cross-elasticity ε_{CL} essentially unchanged. Figure H.3 in Appendix H confirms that the alignment results are similarly robust: the transition from misalignment to near-optimality occurs over the same period under both specifications. The efficiency gains documented above are not an artifact of organizational form shifting.

Discount rates. Present-value calculations require a discount factor, and higher discount rates place more weight on near-term responses where estimation is precise. Figure E.2 varies β from 0.97 to 1. The improvement in alignment is robust across this range: all specifications with $\beta \leq 0.99$ show $\cos\theta$ rising from below 0.5 to nearly perfect alignment.

Estimation methods. The revenue gradient depends on impulse response estimates, which differ across methods. Figure E.3 compares several approaches. The local projection variants (BLP, LP, SLP, and LP-BC), the BVAR, and the SVAR cluster tightly, all showing alignment rising considerably the 2010s. Figure E.4 shows that the same is largely true when including a linear trend.

Welfare weights. The welfare gradient depends on how society values different taxpayers. Figure E.1 varies these weights. Results are almost entirely insensitive to ω_C, the weight on corporate taxpayers—all specifications in the left panel nearly overlap. Only extreme assumptions—placing near-zero weight on personal income taxpayers while placing considerable weight on corporate income—yield meaningfully different results. The asymmetry reflects the larger personal tax base:

its welfare weight matters more for the gradient calculation. However, even that specificaition shows considerable growth in alignment.

Horizon length. The elasticities depend on the length of the horizon cumulated over time. Figure E.5 shows that the upward trend in $\cos\theta$ is consistent across horizon lengths from 20 to 100 quarters under the baseline specification.

Alternative Cholesky Ordering. The proxy SVAR identifies both tax shocks simultaneously, but recovering the individual structural shocks requires a Cholesky decomposition of the 2×2 covariance matrix of the proxy-identified block. The baseline identifies the corporate tax shock using a decomposition that orders the corporate tax variable first, and identifies the personal tax shock using a decomposition that orders the personal tax variable first. This imposes that the shock ordered second has no contemporaneous effect on the tax variable ordered first. As a robustness check, we reverse these orderings: the corporate shock is identified with the personal tax variable ordered first, and vice versa. Similar results under both orderings indicate that the contemporaneous correlation between the two tax shocks is small, so the identifying restriction is not driving the results. The results in Appendix E.3 show that that, quantitatively, this ordering is virtually irrelevant. This is unsurprising given previous results in the literature showing the exact same outcome for a slightly different sample and specification (Mertens and Ravn 2013a).

Filtering choice. The baseline applies a Christiano and Fitzgerald (2003) band-pass filter to bases and rates with a 40-quarter cutoff, retaining variation at periodicities above ten years. This matches the frequency of the policy variables to the horizon over which elasticities are cumulated, and removes cyclical fluctuations orthogonal to the permanent tax structure the framework evaluates so that a recession that temporarily depresses corporate profits does not indicate the tax mix has become misaligned. Figure F.1 shows the alignment metrics are insensitive to this choice. The figure plots $\cos\theta$ and ΔW under cutoffs ranging from 8 to 80 quarters, as well as unfiltered quarterly data. The transition from misalignment in the 1950s to near-perfect alignment by the 1980s is present at every cutoff. Shorter cutoffs produce noisier series—reflecting the cyclical sensitivity of corporate profits—but all specifications converge to the same level. The choice of filtering frequency affects the smoothness of the path but not the destination or the qualitative trajectory.

Seven-variable specification. The baseline uses four variables: tax rates and bases for both instruments. Appendix E shows robustness to a seven-variable specification that adds government spending, GDP, and debt following Mertens and Ravn (2013a). The seven-variable specification yields qualitatively similar point estimates but wider credible intervals, as expected given the additional parameters. The alignment metrics are robust to this choice: $\cos\theta$ in the 2010s exceeds 0.8 under both specifications.

Time-varying elasticities. The baseline holds elasticities fixed at full-sample estimates. As shown in Section 3 and Appendix C, the cross-elasticity is the product of operating leverage and the output response to personal tax cuts. The first component, operating leverage, is directly observable from the national accounts and varies modestly over the postwar period between 2 and 3.5, which bounds the scope for time variation in the cross-elasticity to changes in the output multiplier alone. Direct evidence on time variation in the transmission of tax shocks supports the fixed-elasticity baseline. Mumtaz and Petrova (2023), estimating a time-varying proxy SVAR with the same narrative identification used here, find that changes in the output response to tax shocks largely reflect changes in shock volatility rather than shifts in the transmission mechanism. Similarly, De Wind (2014) finds only a moderate decline in the aggregate tax multiplier. For the cross-elasticity to have been small enough to eliminate early misalignment, either leverage or the output multiplier would need to have been several times smaller in the 1950s than their full-sample values, and neither the observable variation in leverage nor the existing evidence on multiplier stability supports changes of that magnitude.

5 Conclusion

This paper develops a geometric framework for evaluating tax reform in general equilibrium. The angle between the welfare gradient and the revenue gradient measures alignment with the Ramsey optimum: when these vectors are collinear, no revenue-neutral reform can improve welfare; when they are orthogonal, the system is maximally inefficient. Computing this angle requires only two sufficient statistics, tax bases and the general equilibrium response of revenue to tax changes, without specifying preferences, production, or market structure.

Applying this framework to the U.S. federal income tax system from 1950 to 2025 reveals that alignment improved dramatically over the postwar period. In the early 1950s, welfare and revenue gradients were substantially misaligned, and a small reform in the optimal direction would have yielded large welfare gains. By the 2010s, the system had moved to near-optimality on the personal-corporate margin. The decomposition shows that tax policy changes, including investment incentives, accelerated depreciation, and eventually statutory rate cuts, drove this convergence, not structural shifts in the economy. A corollary is that every major reform since 1980, including the Tax Reform Act of 1986 and the Tax Cuts and Jobs Act, has had no measurable welfare effect on this margin. Whatever case exists for those reforms must rest on distribution, timing, or margins outside the personal and corporate income tax system.

Near-perfect alignment means the entire welfare gradient now points along the revenue direction: the remaining question is not how to divide the tax burden between personal and corporate income, but how much total revenue to collect. The framework quantifies the welfare stake in that question but cannot resolve it without spending-side evidence. The efficiency gains documented here are also not permanent. They emerged from policy choices that reduced effective corporate taxation, and they can be reversed by policy choices that raise it. U.S. debt-to-GDP is high, and stabilizing it may eventually require raising revenue, cutting spending, or both. If the adjustment falls disproportionately on corporate taxes, whether through rate increases, base broadening, or the expiration of investment incentives, the alignment gains of the past 70 years could unwind rapidly. Alternatively, the U.S. could turn to other margins to raise revenue. Tariffs, consumption taxes, payroll taxes, and wealth taxes all lie outside the current empirical analysis, but the framework extends naturally to these settings, as it does to state and local tax systems, environmental policy, or social insurance: anywhere policymakers face multi-instrument problems with general equilibrium spillovers. The sufficient statistics it requires, welfare costs and revenue responses, can be estimated from reduced-form evidence in any of these domains, diagnosing both the composition and the scale of remaining inefficiency.

References

Ahmad, Ehtisham, and Nicholas Stern. 1984. "The theory of reform and indian indirect taxes." *Journal of Public Economics* 25, no. 3 (December): 259–298. ISSN: 00472727. https://doi.org/10.1016/0047-2727(84)90057-4.

Atkinson, A. B., and N. H. Stern. 1974. "Pigou, Taxation and Public Goods." *The Review of Economic Studies* 41, no. 1 (January): 119. ISSN: 00346527. https://doi.org/10.2307/2296403.

Bachas, Pierre, and Mauricio Soto. 2021. "Corporate Taxation under Weak Enforcement." *American Economic Journal: Economic Policy* 13, no. 4 (November): 36–71. ISSN: 1945-7731. https://doi.org/10.1257/pol.20180564.

Badel, Alejandro, Mark Huggett, and Wenlan Luo. 2020. "Taxing Top Earners: a Human Capital Perspective." *The Economic Journal* 130, no. 629 (July): 1200–1225. ISSN: 0013-0133. https://doi.org/10.1093/ej/ueaa021.

Barnichon, Regis, and Christian Brownlees. 2019. "Impulse Response Estimation by Smooth Local Projections." *The Review of Economics and Statistics* 101, no. 3 (July): 522–530. ISSN: 0034-6535. https://doi.org/10.1162/rest{_}a{_}00778.

Barnichon, Régis, and Geert Mesters. 2023. "A Sufficient Statistics Approach for Macro Policy." *American Economic Review* 113, no. 11 (November): 2809–2845. ISSN: 0002-8282. https://doi.org/10.1257/aer.20220581.

Barro, Robert J. 1979. "On the Determination of the Public Debt." *Journal of Political Economy* 87, no. 5, Part 1 (October): 940–971. ISSN: 0022-3808. https://doi.org/10.1086/260807.

Bastani, Spencer. 2025. "The marginal value of public funds: a brief guide and application to tax policy." *International Tax and Public Finance* 32, no. 4 (August): 919–956. ISSN: 0927-5940. https://doi.org/10.1007/s10797-024-09860-x.

Bhandari, Anmol, Jaroslav Borovička, and Yuki Yao. 2025. *Robust Bounds on Optimal Tax Progressivity.* Technical report. Cambridge, MA: National Bureau of Economic Research, September. https://doi.org/10.3386/w34218.

Buettner, Thiess. 2003. "Tax base effects and fiscal externalities of local capital taxation: evidence from a panel of German jurisdictions." *Journal of Urban Economics* 54, no. 1 (July): 110–128. ISSN: 00941190. https://doi.org/10.1016/S0094-1190(03)00039-1.

Bukovina, Jaroslav, Tomáš Lichard, Ján Palguta, and Branislav Žúdel. 2025. "Corporate Minimum Tax and the Elasticity of Taxable Income: Evidence from Administrative Tax Records." *American Economic Journal: Economic Policy* 17, no. 2 (May): 358–387. ISSN: 1945-7731. https://doi.org/10.1257/pol.20230108.

Caballero, Ricardo J., Eduardo M.R.A. Engel, and John Haltiwanger. 1997. "Aggregate Employment Dynamics: Building from Microeconomic Evidence." *American Economic Review* 87 (1): 115–137.

Caravello, Tomás, Alisdair McKay, and Christian Wolf. 2024. *Evaluating Monetary Policy Counterfactuals: (When) Do We Need Structural Models?* Technical report. Cambridge, MA: National Bureau of Economic Research, September. https://doi.org/10.3386/w32988.

Chamley, Christophe. 1986. "Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives." *Econometrica* 54 (3): 607–622.

Chari, V.V., Juan Pablo Nicolini, and Pedro Teles. 2020. "Optimal capital taxation revisited." *Journal of Monetary Economics* 116 (December): 147–165. ISSN: 03043932. https://doi.org/10.1016/j.jmoneco.2019.09.015.

Chetty, Raj. 2009. "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics* 1, no. 1 (September): 451–488. ISSN: 1941-1383. https://doi.org/10.1146/annurev.economics.050708.142910.

Christiano, Lawrence J., and Terry J. Fitzgerald. 2003. "The Band Pass Filter." *International Economic Review* 44, no. 2 (May): 435–465. ISSN: 0020-6598. https://doi.org/10.1111/1468-2354.t01-1-00076.

Cloyne, James, Joseba Martinez, Haroon Mumtaz, and Paolo Surico. 2022. *Short-Term Tax Cuts, Long-Term Stimulus.* Technical report. Cambridge, MA: National Bureau of Economic Research, July. https://doi.org/10.3386/w30246.

Conesa, Juan Carlos, Sagiri Kitao, and Dirk Krueger. 2009. "Taxing Capital? Not a Bad Idea After All!" *American Economic Review* 99, no. 1 (February): 25–48. ISSN: 0002-8282. https://doi.org/10.1257/aer.99.1.25.

De Wind, Joris. 2014. "Time variation in the dynamic effects of unanticipated changes in tax policy."

Diamond, Peter, and Emmanuel Saez. 2011. "The Case for a Progressive Tax: From Basic Research to Policy Recommendation." *Journal of Economic Perspectives* 25, no. 4 (November): 165–190. ISSN: 0895-3309. https://doi.org/10.1257/jep.25.4.165.

Diamond, Peter A., and James A. Mirrlees. 1971. "Optimal Taxation and Public Production I: Production Efficiency." *American Economic Review* 61 (1): 8–27.

Diewert, W.E. 1978. "Optimal tax perturbations." *Journal of Public Economics* 10, no. 2 (October): 139–177. ISSN: 00472727. https://doi.org/10.1016/0047-2727(78)90034-8.

Dixit, Avinash. 1975. "Welfare effects of tax and price changes." *Journal of Public Economics* 4, no. 2 (February): 103–123. ISSN: 00472727. https://doi.org/10.1016/0047-2727(75)90014-6.

———. 1979. "Price changes and optimum taxation in a many-consumer economy." *Journal of Public Economics* 11, no. 2 (March): 143–157. ISSN: 00472727. https://doi.org/10.1016/0047-2727(79)90001-X.

Doan, Thomas, Robert Litterman, and Christopher Sims. 1984. "Forecasting and conditional projection using realistic prior distributions." *Econometric Reviews* 3, no. 1 (January): 1–100. ISSN: 0747-4938. https://doi.org/10.1080/07474938408800053.

Dyrda, Sebastian, and Benjamin Pugsley. 2024. "The Rise of Pass-Throughs: an Empirical Investigation." *The Economic Journal* 135, no. 665 (December): 387–403. ISSN: 0013-0133. https://doi.org/10.1093/ej/ueae062.

Favero, Carlo, and Francesco Giavazzi. 2012. "Measuring Tax Multipliers: The Narrative Method in Fiscal VARs." *American Economic Journal: Economic Policy* 4, no. 2 (May): 69–94. ISSN: 1945-7731. https://doi.org/10.1257/pol.4.2.69. https://pubs.aeaweb.org/doi/10.1257/pol.4.2.69.

Feldstein, Martin. 1976. "On the theory of tax reform." *Journal of Public Economics* 6, nos. 1-2 (July): 77–104. ISSN: 00472727. https://doi.org/10.1016/0047-2727(76)90042-6.

Finkelstein, Amy, and Nathaniel Hendren. 2020. "Welfare Analysis Meets Causal Inference." *Journal of Economic Perspectives* 34, no. 4 (November): 146–167. ISSN: 0895-3309. https://doi.org/10.1257/jep.34.4.146.

Geweke, J. 1993. "Bayesian treatment of the independent student-t linear model." *Journal of Applied Econometrics* 8, no. S1 (December): S19–S40. ISSN: 08837252. https://doi.org/10.1002/jae.3950080504.

Giannone, Domenico, Michele Lenza, and Giorgio E. Primiceri. 2015. "Prior Selection for Vector Autoregressions." *Review of Economics and Statistics* 97, no. 2 (May): 436–451. ISSN: 0034-6535. https://doi.org/10.1162/REST{_}a{_}00483.

Guesnerie, Roger. 1977. "On the direction of tax reform." *Journal of Public Economics* 7, no. 2 (April): 179–202. ISSN: 00472727. https://doi.org/10.1016/0047-2727(77)90045-7.

Hendren, Nathaniel. 2016. "The Policy Elasticity." *Tax Policy and the Economy* 30, no. 1 (January): 51–89. ISSN: 0892-8649. https://doi.org/10.1086/685593.

Hendren, Nathaniel, and Ben Sprung-Keyser. 2020. "A Unified Welfare Analysis of Government Policies." *The Quarterly Journal of Economics* 135, no. 3 (August): 1209–1318. ISSN: 0033-5533. https://doi.org/10.1093/qje/qjaa006.

Herbst, Edward P., and Benjamin K. Johannsen. 2024. "Bias in local projections." *Journal of Econometrics* 240, no. 1 (March): 105655. ISSN: 03044076. https://doi.org/10.1016/j.jeconom.2024.105655.

Jacobs, Bas. 2018. "The marginal cost of public funds is one at the optimal tax system." *International Tax and Public Finance* 25, no. 4 (August): 883–912. ISSN: 0927-5940. https://doi.org/10.1007/s10797-017-9481-0.

Jorda, Oscar. 2005. "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review* 95, no. 1 (February): 161–182. ISSN: 0002-8282. https://doi.org/10.1257/0002828053828518. https://pubs.aeaweb.org/doi/10.1257/0002828053828518.

Kleven, Henrik, Claus Kreiner, Kristian Larsen, and Jakob Søgaard. 2025. "Micro versus Macro Labor Supply Elasticities: The Role of Dynamic Returns to Effort." *American Economic Review* 115, no. 9 (September): 2849–2890. ISSN: 0002-8282. https://doi.org/10.1257/aer.20240554.

Koby, Yann, and Christian K. Wolf. 2020. "Aggregation in Heterogeneous-Firm Models: Theory and Measurement."

Li, Dake, Mikkel Plagborg-Møller, and Christian K. Wolf. 2024. "Local Projections vs. VARs: Lessons From Thousands of DGPs." *Journal of Econometrics* 244 (2). https://doi.org/https://doi.org/10.1016/j.jeconom.2024.105722.

Lucas, Robert E. 1990. "Supply-Side Economics: An Analytical Review." *Oxford Economic Papers* 42 (2): 293–316.

Mankiw, N. Gregory, Matthew Weinzierl, and Danny Yagan. 2009. "Optimal Taxation in Theory and Practice." *Journal of Economic Perspectives* 23, no. 4 (November): 147–174. ISSN: 0895-3309. https://doi.org/10.1257/jep.23.4.147.

Mayshar, Joram. 1990. "On measures of excess burden and their application." *Journal of Public Economics* 43, no. 3 (December): 263–289. ISSN: 00472727. https://doi.org/10.1016/0047-2727(90)90001-X.

McKay, Alisdair, and Christian K. Wolf. 2023. "What Can Time-Series Regressions Tell Us About Policy Counterfactuals?" *Econometrica* 91 (5): 1695–1725. ISSN: 0012-9682. https://doi.org/10.3982/ECTA21045.

Mertens, Karel, and José Luis Montiel Olea. 2018. "Marginal Tax Rates and Income: New Time Series Evidence*." *The Quarterly Journal of Economics* 133, no. 4 (November): 1803–1884. ISSN: 0033-5533. https://doi.org/10.1093/qje/qjy008.

Mertens, Karel, and Morten O. Ravn. 2012. "Empirical evidence on the aggregate effects of anticipated and unanticipated US tax policy shocks." *American Economic Journal: Economic Policy* 103 (4): 145–181.

———. 2013a. "The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States." *American Economic Review* 103, no. 4 (June): 1212–1247. ISSN: 0002-8282. https://doi.org/10.1257/aer.103.4.1212. https://pubs.aeaweb.org/doi/10.1257/aer.103.4.1212.

———. 2013b. "The Dynamic Effects of Personal and Corporate Income Taxes in the United States." *American Economic Review* 103:1212–1247.

Mumtaz, Haroon, and Katerina Petrova. 2023. "Changing Impact of Shocks: A Time-Varying Proxy SVAR Approach." *Journal of Money, Credit and Banking* 55, nos. 2-3 (March): 635–654. ISSN: 0022-2879. https://doi.org/10.1111/jmcb.12946.

Raimondos-Møller, Pascalis, and Alan D. Woodland. 2014. "Steepest ascent tariff reform." *Economic Theory* 55, no. 1 (January): 69–99. ISSN: 0938-2259. https://doi.org/10.1007/s00199-013-0741-7.

Ramsey, F. P. 1927. "A Contribution to the Theory of Taxation." *The Economic Journal* 37, no. 145 (March): 47. ISSN: 00130133. https://doi.org/10.2307/2222721.

Romer, Christina D, and David H Romer. 2010. "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks." *American Economic Review* 100 (3): 763–801.

Saez, E. 2001. "Using Elasticities to Derive Optimal Income Tax Rates." *The Review of Economic Studies* 68, no. 1 (January): 205–229. ISSN: 0034-6527. https://doi.org/10.1111/1467-937X.00166.

Saez, Emmanuel, Joel Slemrod, and Seth H Giertz. 2012. "The Elasticity of Taxable Income with Respect to Marginal Tax Rates: A Critical Review." *Journal of Economic Literature* 50, no. 1 (March): 3–50. ISSN: 0022-0515. https://doi.org/10.1257/jel.50.1.3.

Saez, Emmanuel, and Stefanie Stantcheva. 2016. "Generalized Social Marginal Welfare Weights for Optimal Tax Theory." *American Economic Review* 106, no. 1 (January): 24–45. ISSN: 0002-8282. https://doi.org/10.1257/aer.20141362.

Sims, Christopher A. 1993. "A Nine-Variable Probabilistic Macroeconomic Forecasting Model." In *Business Cycles, Indicators and Forecasting,* edited by James H. Stock and Mark W. Watson. National Bureau of Economic Research.

Smith, Matthew, Danny Yagan, Owen Zidar, and Eric Zwick. 2022. "The Rise of Pass-Throughs and the Decline of the Labor Share." *American Economic Review: Insights* 4, no. 3 (September): 323–340. ISSN: 2640-205X. https://doi.org/10.1257/aeri.20210268.

Stiglitz, J. E., and P. Dasgupta. 1971. "Differential Taxation, Public Goods, and Economic Efficiency." *The Review of Economic Studies* 38, no. 2 (April): 151. ISSN: 00346527. https://doi.org/10.2307/2296778.

Straub, Ludwig, and Iván Werning. 2020. "Positive Long-Run Capital Taxation: Chamley-Judd Revisited." *American Economic Review* 110, no. 1 (January): 86–119. ISSN: 0002-8282. https://doi.org/10.1257/aer.20150210.

Tirole, J., and R. Guesnerie. 1981. "Tax reform from the gradient projection viewpoint." *Journal of Public Economics* 15, no. 3 (June): 275–293. ISSN: 00472727. https://doi.org/10.1016/0047-2727(81)90012-8.

Vergara, Damian, and Dustin Swonder. 2025. "A Sufficient Statistics Approach to Optimal Corporate Taxes."

Winberry, Thomas. 2021. "Lumpy Investment, Business Cycles, and Stimulus Policy." *American Economic Review* 111, no. 1 (January): 364–396. ISSN: 0002-8282. https://doi.org/10.1257/aer.20161723.

Appendices

Online Appendix

A Data

The variables and their construction are an extension of Mertens and Ravn (2013a) and Cloyne et al. (2022) from 1947Q1-2019Q4:

- **Personal Income Tax Base.** The personal income tax base is National Income and Product Accounts (NIPA) personal income (Table 2.1 Line 1) plus contributions for government social insurance (3.2 Line 11) less transfers (2.1 Line 17).
- **Personal Income Tax Revenue.** This comes from summing Table 3.2 Lines 3 and 11.
- **Average Personal Income Tax Rate.** The APITR is revenue divided by the lagged base.
- **Corporate Income Tax Base.** The corporate tax base is table 1.2 Line 13 (corporate profits with inventory valuation and capital consumption adjustments) less Federal Reserve Bank profits (Historical Tables 6.16 B-C-D).
- **Corporate Income Tax Revenue.** Corporate revenue is federal taxes on corporate income excluding Federal Reserve banks (Table 3.2 Line 8).
- **Average Corporate Income Tax Rate.** The ACITR is revenue divided by the lagged base.
- **Government Spending.** Real federal government consumption expenditures and gross investment (1.1.3 Line 23).
- **GDP.** Real GDP is from NIPA Table 1.1.3 Line 1.
- **Debt.** Debt is spliced together using Federal debt held by the public from Favero and Giavazzi (2012) and the series FYGFDPUN from FRED.
- **Narrative Instruments.** See Cloyne et al. (2022) for an extension to the Mertens and Ravn (2013a) corporate and personal income tax shock series.

All series in levels are deflated by the GDP deflator (NIPA Table 1.1.9 Line 1) and population, then log-transformed. Population is total population over age 16 from the Bureau of Labor Statistics (CNP16OV in FRED). Because this series starts in 1948, we use a simple linear extrapolation to extend back four quarters to 1947Q1.

B Estimation Details

This appendix provides details on the two-stage Bayesian local projection procedure used to estimate impulse response functions. Many of the estimation details follow Cloyne et al. (2022), Appendix G.

B.1 Stage 1: Bayesian VAR with Hierarchical Priors

The first stage estimates a Bayesian VAR to obtain draws from the posterior distribution of the impact matrix A_0. The VAR is:

$$Z_t = c + B_1 Z_{t-1} + \cdots + B_p Z_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma) \tag{A.1}$$

where Z_t is the vector of endogenous variables and $p = 4$ lags.

Priors. Following Giannone, Lenza, and Primiceri (2015), I treat the hyperparameters governing prior tightness as random and estimate them jointly with the VAR coefficients. The prior combines three components:

- *Minnesota prior*: The prior mean for the coefficient on the first own lag is set to the AR(1) coefficient from a preliminary regression; all other coefficients have prior mean zero. The prior variance for the coefficient on lag ℓ of variable j in equation i is:

  $$V_{ij\ell} = \frac{\lambda^2}{\ell^2} \cdot \frac{s_i^2}{s_j^2} \tag{A.2}$$

  where s_i and s_j are residual standard deviations from preliminary AR(1) regressions and λ controls overall tightness. The prior on λ is $\lambda \sim N^+(0.2, 0.4^2)$ truncated to $[10^{-4}, 5]$.

- *Sum-of-coefficients prior*: This prior, introduced by Doan, Litterman, and Sims (1984), shrinks the sum of lag coefficients toward unity for persistent variables, helping capture unit root behavior.

- *Single-unit-root prior*: This prior implements the dummy initial observation approach of Sims (1993), providing additional shrinkage toward a random walk.

The hyperparameters governing the sum-of-coefficients and single-unit-root priors are also treated as random, with priors $\Gamma(1, 1)$ truncated to $[10^{-4}, 50]$.

Posterior simulation. The posterior is approximated via Markov chain Monte Carlo. Conditional on hyperparameters, the posterior for VAR coefficients and the covariance matrix Σ is Normal-inverse-Wishart and can be sampled directly. The hyperparameters are updated using a Metropolis-Hastings step. I use 20,000 iterations with 10,000 discarded as burn-in.

B.2 Stage 1b: Proxy Identification of A_0

For each draw of the VAR coefficients and covariance matrix from the Stage 1 posterior, I apply the proxy identification procedure of Mertens and Ravn (2013a) to obtain a draw of the impact matrix A_0.

Let $u_t = A_0\varepsilon_t$ where ε_t contains the structural shocks with $\mathrm{Var}(\varepsilon_t) = I$. Partition the variables so that the first k correspond to the tax rates (instrumented by the narrative measures m_t) and the remaining $n-k$ are non-tax variables. The reduced-form covariance matrix $\Sigma = A_0A_0'$ can be partitioned conformably:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \tag{A.3}$$

The proxy relevance condition $E[m_t\varepsilon_{1,t}'] = \Phi \neq 0$ and exogeneity condition $E[m_t\varepsilon_{2,t}'] = 0$ identify the subspace spanned by the tax shocks. Regressing the non-tax residuals on the tax residuals, using the proxies as instruments, recovers the ratio $\beta_{21}\beta_{11}^{-1}$. The remaining elements of A_0 corresponding to the tax shocks follow from the algebra in Mertens and Ravn (2013a).

To separate the personal and corporate tax shocks within the identified subspace, I apply a Cholesky factorization. When estimating the impulse response to shock j, I order tax j last. This imposes that shock j has no direct contemporaneous effect on the other tax rate.

Each draw of A_0 is normalized so that the impact effect on the relevant tax rate equals -1 percentage point (a tax cut).

B.3 Stage 2: Bayesian Local Projections

The second stage estimates local projections at each horizon $h \geq 1$. For outcome variable y (e.g., a tax rate or log tax base), the local projection is:

$$y_{t+h} = c^{(h)} + B_1^{(h)} Z_{t-1} + \sum_{\ell=2}^{p} b_\ell^{(h)} Z_{t-\ell} + u_{t+h} \tag{A.4}$$

where $p = 4$ lags. Let $\omega^{(h)}$ collect the coefficients and let $X_t = (Z_{t-1}', \ldots, Z_{t-p}', 1)'$ denote the regressors.

Priors. The prior for $\omega^{(h)}$ is Normal with mean ω_0 and variance S_0. The prior mean implies that the outcome variable follows an AR(1) process: the element corresponding to the first own lag equals the AR(1) coefficient from a preliminary regression, with all other elements set to zero. The prior variance takes the Minnesota form:

$$S_{0,j\ell} = \frac{\lambda^2}{\ell^2} \cdot \frac{s_y^2}{s_j^2} \tag{A.5}$$

where s_y is the standard deviation of the outcome and s_j is the standard deviation of regressor j. The tightness parameter is $\lambda = 10$, reflecting weak prior information. The prior variance on the intercept is 100.

Non-Gaussian errors. As discussed in Jorda (2005), local projection residuals are generally heteroskedastic when $h > 0$. To accommodate this, I model the errors as a scale mixture of normals. Specifically:

$$u_{t+h}|\phi_t \sim N(0, \sigma^2/\phi_t), \qquad \phi_t \sim \Gamma(\nu/2, \nu/2) \tag{A.6}$$

Integrating out the mixing variable ϕ_t yields a Student-t distribution for u_{t+h} with ν degrees of freedom (Geweke 1993). This formulation accommodates heteroskedasticity of unknown form without imposing a specific structure.

The degrees of freedom ν receives an exponential prior:

$$p(\nu) \propto \exp(-\nu/\nu_0) \tag{A.7}$$

with $\nu_0 = 10$. Small values of ν allow heavy tails; the prior places mass on moderate tail thickness while permitting approximate normality.

Posterior simulation. I approximate the posterior using Gibbs sampling with a Metropolis-Hastings step for ν. The sampler iterates over the following blocks, conditioning on all other parameters:

1. *Mixing weights* ϕ_t: Given the current residuals and ν, each ϕ_t is drawn independently from a Gamma distribution with shape $(\nu + 1)/2$ and rate $(u_{t+h}^2/\sigma^2 + \nu)/2$.

2. *Degrees of freedom* ν: The conditional posterior is nonstandard. I update ν via random walk Metropolis-Hastings, proposing $\nu' = \nu + \eta$ where $\eta \sim N(0, c)$. The proposal is accepted with the usual Metropolis probability. The step size c is tuned to achieve an acceptance rate between 30% and 50%.

3. *Error variance* σ^2: Defining transformed residuals $\tilde{u}_{t+h} = u_{t+h}\sqrt{\phi_t}$, the conditional posterior for σ^2 is inverse Gamma with shape $T/2$ and scale $\sum_t \tilde{u}_{t+h}^2/2$.

4. *Coefficients* $\omega^{(h)}$: Applying the same transformation to the dependent variable and regressors, $\tilde{y}_{t+h} = y_{t+h}\sqrt{\phi_t}$ and $\tilde{X}_t = X_t\sqrt{\phi_t}$, yields a standard conjugate updating formula. The conditional posterior is Normal with variance

$$V = \left(S_0^{-1} + \sigma^{-2}\tilde{X}'\tilde{X}\right)^{-1} \tag{A.8}$$

and mean

$$\bar{\omega} = V\left(S_0^{-1}\omega_0 + \sigma^{-2}\tilde{X}'\tilde{y}\right) \tag{A.9}$$

I run 12,000 iterations, discarding the first 3,000 as burn-in and retaining every third draw, yielding 3,000 posterior draws for inference.

Impulse response construction. At horizon $h = 0$, the impulse response equals the A_0 draw from Stage 1b. At horizons $h \geq 1$, the impulse response is constructed following Jorda (2005):

$$IRF_h = B_1^{(h-1)}A_0 \tag{A.10}$$

where $B_1^{(h)}$ contains the coefficients on Z_{t-1} from the horizon-h local projection. For each posterior draw, I pair the A_0 draw from Stage 1 with the $B_1^{(h)}$ draw from Stage 2, ensuring that uncertainty in the impact matrix propagates through to longer horizons. By construction, at $h = 0$, $IRF_0 = A_0$.

Revenue gradients. As discussed in Section 3, the present-value revenue gradient is constructed from the impulse responses of retention rates and tax bases. For tax instrument j, equation (16) gives:

$$r_j^{PV} = B_{j,0} \cdot A \cdot \left[1 - \frac{\tau_j}{1-\tau_j}\varepsilon_{jj} - \frac{\tau_k}{1-\tau_j}\frac{B_k}{B_j}\varepsilon_{kj}\right], \tag{A.11}$$

where $A = \sum_{h=0}^{H} \beta^h$ is the annuity factor. Since the elasticities ε_{ij} are permanent-equivalent (cumulative base responses divided by the cumulative retention rate response κ_j), the revenue gradient uses the common annuity factor rather than instrument-specific persistence. This ensures that g^{PV} and r^{PV} correspond to the same policy experiment: a permanent marginal reform. This calculation is performed for each posterior draw, yielding a posterior distribution for the revenue gradient.

C Rationalizing the Magnitude of ε_{CL}

I find across a variety of methods that the elasticity of the corporate tax base with respect to personal taxes is the largest entry in the 2×2 elasticity matrix. This appendix shows that the magnitude is a predictable consequence of two well-established forces: operating leverage, which amplifies output movements into larger profit movements, and the fact that personal tax cuts generate larger output responses than corporate tax cuts. Together, these explain both the magnitude of the cross-elasicity and the full pattern of elasticities in Table 1.

The intuition begins with a basic feature of national accounts: corporate profits are the residual left after a firm pays its workers and covers costs that are slow to adjust. Consider a firm with \$100 in revenue, \$85 in costs, and \$15 in profits. If revenue rises 1 percent to \$101 but costs do not immediately adjust, profits rise to roughly \$16, which is a 6.7% increase in the corporate tax base. The same 1 percent shock to revenue produced a 6.7 percent change in profits because profits are a thin margin between two large numbers. This is operating leverage: the smaller the profit margin, the larger the amplification.

Operating Leverage. Corporate output—or corporate gross value added—decomposes as

$$Y = wL + \pi + F,$$

where wL is the wage bill, π is pre-tax corporate profits, and F is the remainder of costs, including interest payments, taxes, depreciation, and so on. For convenience, define the labor share as $\gamma = wL/Y$ and the profit share as $s_\pi = \pi/Y$. When output changes by dY and F does not adjust, the change in profits is $d\pi = dY - d(wL)$. If wages are fully flexible and the labor share is roughly constant at the margin, then $d(wL) = \gamma\, dY$ and

$$d \log \pi = \frac{1-\gamma}{s_\pi}\, d \log Y.$$

The ratio $(1-\gamma)/s_\pi$ is the *frictionless* operating leverage. It is a lower bound on the true leverage because it assumes wages absorb their full share of output movements within the horizon. When wages are sticky, compensation adjusts by less than $\gamma\, dY$ in the short run, so a larger share of any output gain initially accrues to profits. This raises effective leverage above $(1-\gamma)/s_\pi$ and simultaneously attenuates the passthrough of output to the personal tax base below one. The same friction that makes the cross-elasticity larger than the frictionless prediction makes the personal base elasticities smaller. It exceeds one whenever $F > 0$ since non-labor income must cover both fixed costs F and profits π, so the percentage change in profits exceeds the percentage change in output by a factor of $(1-\gamma)/s_\pi$.[15] This is a lower bound on effective leverage because it assumes the labor bill freely adjusts. Unlike profits, wages have no leverage. When compensation adjusts freely, $d\log(wL) = d\log Y$, so a percent increase in output fully passes through to the personal tax base. When wages are sticky, the passthrough is attenuated further below one. In other words, the same friction that raises leverage on the corporate tax base lowers passthrough on the personal income tax base.

Tax Cuts and the Response of Output. The leverage result shows that the elasticity of the corporate base with respect to any shock ε_{CX} decomposes into the response of output to that shock, amplified by the factor $(1-\gamma)/s_\pi$:

$$\varepsilon_{CL} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{CC} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YC}$$

where ε_{YL} is the elasticity of output with respect to the log personal retention rate. To evaluate the contribution of leverage to the response of the corporate base, we need to know both ε_{YL} and ε_{YC}. These are not directly estimated in the baseline specification, but Appendix G extends the baseline to include GDP, debt, and government spending—exactly the specification in Mertens and Ravn (2013b). That allows direct estimation of both ε_{YL} and ε_{YC}. Across estimation methodologies, I find that both ε_{YL} and ε_{YC} are robustly positive and $\varepsilon_{YL} \approx 2 \times \varepsilon_{YC}$. Under the Bayesian local projection, $\varepsilon_{YL} = 2.1$ and $\varepsilon_{YC} = 1.0$. The asymmetry is intuitive. Personal tax cuts raise labor supply through a higher after-tax wage, which immediately raises output. Corporate tax cuts lower the user cost of capital, but capital accumulation is slow and gradual, and the direct effect is confined to the corporate sector's investment margin. The output response to corporate tax cuts is therefore smaller and more back-loaded.

Since leverage applies equally to both shocks, the full elasticity matrix is determined by two objects: the leverage factor $(1-\gamma)/s_\pi$ and the output elasticities. For the corporate base, the predictions are given above. For the personal base, there is no leverage since wages should scale appropriately with output, so

$$\varepsilon_{LL} \approx \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{LC} \approx \varepsilon_{YC}.$$

Both γ and s_π are directly observable from NIPA Table 1.14 (Gross Value Added of Domestic Corporate Business): γ is compensation of employees divided by GVA, and s_π is corporate profits

15. All that we require for the argument to work is that some portion of the cost structure is predetermined. This is almost trivially satisfied. Capital is subject to fixed adjustment costs, with the majority of firms holding their capital stock fixed in any given quarter (Winberry 2021; Koby and Wolf 2020). Employment adjusts in infrequent, discrete changes rather than continuously (Caballero, Engel, and Haltiwanger 1997). Leases, debt service, and administrative overhead are fixed by construction.

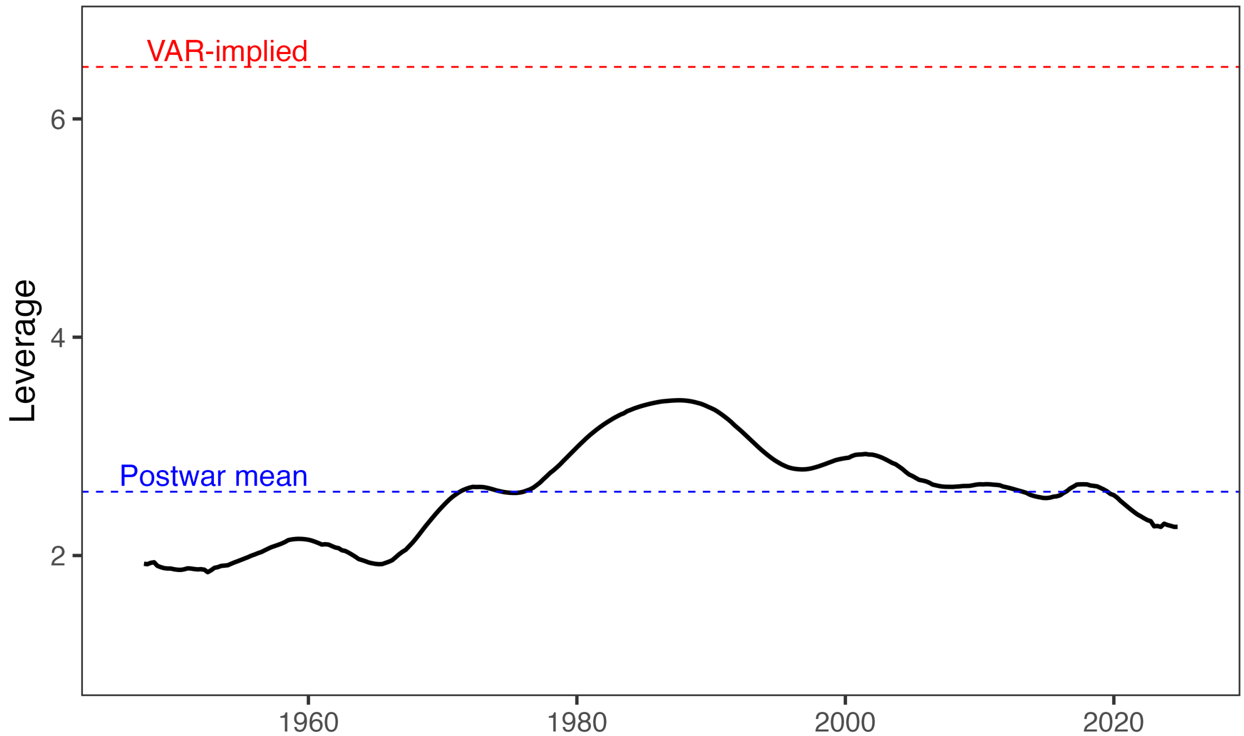
with inventory valuation and capital consumption adjustments divided by GVA. Figure C.1 plots the leverage series over time. It ranges from approximately 2 to 3.5, with a postwar mean of 2.6; the empirical leverage from the extended specification ($\varepsilon_{CL}/\varepsilon_{YL} = 6.4$) lies above it at every date, consistent with wage stickiness raising effective leverage above the frictionless benchmark. Leverage was lowest in the early postwar period and peaked in the 1980s. Table C.1 compares the frictionless predictions to the data, using the postwar mean leverage of 2.6.

Table C.1: Frictionless Predictions vs. Data

	Frictionless	Data (4-var)	Data (7-var)
ε_{CC}	$2.6 \times 1.0 \approx 2.6$	4.25	4.00
ε_{CL}	$2.6 \times 2.1 \approx 5.5$	9.03	13.59
ε_{LC}	$1 \times 1.0 \approx 1.0$	−0.11	0.65
ε_{LL}	$1 \times 2.1 \approx 2.1$	0.98	1.23

The frictionless model matches the qualitative ordering $\varepsilon_{CL} > \varepsilon_{CC} > \varepsilon_{LL} > \varepsilon_{LC}$ and rationalizes the cross-elasticity to within a factor of 1.6, using only national accounts arithmetic and the output elasticities from the extended specification. Moreover, the frictionless prediction of 5.5 is a lower bound. The empirical leverage from the seven-variable specification—$\varepsilon_{CL}/\varepsilon_{YL} = 6.4$—times the output elasticity of 2.1 gives 13.4, which exceeds the four-variable estimate of 9. The baseline estimate therefore sits between the frictionless floor (5.5) and the sticky-wage ceiling implied by the extended specification (13.4), exactly where a model with partial wage adjustment would place it. The systematic pattern in the residuals confirms the wage stickiness channel introduced above. The frictionless model overpredicts personal base elasticities and underpredicts corporate base elasticities in every entry—exactly the signature of slow wage adjustment. Empirical leverage exceeds the frictionless benchmark for both shocks: 6.4 versus 2.6 for the personal tax shock and 4.0 versus 2.6 for the corporate tax shock. The gap is larger for the personal shock because its output response is more front-loaded, so high short-run leverage has more to amplify. A model with partial wage adjustment would close the remaining gap between predictions and data. The frictionless benchmark is not intended as a complete account; it is intended to show that the cross-elasticity is a predictable consequence of operating leverage and the output multiplier asymmetry, not of exotic behavioral responses.

Figure C.1: Leverage in the National Income and Product Accounts

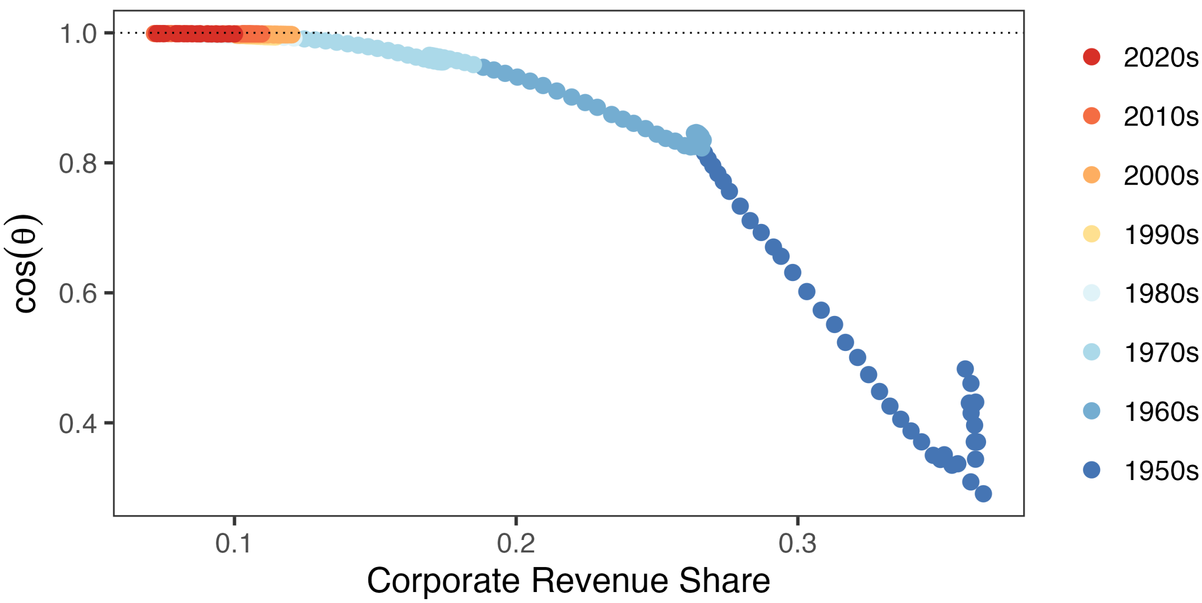


Note: Leverage is computed using Table 1.14 from the National Income and Product Accounts. I compute the compensation share γ as compensation of employees (line 7) divided by gross value added of corporate business (line 1), and the profit share s_π as corporate profits with inventory valuation and capital consumption adjustments (line 14) divided by gross value added of corporate business. Raw leverage is $(1-\gamma)/s_\pi$. I remove fluctuations at business cycle frequencies (2–40 quarters) using the Christiano and Fitzgerald (2003) filter and plot the resulting trend.

D Additional Alignment Results

Figure D.1 shows that alignment tracks the corporate share of income tax revenue almost monotonically, motivating the decomposition analysis in Section 4.

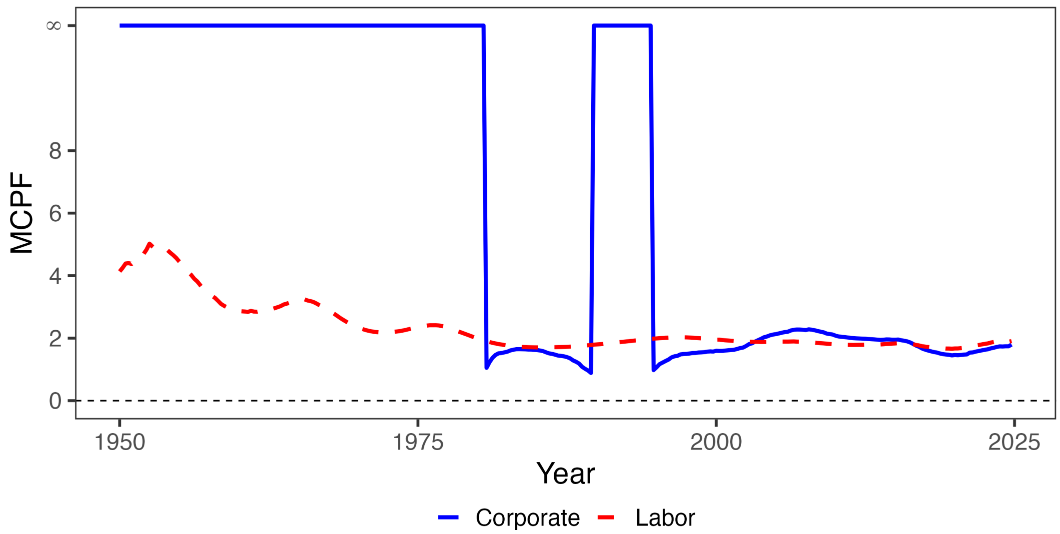
Figure D.1: System Efficiency and the Corporate Revenue Share



Note: Each point represents a quarter, colored by decade. The horizontal axis plots the corporate share of federal income tax revenue; the vertical axis plots $\cos\theta$, the alignment between welfare and revenue gradients. Higher values indicate greater alignment with the Ramsey optimum.

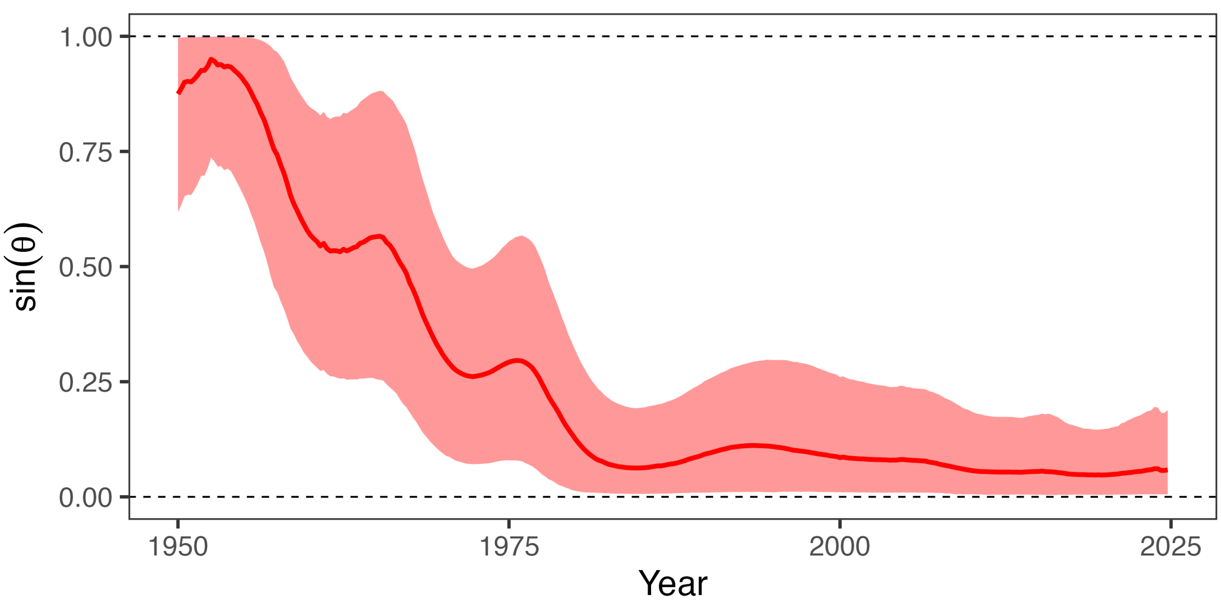
Figure D.2 plots the marginal value of public funds for each instrument. The corporate MVPF starts infinite in the early postwar period—indicating corporate taxes were on the wrong side of the Laffer curve—and falls toward convergence with the personal MVPF by the 1980s. This is the same convergence visible in $\cos\theta$, expressed in the classical language of optimal taxation.

Figure D.2: The Marginal Cost of Public Funds



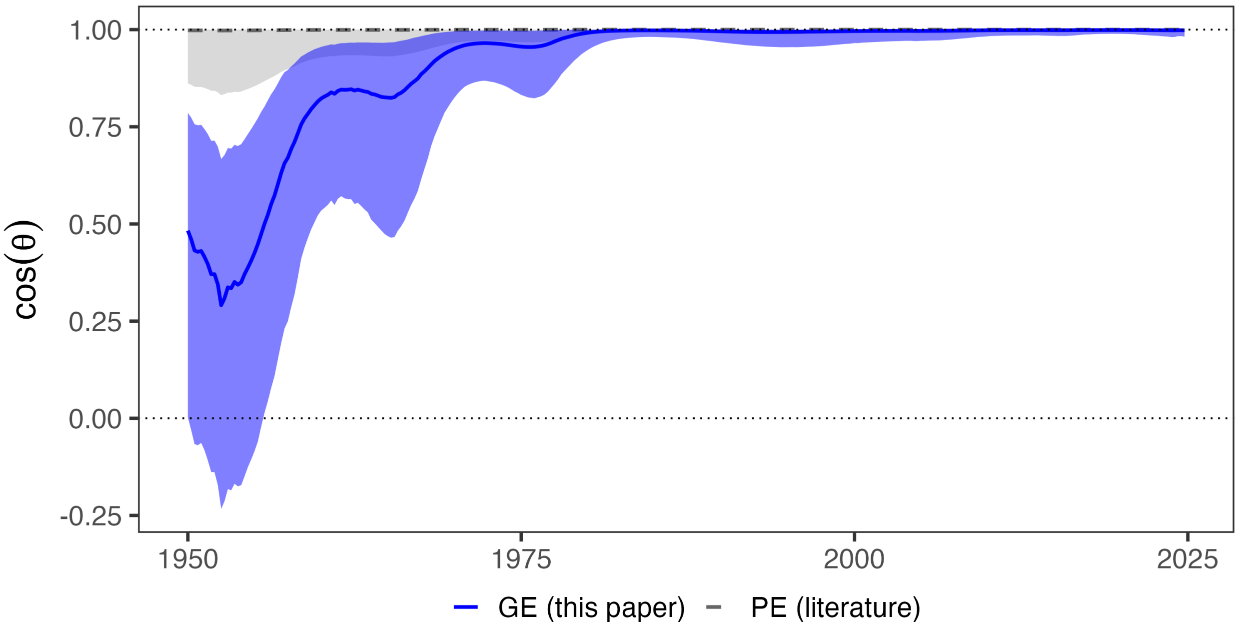
Note: This figure plots the marginal value of public funds for corporate and personal income taxes. Following Finkelstein and Hendren (2020), I denote the MVPF as infinite if it is on the wrong side of the Laffer curve.

Figure D.3: The Share of Welfare Gains Available Revenue-Neutrally



Note: $\sin\theta$ measures the share of welfare gains available from revenue-neutral reform. Shaded regions show 90% credible intervals.

Figure D.4: Partial vs. General Equilibrium Alignment

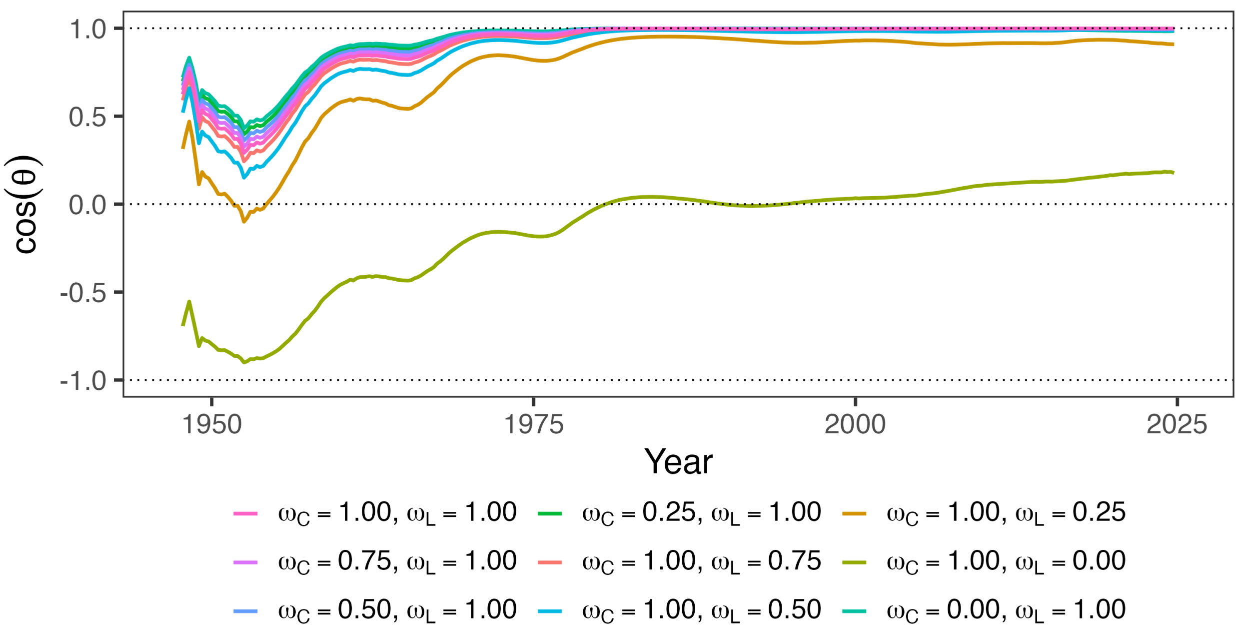


Note: Solid blue line shows $\cos\theta$ from general equilibrium estimates with cross-base spillovers. Dashed gray line and shaded band show $\cos\theta$ using partial equilibrium own-elasticities from the literature with cross-elasticities set to zero. The PE band spans all combinations of $\varepsilon_{CC} \in [0.08, 4.79]$ and $\varepsilon_{LL} \in [0.12, 0.40]$.

E Four-Variable System Robustness

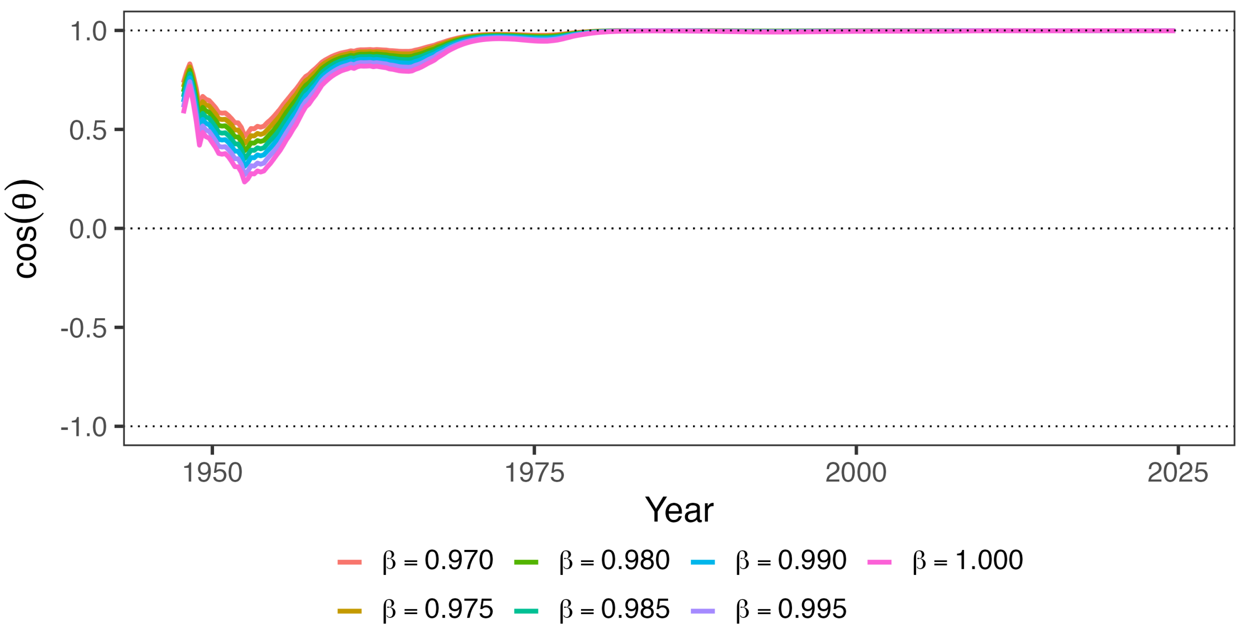
E.1 Robustness of Alignment Statistics

Figure E.1: Robustness Across Welfare Weights



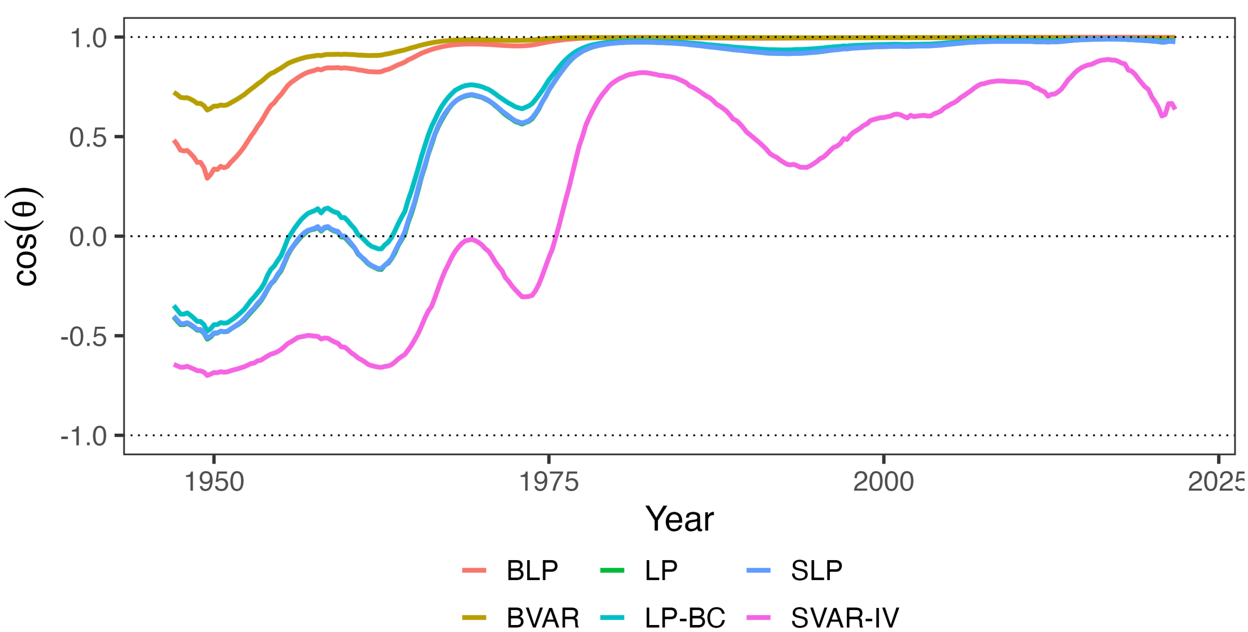
Note: Each panel varies one welfare weight while holding the other at 1. Left panel varies ω_C (weight on corporate taxpayers) with $\omega_L = 1$; right panel varies ω_L (weight on personal taxpayers) with $\omega_C = 1$. The utilitarian baseline has $\omega_C = \omega_L = 1$.

Figure E.2: Robustness Across Discount Factors



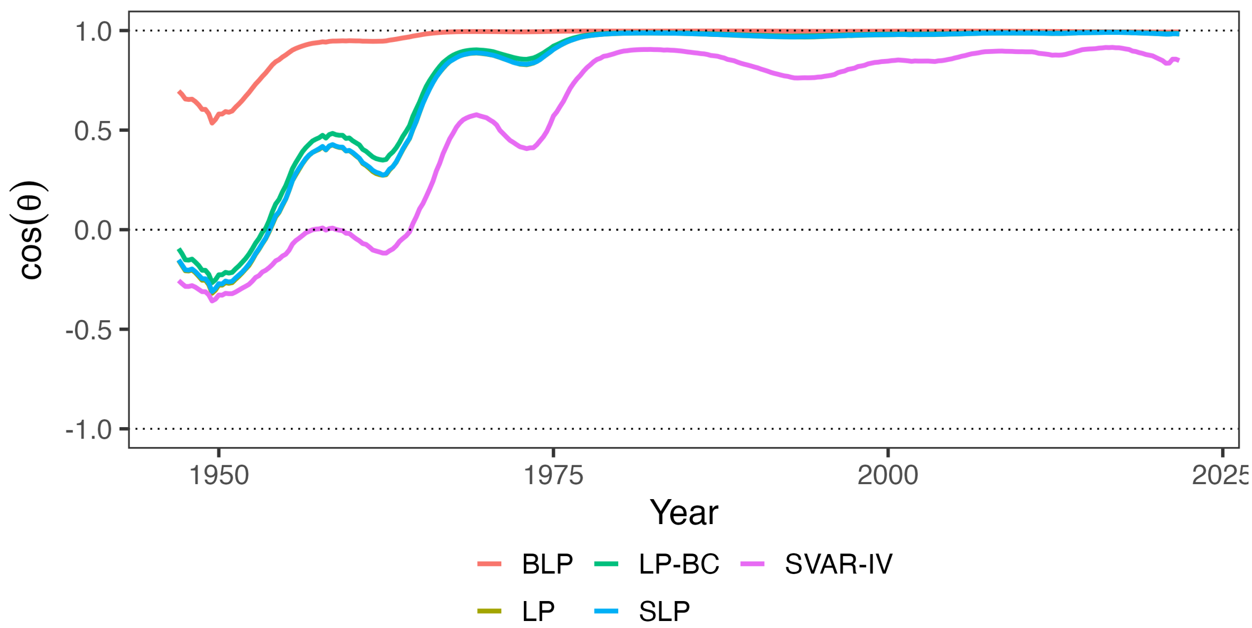
Note: This figure plots alignment $\cos\theta$ across different discount rate regimes. Under the baseline, $\beta = 0.9926$.

Figure E.3: Robustness Across Estimation Methodologies



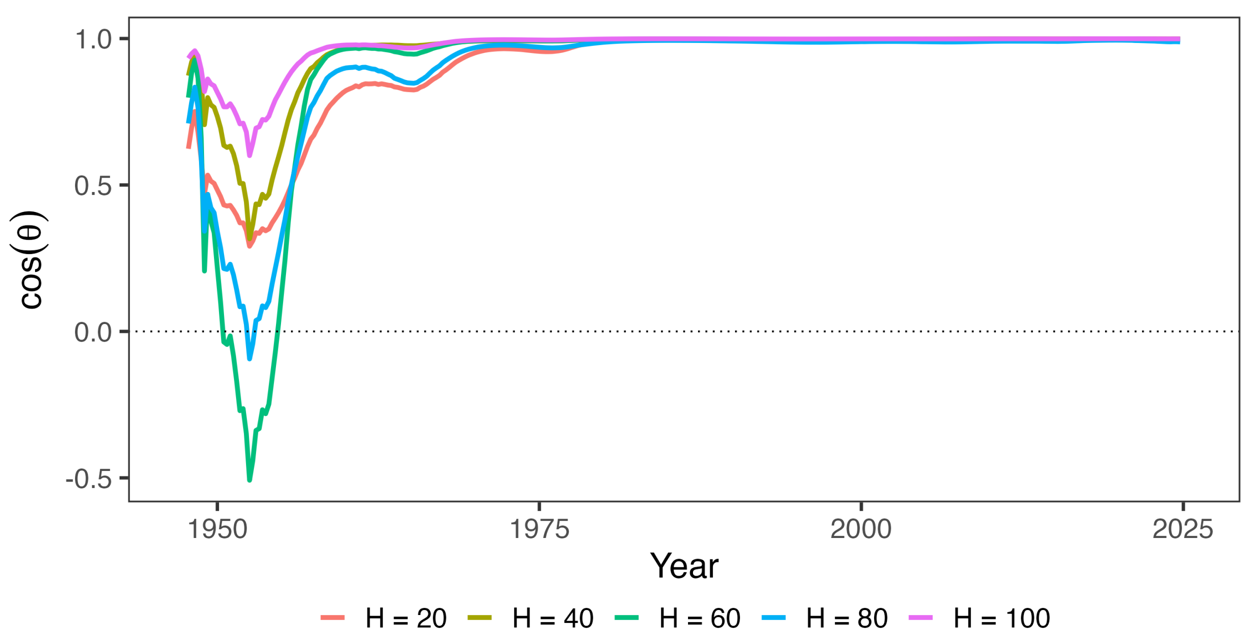
Note: This figure plots alignment $\cos\theta$ across estimation methodologies.

Figure E.4: Robustness with Linear Trend



Note: This figure plots alignment $\cos\theta$ across estimation methodologies and adds a linear trend. I do not include the BVAR because it takes care of trends by construction.

Figure E.5: Robustness Across Horizon Specifications



Note: This figure plots alignment $\cos\theta$ for select models in which vary the horizon length, holding fixed $\beta = 0.9926$.

E.2 Underlying Elasticities and CIRFs

This appendix shows robustness for the own and cross-revenue effects of corporate and personal income tax shocks. Table 1 in the main text shows the present value of each, while this Appendix shows the IRFs that generate the present value results. All specifications use the same set of variables. The results are consistent across local projections specifications but somewhat noisier with the VAR specifications.

Local projections. This variant uses standard local projections with the Mertens-Ravn structural shocks as instruments. Again, the results are similar, though somewhat smaller. I propagate uncertainty about A_0 using the wild bootstrap proxy SVAR.

Smooth LP. Because the local projections results are lumpy, I show another variant which penalizes deviations across horizons using Barnichon and Brownlees (2019) in Figure E.8.

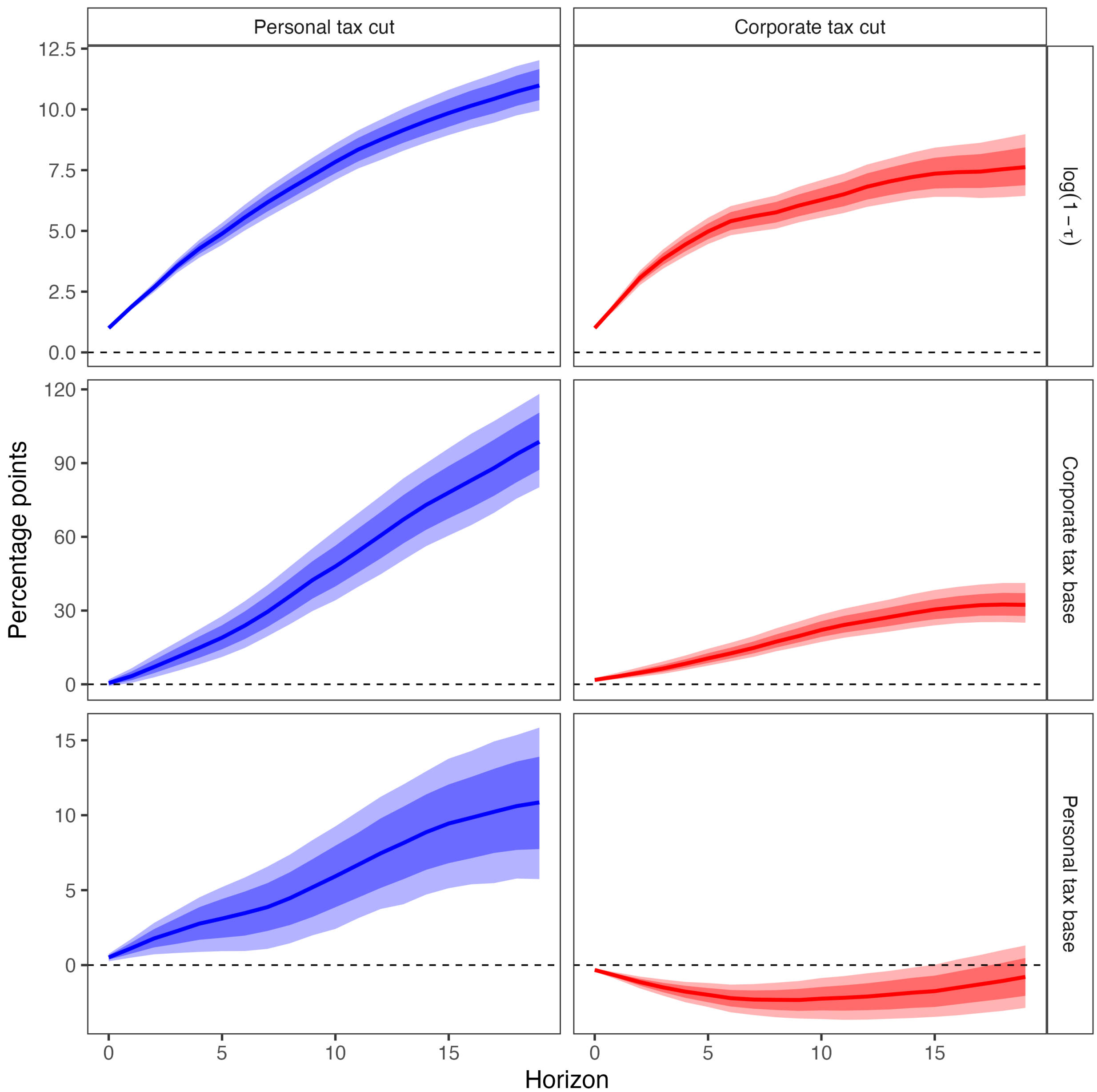
Bias-Corrected LP. Figure E.9 shows the bias-corrected discounted cumulative IRF from Herbst and Johannsen (2024).

Bayesian VAR. The main results come from a Bayesian local projection with the instruments generated from a Bayesian VAR with four lags (selected by the BIC). Figure E.10 shows the impulse responses generated by the Bayesian VAR along with 68% and 90% credible intervals. The IRFs are qualitatively similar, though noticeably larger than the baseline BLP.

Proxy SVAR. Figure E.11 shows the impulse responses generated by corporate and personal income tax shocks in a frequentist structural vector autoregression. The specification is the exact same as in the baseline Mertens and Ravn (2013b); it is just extended to 2019Q4. Following their lead, I employ a wild bootstrap. The results are less precise but otherwise qualitatively and quantitatively quite similar to the baseline.

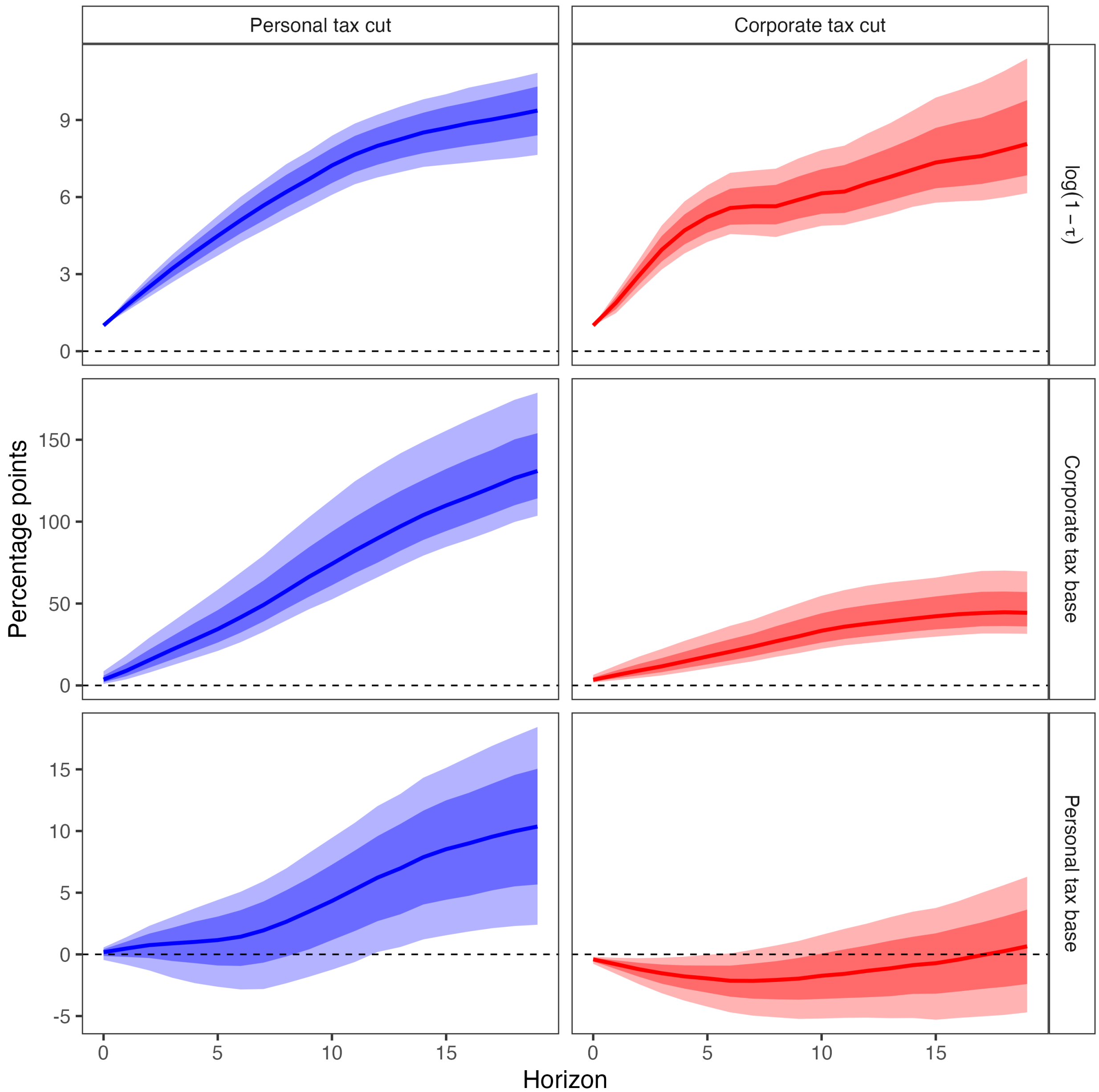
Linear Trends. Figures E.12-E.16 repeat the above specifications but add linear trends.

Figure E.6: Dynamic Effects of Corporate and Personal Income Tax Cuts on Tax Rates and Bases



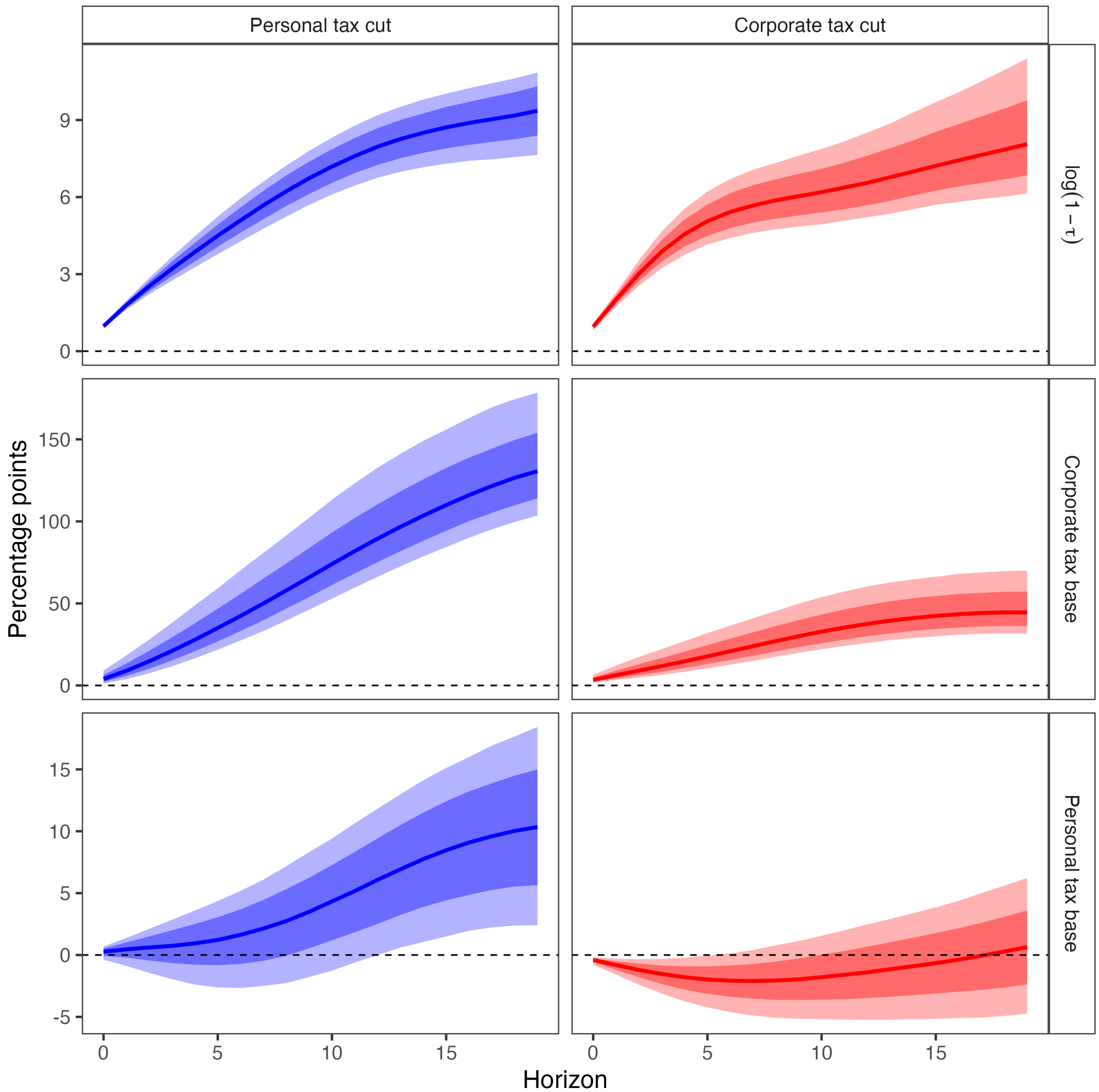
Note: Discounted cumulative impulse responses ($\beta = 0.9926$) to a unit increase in the log retention rate $\log(1 - \tau)$. Sample: 1947Q1–2019Q4. Posterior medians with 68% and 90% credible intervals.

Figure E.7: Dynamic Effects of Narrative Tax Shocks on Revenue (LP)



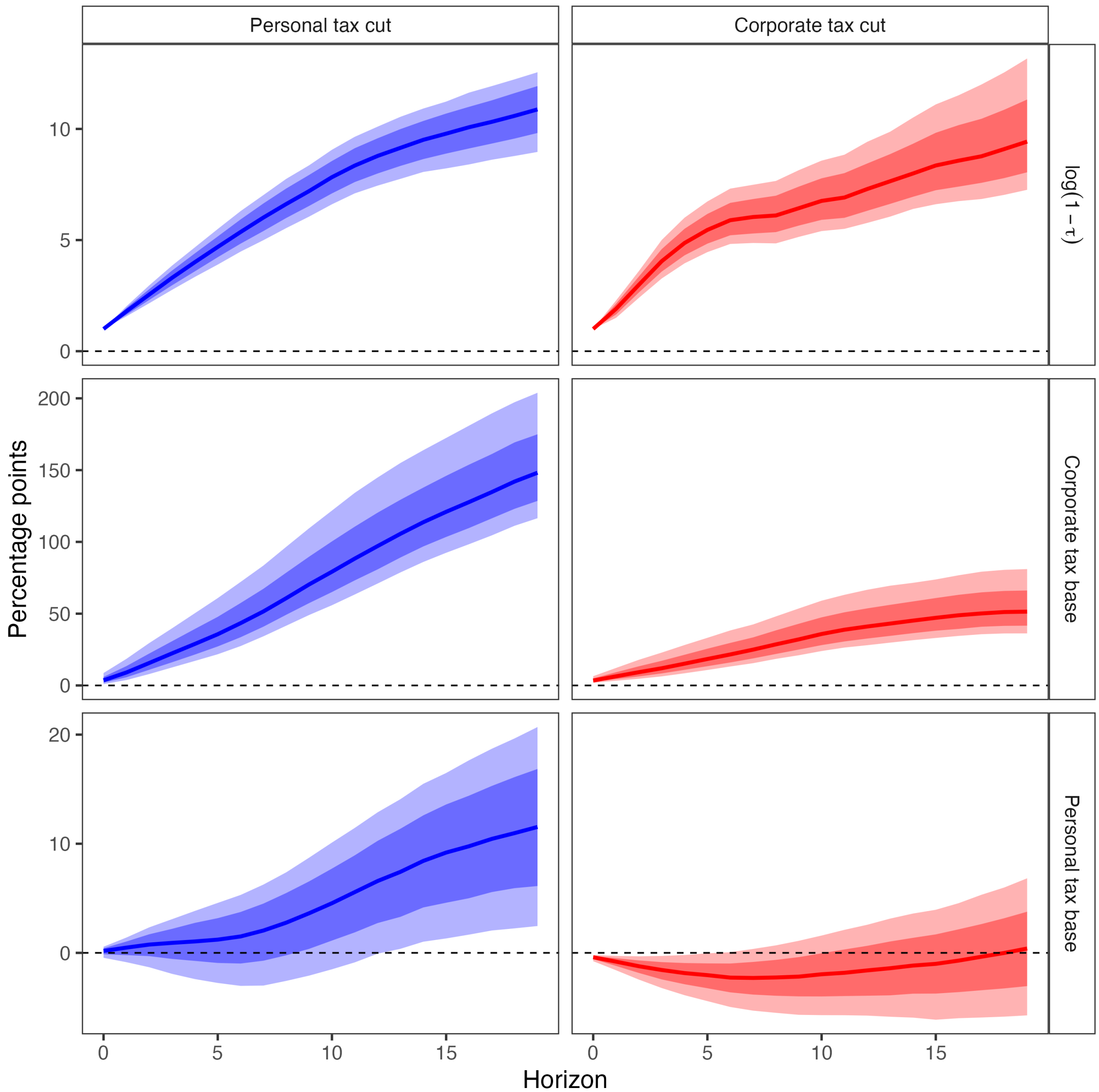
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

Figure E.8: Dynamic Effects of Narrative Tax Shocks on Revenue (Smooth LP)



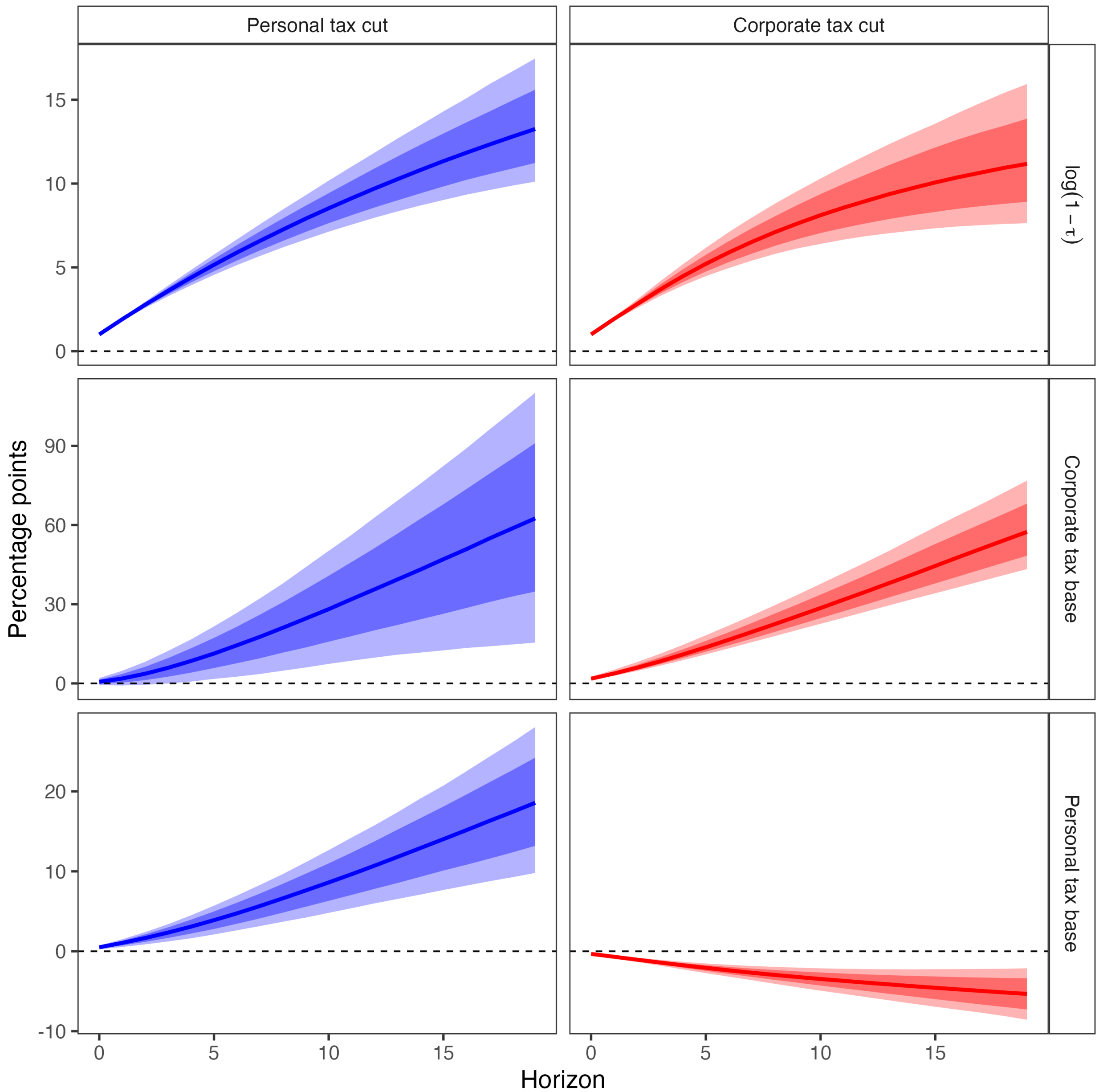
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

Figure E.9: Dynamic Effects of Narrative Tax Shocks on Revenue (Bias-Corrected LP)



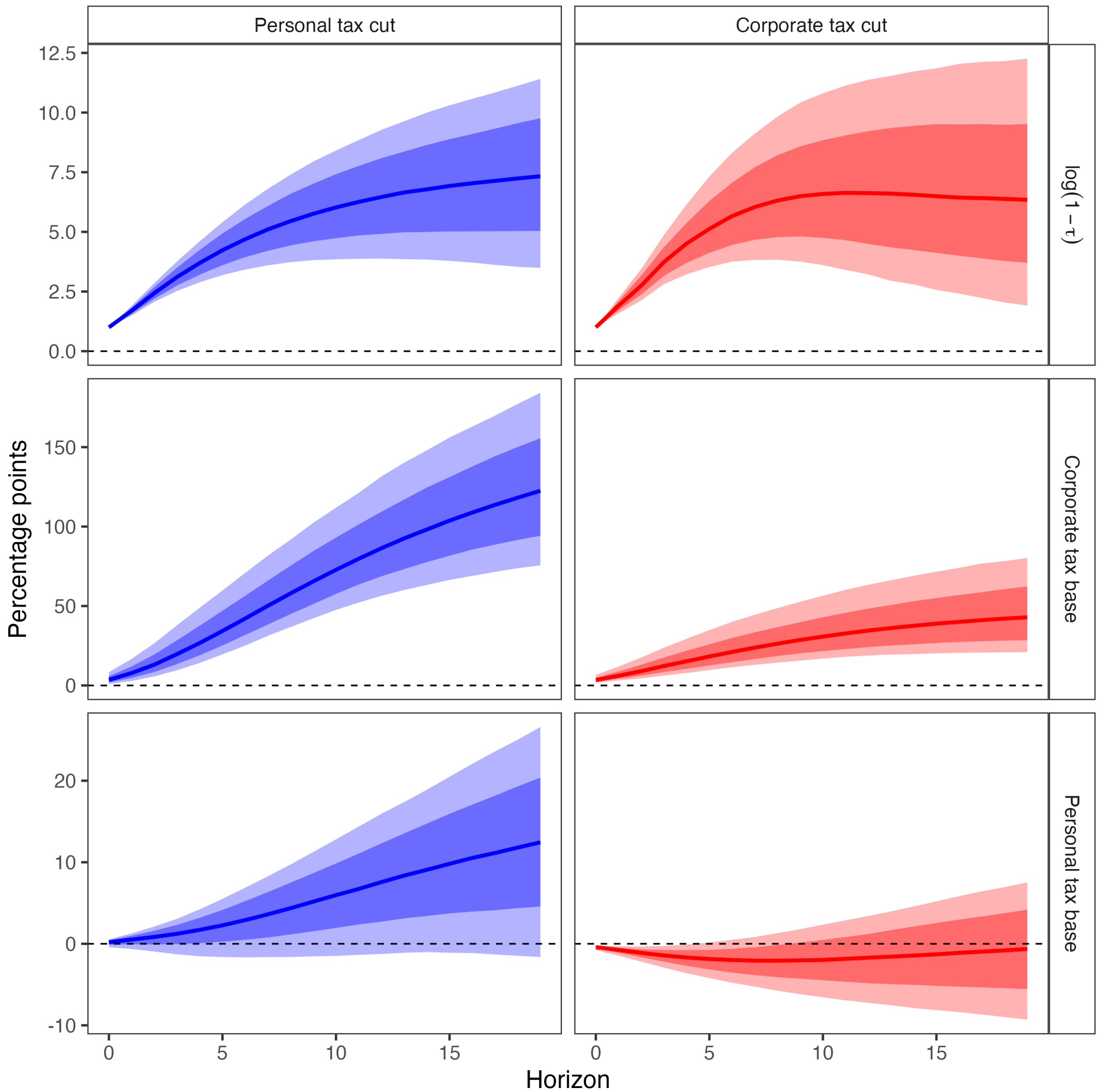
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

Figure E.10: Dynamic Effects of Narrative Tax Shocks on Revenue (Bayesian VAR)



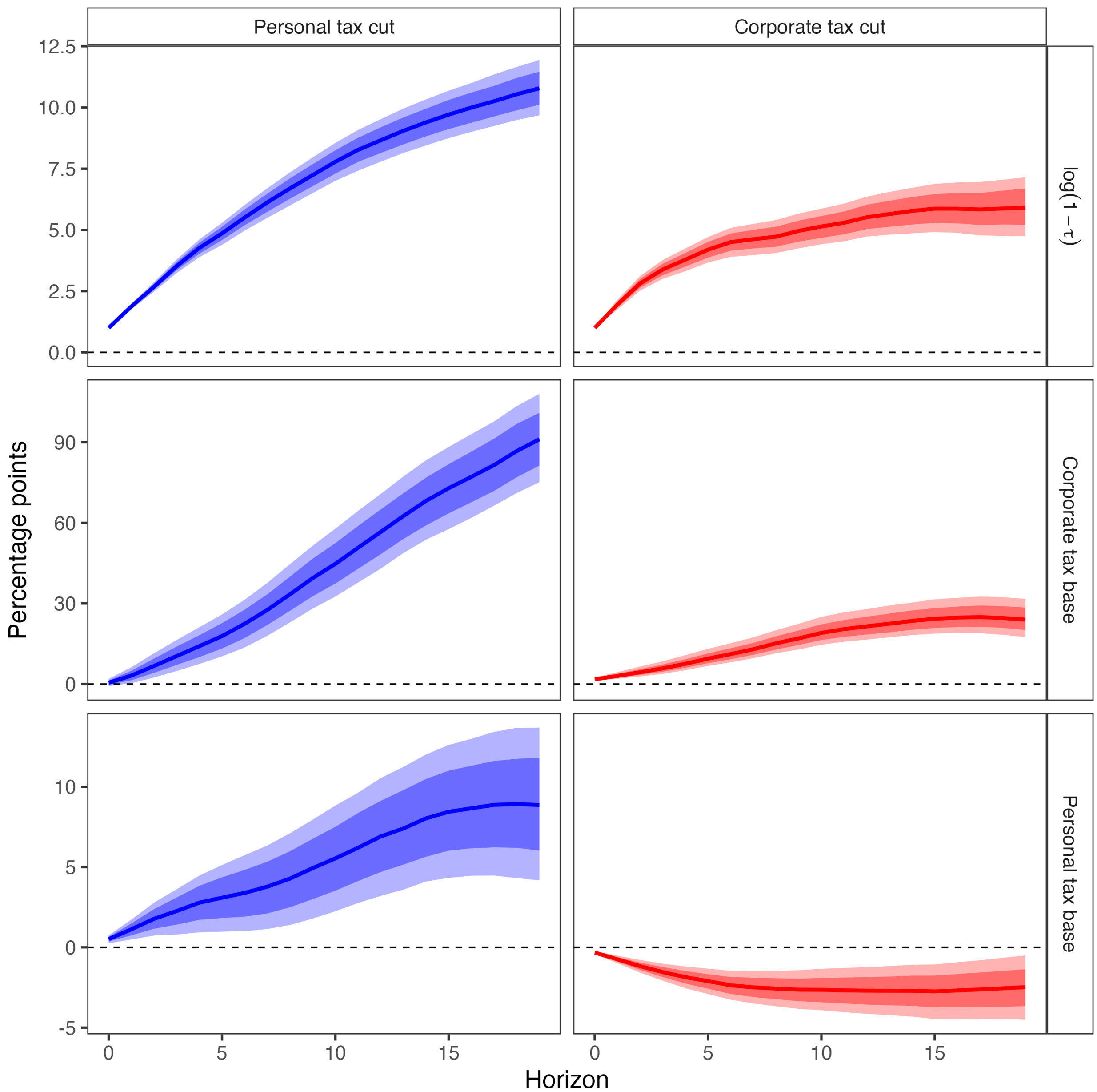
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

Figure E.11: Dynamic Effects of Narrative Tax Shocks on Revenue (Proxy SVAR)



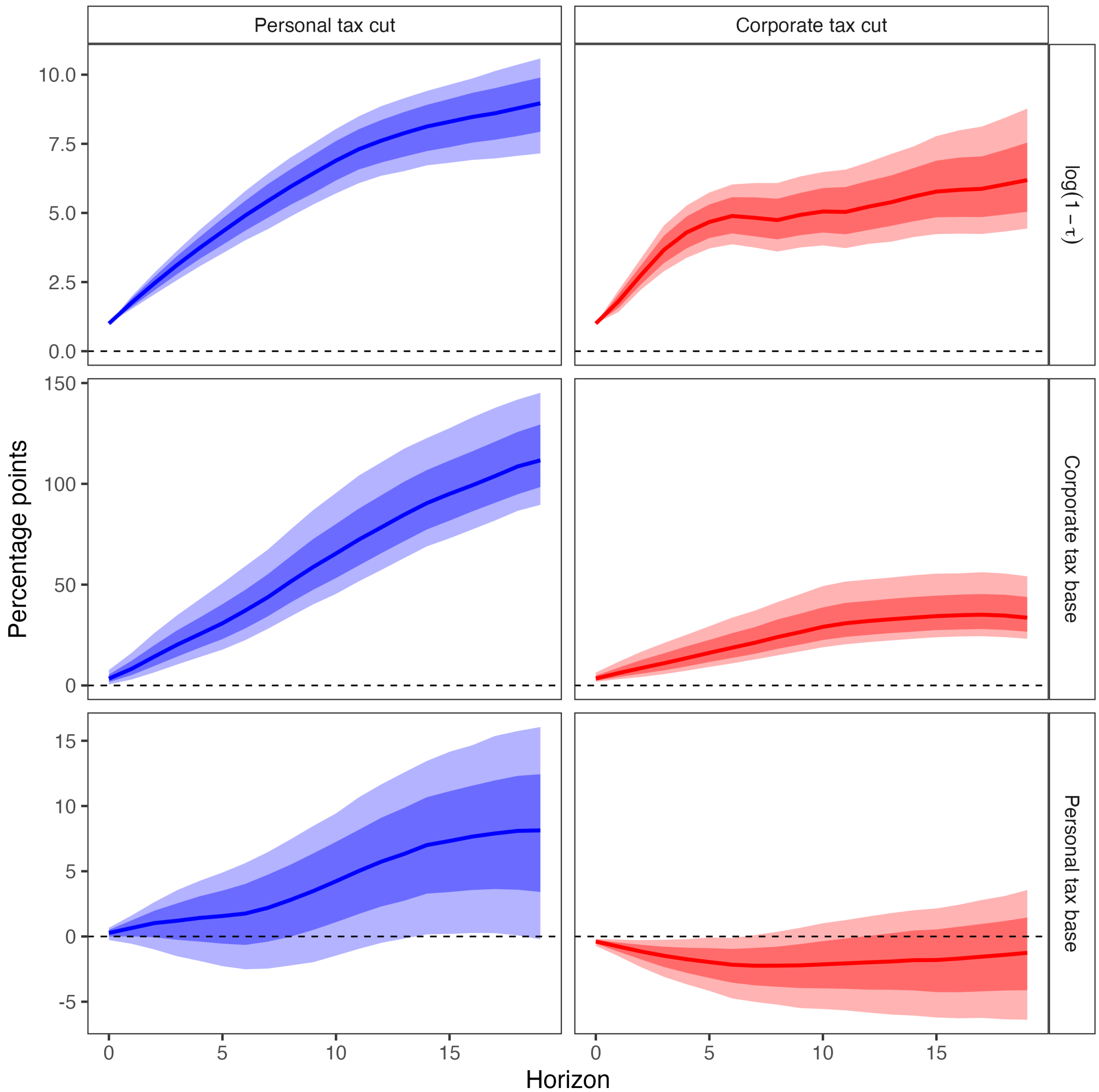
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

Figure E.12: Dynamic Effects of Corporate and Personal Income Tax Cuts on Tax Rates and Bases (BLP with Linear Trend)



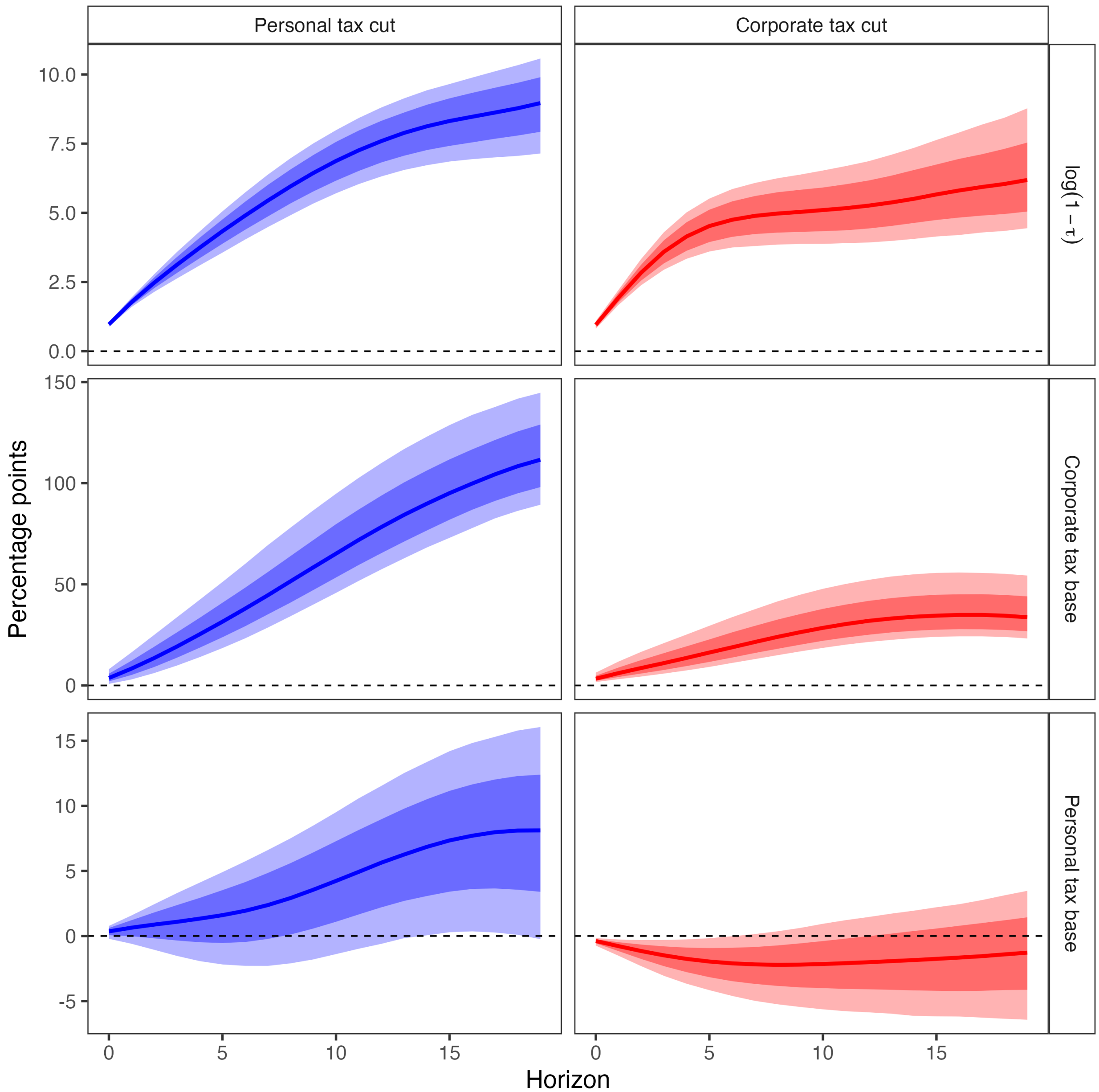
Note: Discounted cumulative impulse responses ($\beta = 0.9926$) to a unit increase in the log retention rate $\log(1-\tau)$. Sample: 1947Q1–2019Q4. Posterior medians with 68% and 90% credible intervals. This specification has a linear trend.

Figure E.13: Dynamic Effects of Narrative Tax Shocks on Revenue (LP with Linear Trend)



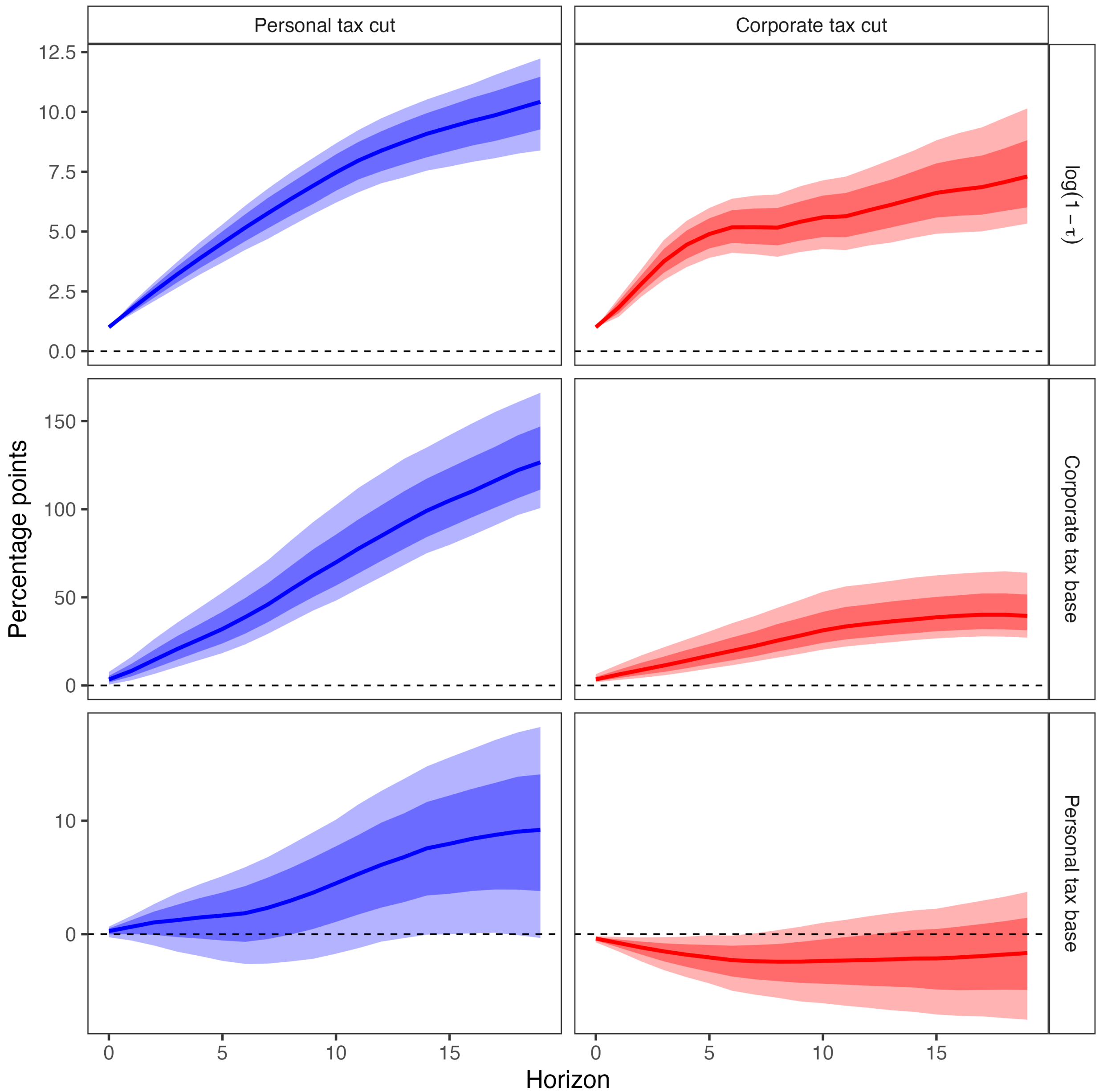
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$. This specification has a linear trend.

Figure E.14: Dynamic Effects of Narrative Tax Shocks on Revenue (Smooth LP with Linear Trend)



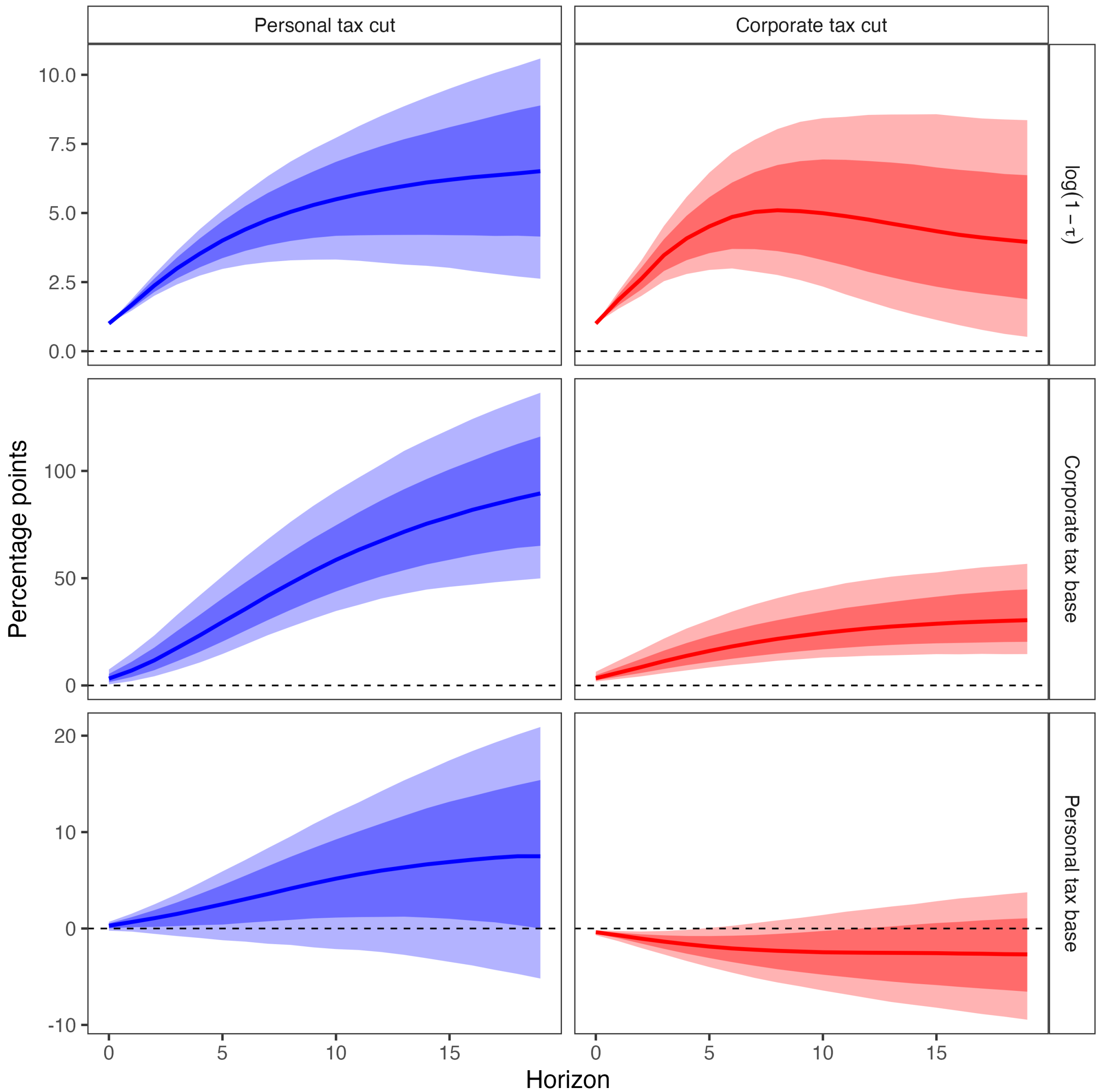
Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$. This specification has a linear trend.

Figure E.15: Dynamic Effects of Narrative Tax Shocks on Revenue (Bias-Corrected LP with Linear Trend)



Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$. This specification has a linear trend.

Figure E.16: Dynamic Effects of Narrative Tax Shocks on Revenue (Proxy SVAR with Linear Trend)



Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$. This specification has a linear trend.

Table E.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates (with linear trend)

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	4.10	-0.42	8.49	0.81
	(2.71, 6.13)	(-0.75, -0.09)	(6.68, 10.53)	(0.35, 1.29)
LP	5.47	-0.21	12.37	0.90
	(3.75, 8.36)	(-0.92, 0.63)	(8.99, 19.17)	(0.02, 2.09)
SLP	5.49	-0.21	12.34	0.90
	(3.77, 8.38)	(-0.92, 0.62)	(8.99, 19.17)	(0.02, 2.09)
LP-BC	5.44	-0.23	12.06	0.88
	(3.75, 8.39)	(-0.91, 0.57)	(8.74, 18.65)	(0.00, 2.04)
SVAR-IV	7.25	-0.64	13.65	1.13
	(1.93, 35.95)	(-3.90, 1.65)	(7.97, 30.38)	(-0.67, 6.38)

Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the four-variable system comprised solely of tax bases and rates.

E.3 Re-ordering the Shocks

Separating the two structural tax shocks from the proxy-identified block requires a Cholesky decomposition of the 2×2 residual covariance matrix. In the baseline specification, when estimating the corporate tax shock we place the corporate tax rate first in the ordering, and when estimating the personal tax shock we place the personal tax rate first. The variable ordered second is restricted to have no contemporaneous response to the other shock.

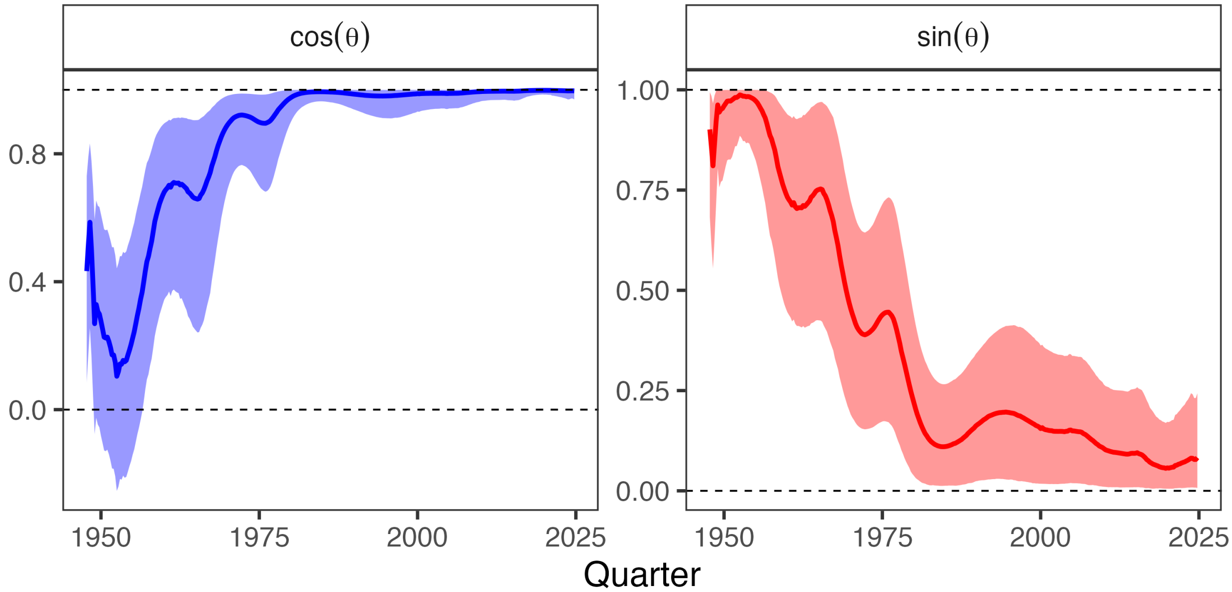
Here we reverse both orderings. The stability of the impulse responses across specifications suggests that the two shocks exhibit minimal contemporaneous correlation, and hence the triangularization assumption plays little role in shaping our conclusions.

Table E.2: Present Value Elasticities of Tax Bases with Respect to Retention Rates (Alternative Ordering)

	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	5.16	-0.05	9.62	0.96
	(3.66, 7.16)	(-0.35, 0.26)	(8.09, 11.43)	(0.48, 1.49)
LP	3.24	-0.06	11.72	1.04
	(-0.10, 6.24)	(-0.56, 0.47)	(9.93, 14.09)	(0.40, 1.89)
SLP	3.25	-0.06	11.71	1.05
	(-0.09, 6.30)	(-0.56, 0.47)	(9.90, 14.02)	(0.40, 1.88)
LP-BC	3.29	-0.09	11.42	1.02
	(0.02, 6.17)	(-0.58, 0.42)	(9.65, 13.74)	(0.39, 1.84)
BVAR	5.80	-0.42	5.64	1.30
	(3.28, 10.02)	(-0.75, -0.11)	(2.50, 9.10)	(0.56, 2.23)
SVAR-IV	4.05	-0.28	14.92	1.85
	(-0.53, 11.16)	(-1.97, 0.91)	(9.56, 28.13)	(0.15, 6.43)

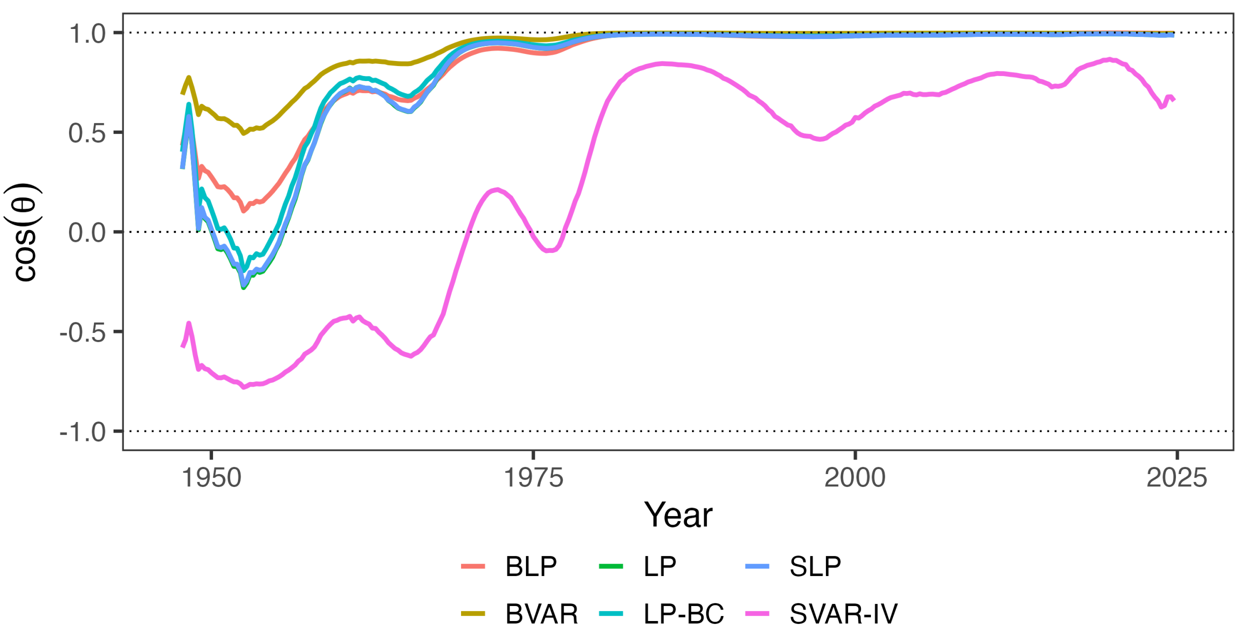
Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1-\tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the four-variable system comprised solely of tax bases and rates, but with the ordering of the shocks reversed.

Figure E.17: The Evolution of Tax System Efficiency (Alternative Ordering)



Note: Left panel plots $\cos\theta$, the alignment between welfare and revenue gradients; $\cos\theta = 1$ indicates the Ramsey optimum. Right panel plots $\sin^2\theta$, the share of potential welfare gains available through revenue-neutral reform. Shaded regions show 90% credible intervals. This figure alters the ordering of the shocks for the Bayesian LP.

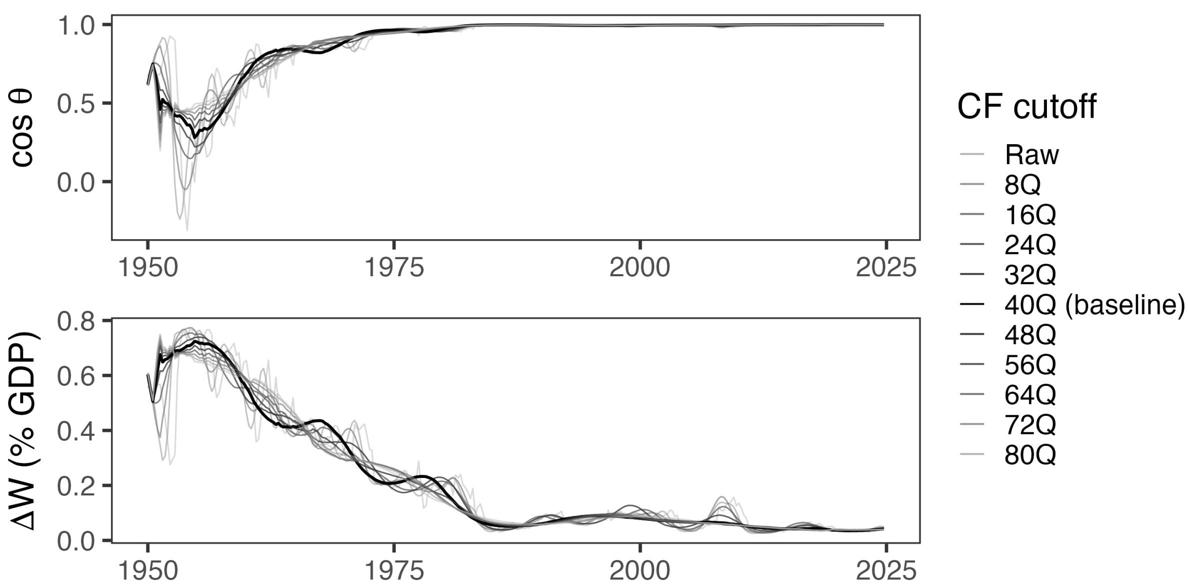
Figure E.18: Robustness Across Estimation Methodologies (Alternative Ordering)



Note: This figure plots alignment $\cos\theta$ across estimation methodologies while switching the order of the shocks.

F Robustness to Filtering Choice

Figure F.1: Robustness to Filtering Choice



Note: Alignment metrics under alternative Christiano-Fitzgerald band-pass filter cutoffs. "Raw" uses unfiltered quarterly data; other series retain variation at periodicities above the indicated cutoff. The baseline specification (40Q, black) is highlighted. Top panel shows $\cos\theta$; bottom panel shows welfare gains from optimal reform ΔW as a share of GDP.

G Seven-Variable System Robustness

The baseline specification uses four variables: personal and corporate tax rates and their corresponding bases. This appendix explains why the four-variable specification is appropriate for estimating revenue gradients, and shows robustness to a seven-variable specification that adds output, government spending, and debt.

The revenue gradient in the projection framework is the total derivative of present-value revenue with respect to the tax rate:

$$r_j = \frac{\partial}{\partial \tau_j} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t R_t,$$

where $R_t = \tau_L B_L + \tau_C B_C$ is total income tax revenue. The key object is $\frac{\partial B_k}{\partial \tau_j}$, the total derivative of base k with respect to tax rate j. This total derivative includes all channels through which a tax change affects the base: direct behavioral responses, income shifting between bases, and general equilibrium effects through output, investment, and employment. The theory does not ask for $\frac{\partial B_k}{\partial \tau_j}\Big|_Y$, the partial derivative holding output constant. It asks for the full response that a policymaker would observe if they changed the tax rate.

The four-variable specification delivers this total derivative. When we estimate the impulse response of the corporate base to a personal tax shock, the response captures all channels, including any effects operating through GDP. We are not conditioning on GDP, so the full reduced-form relationship between tax shocks and bases is preserved.

A seven-variable specification that includes GDP, government spending, and debt estimates a different object. By including lagged GDP as a regressor in the local projection, we ask: what is the effect of a tax shock on the corporate base, controlling for GDP? This partials out GDP-mediated effects. If a personal tax cut raises GDP, and higher GDP raises corporate profits, then including GDP in the system absorbs some of the variation we want to attribute to the tax shock. The resulting elasticity is a conditional object that understates the total revenue response.

One might worry that omitting GDP leaves the tax shock correlated with other structural disturbances, biasing the estimates. But narrative identification does not require controlling for other variables. The Mertens-Ravn proxy SVAR achieves identification through moment conditions requiring that the narrative instruments correlate with structural tax shocks and are uncorrelated with other structural shocks. These conditions are satisfied by construction: Romer and Romer (2010) selected tax changes legislated for reasons unrelated to current economic conditions. This selection is what ensures exogeneity, not the inclusion of control variables. Adding variables to the VAR does not improve identification; it changes what object is being estimated.

Mertens and Ravn (2013a) included output, government spending, and debt because they were interested in output multipliers and fiscal sustainability. For their research question, the seven-variable specification was appropriate. For the present question—what are the revenue effects of tax changes for evaluating Ramsey alignment?—the total elasticity is required, and the additional variables are unnecessary.

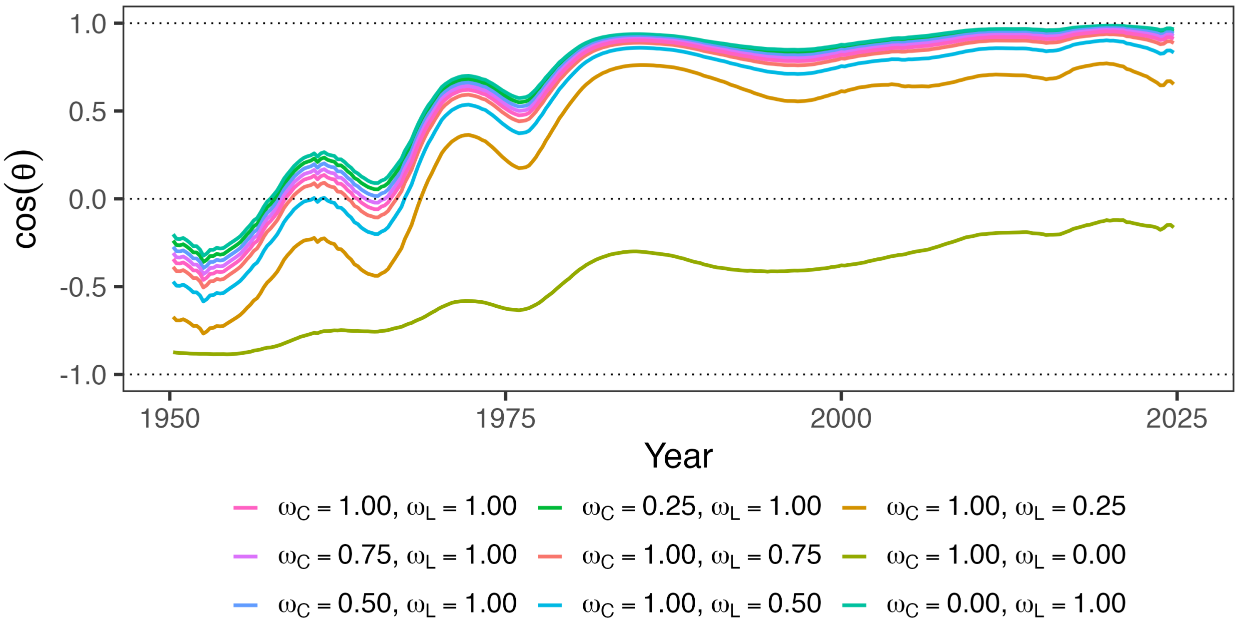
Table G.1 reports elasticities from both specifications. The seven-variable specification yields qualitatively similar point estimates but wider credible intervals, as expected given the additional parameters. The alignment metrics are robust to this choice: $\cos\theta$ in the 2010s exceeds 0.8 under both specifications. The four-variable specification is the appropriate baseline because it estimates the object the theory requires; the seven-variable specification serves as a robustness check demonstrating that results are not sensitive to the conditioning set.

Table G.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates (Seven Variable System)

	Corporate Tax Shock			Personal Tax Shock		
Method	ε_{CC}	ε_{LC}	ε_{CY}	ε_{CL}	ε_{LL}	ε_{LY}
BLP	4.0	0.7	1.0	13.6	1.2	2.1
	(2.6, 6.1)	(0.2, 1.3)	(0.6, 1.7)	(8.5, 30.8)	(0.0, 5.0)	(0.8, 6.7)
LP	5.6	1.3	1.6	12.8	1.0	2.0
	(3.9, 8.6)	(0.7, 2.3)	(1.0, 2.7)	(5.9, 34.7)	(-0.8, 6.3)	(-0.1, 8.7)
SLP	5.6	1.3	1.6	12.8	1.0	2.0
	(3.9, 8.6)	(0.7, 2.3)	(1.0, 2.7)	(5.8, 34.8)	(-0.8, 6.3)	(-0.1, 8.5)
LP-BC	5.7	1.2	1.6	12.5	1.0	2.0
	(3.9, 8.8)	(0.6, 2.2)	(1.0, 2.6)	(5.6, 33.2)	(-0.7, 5.8)	(-0.1, 8.0)
BVAR	9.4	0.3	1.8	11.0	2.4	3.2
	(3.2, 42.0)	(-1.0, 2.8)	(0.3, 8.9)	(3.7, 37.0)	(0.8, 6.8)	(1.4, 10.4)
SVAR-IV	8.0	1.7	2.2	19.6	3.3	4.6
	(3.3, 29.0)	(0.3, 7.5)	(0.8, 8.9)	(-0.2, 94.7)	(-1.4, 25.2)	(-0.9, 29.9)

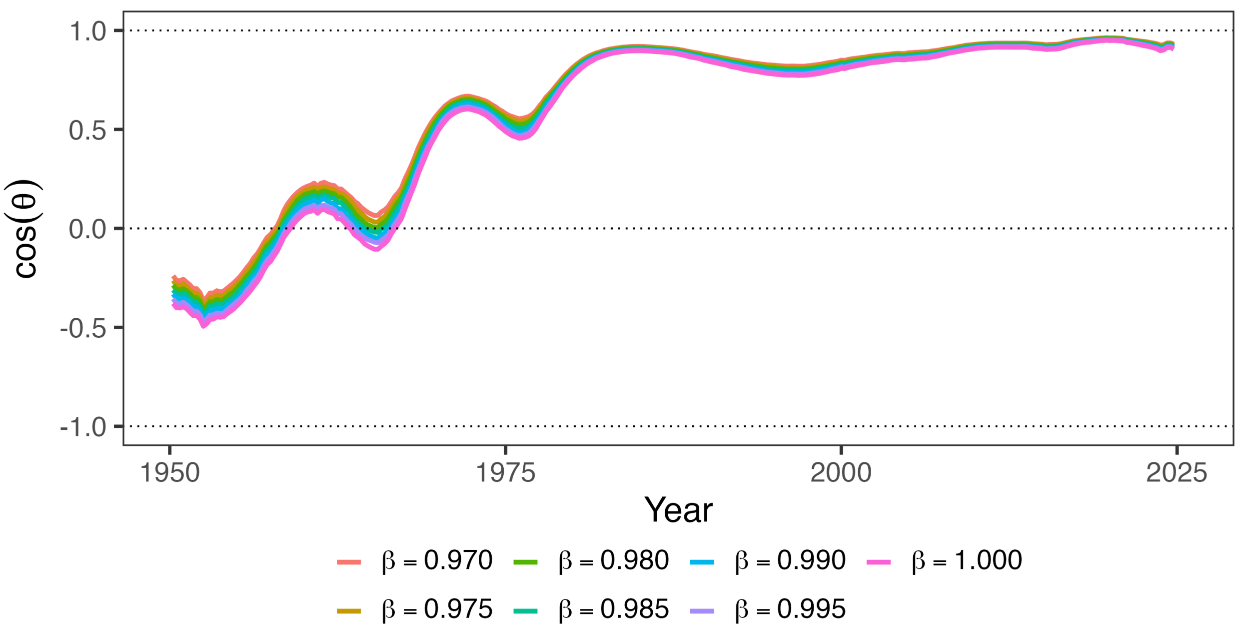
Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1 - \tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the seven-variable system, which adds government spending, debt, and GDP to the four-variable system.

Figure G.1: Robustness Across Welfare Weights



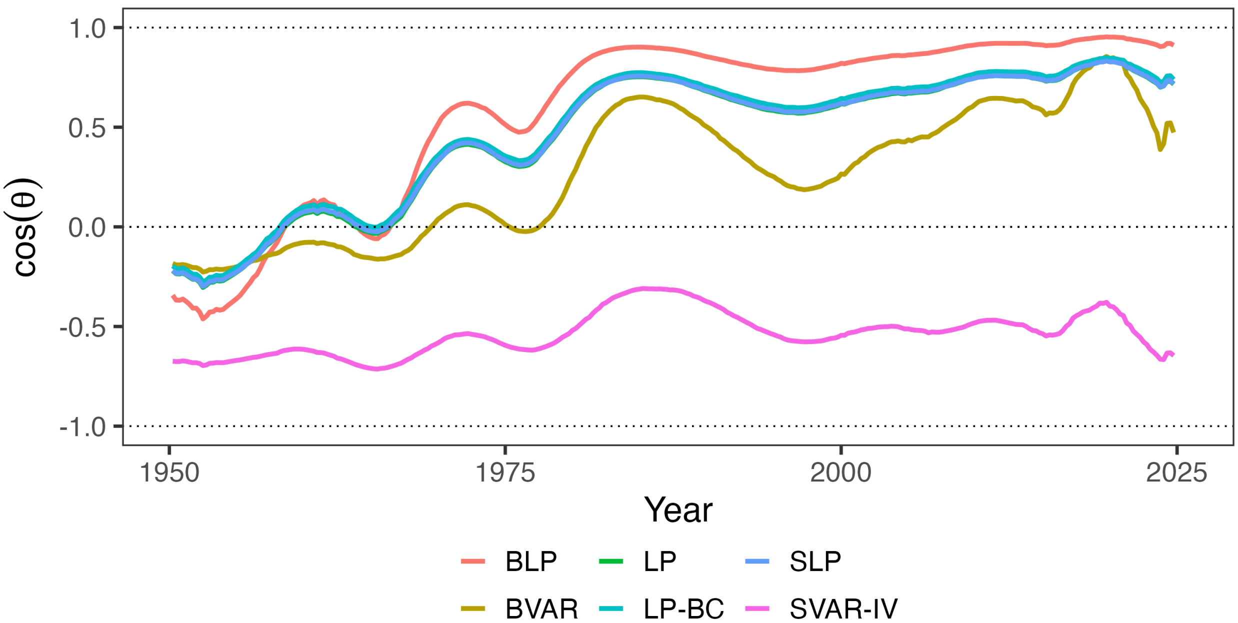
Note: Each panel varies one welfare weight while holding the other at 1. Left panel varies ω_C (weight on corporate taxpayers) with $\omega_L = 1$; right panel varies ω_L (weight on personal taxpayers) with $\omega_C = 1$. The utilitarian baseline has $\omega_C = \omega_L = 1$. Each line corresponds to an estimate with the seven-variable system.

Figure G.2: Robustness Across Discount Factors



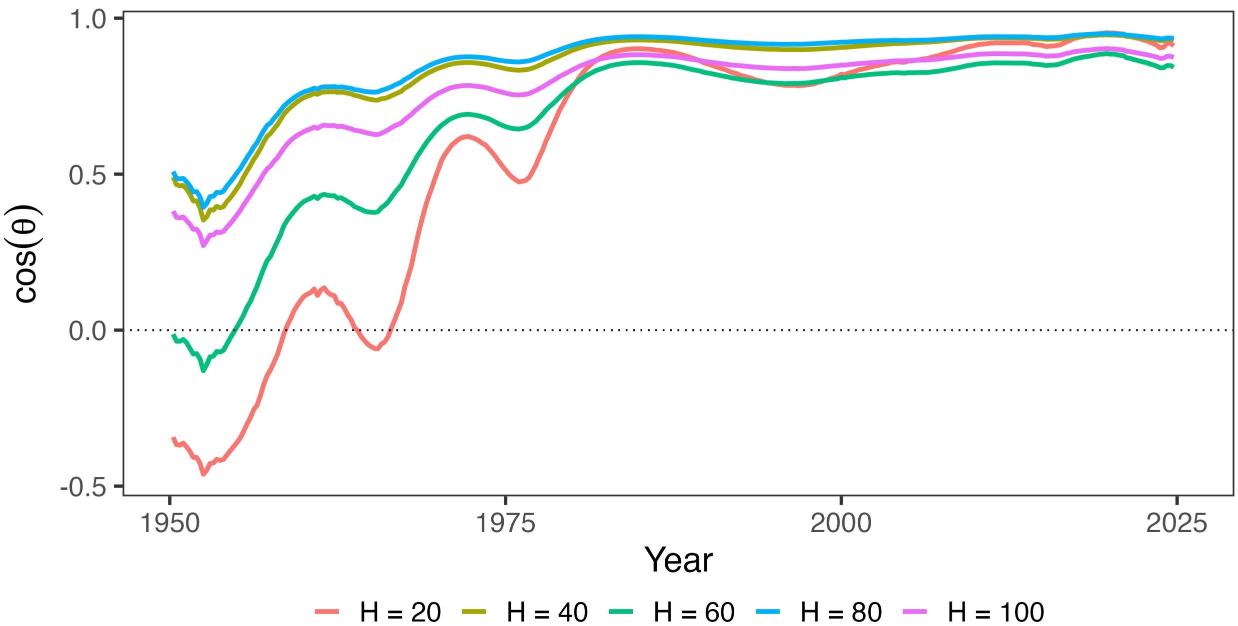
Note: This figure plots alignment $\cos\theta$ across different discount rate regimes for the seven-variable system. Under the baseline, $\beta = 0.9926$.

Figure G.3: Robustness Across Estimation Methodologies



Note: This figure plots alignment $\cos\theta$ across estimation methodologies for the seven-variable system. See Appendix E for further estimation details.

Figure G.4: Robustness Across Horizon Specifications



Note: This figure plots alignment $\cos\theta$ for select models in which vary the horizon length, holding fixed $\beta = 0.9926$. This is for the seven-variable system.

H Robustness to S-Corporation Treatment

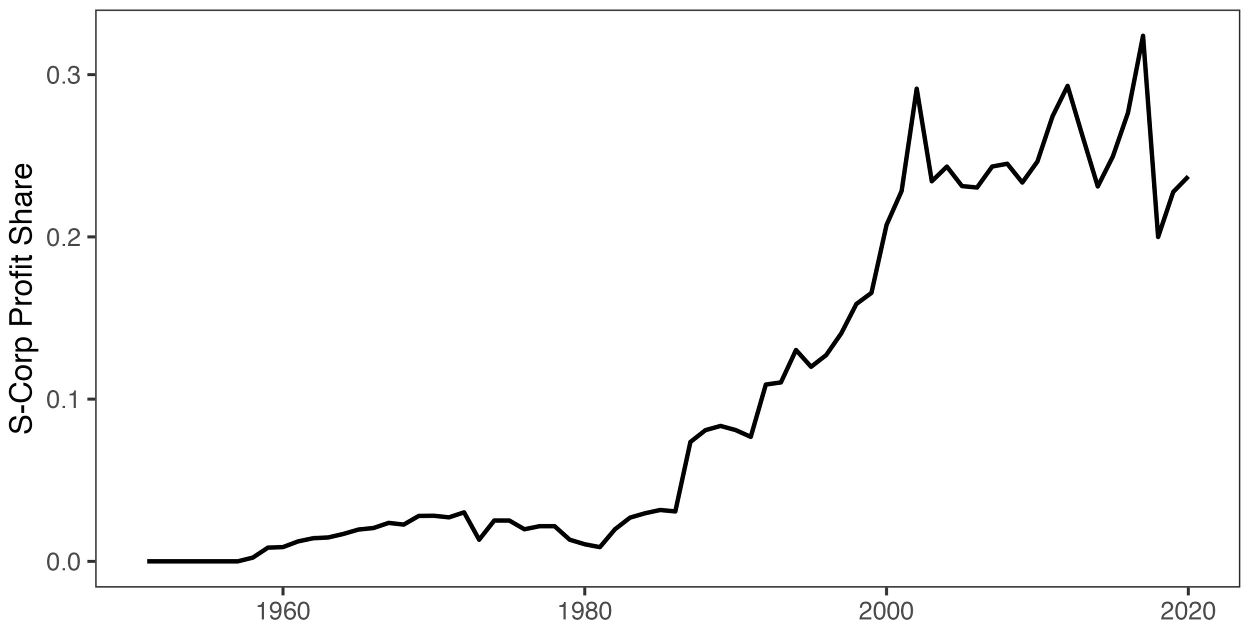
The baseline specification uses NIPA corporate profits as the corporate tax base, which includes both C-corporation and S-corporation income. Since S-corporations are pass-through entities whose income is taxed at the shareholder level rather than at the entity, one might be concerned that the results are driven by organizational form arbitrage rather than traditional general equilibrium channels.

To address this concern, we construct adjusted tax bases that reallocate S-corporation income from the corporate to the personal base. I use the Annual Corporate Reports from the Statistics of Income (SOI) to compile a time series of the profits of C-corporations and S-corporations. I measure profits for the former using the SOI's measure of "Income Subject to Tax" for form 1120 filers (excluding 1120-REIT, 1120-RIC, and 1120-S filers) and profits for the latter as "Net Income (Less Deficit)" for form 1120-S filers. Since Subchapter S was enacted in 1958, allowing qualifying corporations to elect passthrough treatment, I do this for every year from 1958-2020. Toward reallocating NIPA corporate income for S-corps, I compute the S-corporation share of total corporate income as:

$$s_t = \frac{\text{S-corp net income}_t}{\text{S-corp net income}_t + \text{C-corp net income}_t} \tag{A.12}$$

using SOI data for both numerator and denominator to maintain consistency. The resulting series is plotted in Figure H.1. As is well-documented elsewhere, s_t is negligible until after the Tax Reform Act of 1986; since then, S-corps have been large relative to C-corps (Dyrda and Pugsley 2024).

Figure H.1: S-Corporate Share of Corporate Profits



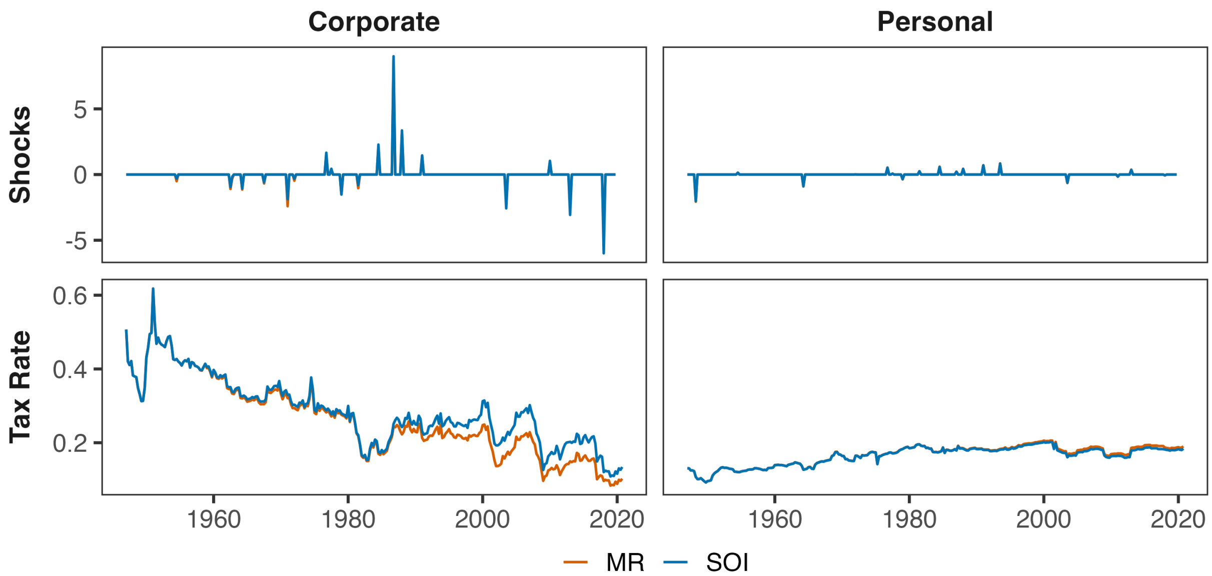
Note: The S-corporate share of corporate profits is computed from the Statistics of Income's Annual Corporate Reports.

NIPA applies a number of conceptual adjustments to corporate profits which are sensible economically, but are not present in the SOI data. To maintain consistency, I assume that these adjustments are proportional, and get quarterly tax bases by applying s_t in year t to NIPA data in every quarter q of that year as:

$$\text{Corporate base}^{adj}_{q,t} = \text{NIPA corporate profits}_{q,t} \times (1 - s_t) \tag{A.13}$$

$$\text{Personal base}^{adj}_{q,t} = \text{NIPA personal income}_{q,t} + \text{NIPA corporate profits}_{q,t} \times s_t \tag{A.14}$$

Figure H.2: Comparing Mertens-Ravn and Reallocated S-Corp Tax Rates and Shocks



Note: After reallocating S-corps to the personal tax base, I recompute average tax rates and tax shocks (as shown above). There is a somewhat sizable difference for the average corporate tax rate, but the results are similar otherwise.

Table H.1 reports the estimated elasticities under the adjusted bases. Several patterns are notable. First, the own-elasticity of the corporate base falls from 4.25 to 3.20 under BLP estimation. This is consistent with the baseline estimate capturing some organizational form arbitrage: when personal taxes change, firms shift between C-corp and S-corp status, and this response inflates the measured elasticity of NIPA corporate profits. Isolating C-corporation income removes this margin, yielding a smaller but still substantial own-elasticity.

Second, the cross-elasticity ε_{CL} remains large—9.11 under BLP, virtually unchanged from the baseline 9.03. This is the key finding: the large cross-base spillover from personal to corporate taxes is not an artifact of S-corporation dynamics. Even when S-corporation income is stripped from the corporate base and added to the personal base, personal tax cuts still generate substantial expansions in C-corporation profits. This suggests the cross-elasticity reflects genuine general equilibrium channels—demand effects, labor-capital complementarity, investment responses—rather than mere organizational form shifting.

Third, the cross-elasticity ε_{LC} becomes more negative (-0.43 versus -0.11 in the baseline), and the personal own-elasticity ε_{LL} rises modestly (1.54 versus 0.98). Both patterns are consistent with the reallocation: S-corporation income, which responds positively to personal tax cuts, has been moved from the corporate base (where it attenuated negative ε_{LC}) to the personal base (where it amplifies positive ε_{LL}).

The qualitative pattern—large asymmetric cross-elasticity, positive own-elasticities—is robust across all estimation methods. The frequentist LP variants yield somewhat larger ε_{CL} estimates (12–13) with wider confidence intervals, while BVAR produces estimates similar to BLP. The key finding that $\varepsilon_{CL} \gg |\varepsilon_{LC}|$ holds uniformly.

Table H.1: Present Value Elasticities of Tax Bases with Respect to Retention Rates (Seven Variable System)

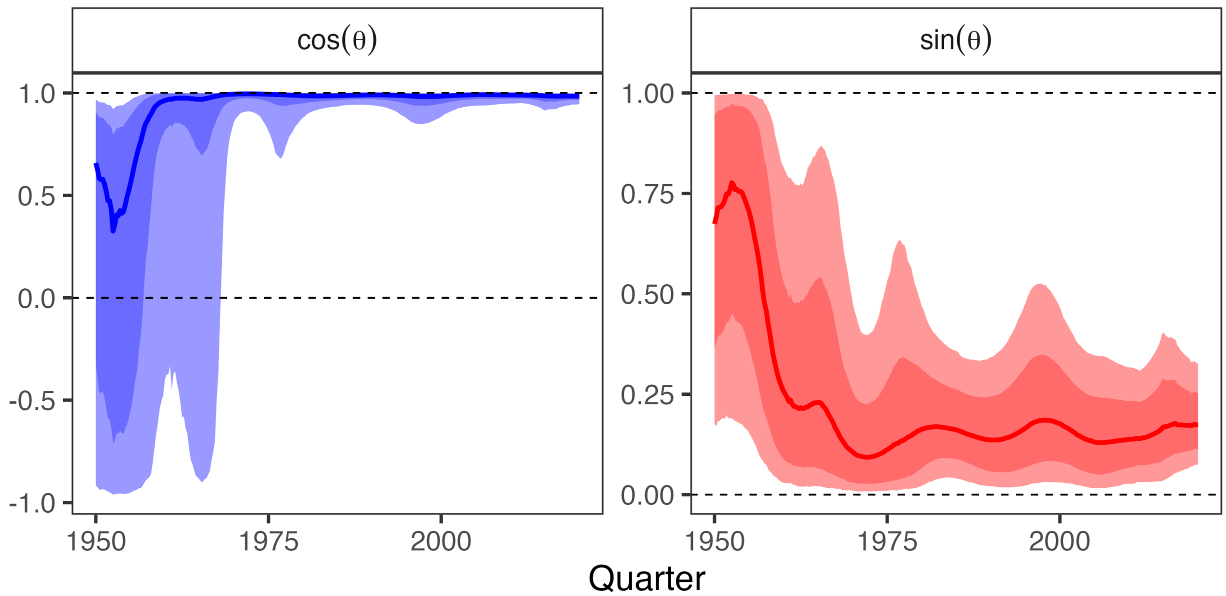
	Corporate Tax Shock		Personal Tax Shock	
Method	ε_{CC}	ε_{LC}	ε_{CL}	ε_{LL}
BLP	3.20	-0.43	9.11	1.54
	(2.31, 4.37)	(-0.65, -0.22)	(6.18, 13.33)	(0.89, 2.24)
LP	2.95	-0.42	12.66	1.52
	(2.29, 3.87)	(-0.67, -0.18)	(8.01, 23.10)	(0.50, 2.61)
SLP	2.96	-0.42	12.67	1.51
	(2.31, 3.88)	(-0.68, -0.18)	(8.01, 23.24)	(0.51, 2.61)
LP-BC	3.06	-0.44	12.37	1.43
	(2.40, 4.00)	(-0.69, -0.20)	(7.86, 22.07)	(0.44, 2.47)
BVAR	8.66	-0.66	8.47	1.20
	(5.53, 14.59)	(-1.11, -0.33)	(3.99, 13.98)	(0.48, 2.08)
SVAR-IV	3.99	-0.72	13.71	1.46
	(1.61, 12.02)	(-2.80, 0.26)	(8.46, 27.14)	(-0.29, 4.84)

Note: Each elasticity ε_{ij} measures the percent change in base i per one percent increase in retention rate $(1-\tau_j)$. Bayesian specifications have 90% credible intervals in parentheses, while frequentist ones are 90% confidence intervals. This table reports elasticities for the seven-variable system, which adds government spending, debt, and GDP to the four-variable system.

Figure H.3 shows that the alignment results are similarly robust. The transition from misalignment ($\cos\theta \approx -1$) to alignment ($\cos\theta \approx 1$) occurs over the same period as in the baseline specification, with $\sin\theta$ remaining bounded well below unity throughout the sample. This confirms that the efficiency gains documented in Section 4 reflect genuine improvements in tax system design rather than artifacts of organizational form shifting between C-corporations and

S-corporations.

Figure H.3: Dynamic Effects of Narrative Tax Shocks on Revenue (Proxy SVAR)



Note: Dynamic effects of a unit increase in each instrument's average log retention rate from 1947Q1-2019Q4. The response is a discounted cumulative IRF with $\beta = 0.9926$.

I Proofs

I.1 MVPF Equalization

The government chooses $\boldsymbol{\tau}$ to minimize social cost subject to a revenue requirement:

$$\min_{\boldsymbol{\tau}} W(\boldsymbol{\tau}) \quad \text{subject to} \quad R(\boldsymbol{\tau}) = G. \tag{A.15}$$

Proposition 2 (Ramsey first-order condition)**.** *Suppose an interior optimum $\boldsymbol{\tau}^* \in int(\mathcal{T})$ exists with $B_i(\boldsymbol{\tau}^*) > 0$ and $r_i(\boldsymbol{\tau}^*) > 0$ for all i, and constraint qualification holds, where i indexes components of $\boldsymbol{\tau} \in \mathbb{R}^{n(T+1)}$, i.e., instrument-date pairs. Then there exists $\lambda \in \mathbb{R}$ such that*

$$g(\boldsymbol{\tau}^*) = \lambda\, r(\boldsymbol{\tau}^*). \tag{A.16}$$

Equivalently, marginal costs of public funds are equalized across instruments:

$$MVPF_i(\boldsymbol{\tau}^*) := \frac{g_i(\boldsymbol{\tau}^*)}{r_i(\boldsymbol{\tau}^*)} = \lambda \quad \textit{for all } i. \tag{A.17}$$

Proof. Form the Lagrangian $\mathcal{L}(\boldsymbol{\tau}, \lambda) = W(\boldsymbol{\tau}) - \lambda(R(\boldsymbol{\tau}) - G)$. Interior first-order conditions give $\nabla_{\boldsymbol{\tau}} \mathcal{L} = 0$, i.e., $g(\boldsymbol{\tau}^*) = \lambda\, r(\boldsymbol{\tau}^*)$. Dividing componentwise by $r_i(\boldsymbol{\tau}^*) > 0$ yields (A.17). □

I.2 Proof of Lemma 1

Proof. The problem is $\max_{d\boldsymbol{\tau}} -g^\top d\boldsymbol{\tau}$ subject to $r^\top d\boldsymbol{\tau} = 0$. Form the Lagrangian:

$$\mathcal{L}(d\boldsymbol{\tau}, \mu) = -g^\top d\boldsymbol{\tau} + \mu r^\top d\boldsymbol{\tau}. \tag{A.18}$$

First-order conditions yield $-g + \mu r = 0$. This cannot be satisfied for finite μ when MVPFs differ (which would require $g = \mu r$, i.e., g parallel to r). The constraint $r^\top d\boldsymbol{\tau} = 0$ implies $r^\top(-g+\mu r) = 0$, giving:

$$\mu = \frac{r^\top g}{r^\top r}. \tag{A.19}$$

Thus the optimal direction is:

$$d\boldsymbol{\tau}^* \propto -\left(g - \frac{r^\top g}{r^\top r} r\right) = -P_{\perp r} g. \tag{A.20}$$

Normalizing to a reform of size α (in percentage points) gives $d\boldsymbol{\tau}^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}$. □

I.3 Interpreting the Alignment Metric

Table I.1: Interpreting the Alignment Metric

$\cos\theta$	θ	$\sin\theta$	Interpretation
1.00	0°	0.00	Ramsey optimum
0.87	30°	0.50	Near optimum; 50% of gains RN
0.50	60°	0.87	Moderate misalignment; 87% of gains RN
0.00	90°	1.00	Orthogonal; all gains RN
−0.50	120°	0.87	Negative alignment; 87% of gains RN
−0.87	150°	0.50	Severe misalignment; 50% of gains RN
−1.00	180°	0.00	Perfect anti-alignment

Note: Revenue-neutral potential $\sin\theta$ is maximized at $\theta = 90°$ and symmetric around it. For example, $\theta = 60°$ and $\theta = 120°$ both yield $\sin\theta \approx 0.87$ because the perpendicular component $\|g\|\sin\theta$ has equal magnitude in both cases.

J Evaluating Deficit-Financed Reforms

Real-world fiscal reforms rarely occur under strict revenue neutrality. Issuing government debt is equivalent to choosing the timing of taxation: the transversality condition ensures all debt must eventually be repaid, so deficit financing defers the tax burden rather than eliminating it. The question "should we issue more debt?" reduces to "should we lower taxes today and raise them tomorrow?" This appendix extends the projection framework to evaluate such intertemporal reforms.

J.1 The Dynamic Ramsey Problem with Dual Discounting

Imposing the transversality condition, the government chooses the tax path $\{\boldsymbol{\tau}_t\}_{t=0}^{\infty}$ to maximize:

$$\max_{\{\boldsymbol{\tau}_t\}} \sum_{t=0}^{\infty} \beta^t W_t(\boldsymbol{\tau}) \quad \text{subject to} \quad \sum_{t=0}^{\infty} q^t R_t(\boldsymbol{\tau}) = \sum_{t=0}^{\infty} q^t G_t + D_0, \tag{A.21}$$

where $\beta = (1+\rho)^{-1}$ is the social discount factor, $q = (1 + r_{\text{debt}})^{-1}$ is the fiscal discount factor, and D_0 is initial debt. The debt path $\{D_t\}$ is residual, determined by the budget identity $D_t = D_{t-1}(1 + r_{\text{debt}}) + G_t - R_t$. Debt is not a choice variable; the transversality condition eliminates it as a degree of freedom. The only choice is the tax path.

Because changing $\tau_{i,t}$ affects bases at all dates, the first-order condition involves the full intertemporal derivatives:

$$\underbrace{\sum_{s=0}^{\infty} \beta^s \frac{\partial W_s}{\partial \tau_{i,t}}}_{g_{i,t}} = \lambda \underbrace{\sum_{s=0}^{\infty} q^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\tilde{r}_{i,t}} \quad \text{for all } i, t, \tag{A.22}$$

where λ is the multiplier on the intertemporal budget constraint. The left-hand side is the welfare gradient $g_{i,t}$ from Section 2.1. The right-hand side is the revenue gradient discounted at the fiscal rate q rather than the social rate β. Rearranging:

$$\frac{g_{i,t}}{\tilde{r}_{i,t}} = \lambda \quad \text{for all } i, t. \tag{A.23}$$

When $r_{\text{debt}} = \rho$, $\tilde{r}_{i,t} = r_{i,t}$ and this reduces to MVPF equalization from Corollary 1. When $r_{\text{debt}} < \rho$ (as in recent decades), the fiscal constraint underweights future costs relative to social welfare, favoring back-loaded taxation. When $r_{\text{debt}} = \rho$, this recovers the tax smoothing result derived in Appendix K.2.

For local reforms around baseline $\bar{\boldsymbol{\tau}}$, the projection formula from Section 2.2 carries over with one modification: the revenue gradient must be discounted at q, not β.

$$d\boldsymbol{\tau}^* = -P_{\perp \tilde{r}}\, g, \tag{A.24}$$

where $P_{\perp \tilde{r}} = I - \tilde{r}(\tilde{r}^\top \tilde{r})^{-1}\tilde{r}^\top$. The welfare gradient uses social discounting because we evaluate welfare from society's perspective; the revenue constraint uses fiscal discounting because the government's ability to borrow and repay is governed by market interest rates. When the two rates coincide, equation (A.24) reduces to $d\boldsymbol{\tau}^* = -P_{\perp r}\, g$ from Section 2.

J.2 Bounding the Cost of Deferral

The Ramsey problem delivers the globally optimal tax path. In practice, we rarely observe complete reform paths—a deficit-financed tax cut specifies changes at $t = 0$ but leaves the repayment path implicit. Evaluating such incomplete reforms requires comparing the welfare cost of deferral against immediate revenue-neutral financing, without knowing when or how the debt will be repaid.

A deficit-financed reform at $t = 0$ creates a present-value fiscal gap $\text{FG} = \sum_{h=0}^{H} q^h [r_{C,h}\Delta\tau_C(h) + r_{L,h}\Delta\tau_L(h)]$. This gap must be closed by raising taxes in future periods. A repayment rule $\mathcal{P}$

specifies the path $\{\Delta\tau_{k,t}\}_{t\geq 1}$ satisfying:

$$\sum_{t\geq 1}\sum_{k} q^t r_{k,t}\Delta\tau_{k,t} = \mathrm{FG}. \tag{A.25}$$

The marginal value of public funds for deferral under rule $\mathcal{P}$ is a weighted average of future MVPFs:

$$\mathrm{MVPF}(\mathcal{P}) = \frac{\sum_{t\geq 1}\sum_k \beta^t g_{k,t}\Delta\tau_{k,t}}{\mathrm{FG}} = \sum_{t,k}\pi_{t,k}(\mathcal{P})\left(\frac{\beta}{q}\right)^t \mathrm{MVPF}_{k,t}, \tag{A.26}$$

where $\pi_{t,k}(\mathcal{P}) = q^t r_{k,t}\Delta\tau_{k,t}/\mathrm{FG}$ are repayment shares summing to one and $\mathrm{MVPF}_{k,t} = g_{k,t}/r_{k,t}$. The discount wedge $(\beta/q)^t$ appears because welfare and revenue are discounted at different rates. When $r_{\text{debt}} < \rho$, this wedge is below one for $t > 0$, mechanically lowering the effective cost of deferral: low borrowing rates make deferral cheap.

The repayment rule $\mathcal{P}$ is unobserved. But restricting attention to rules that only raise taxes ($\Delta\tau_{k,t} \geq 0$), the weights $\pi_{t,k}$ are non-negative and sum to one, so $\mathrm{MVPF}(\mathcal{P})$ is a convex combination of the atoms $a_{t,k} := (\beta/q)^t\mathrm{MVPF}_{k,t}$. For any repayment horizon $\mathcal{T}$ and set of instruments $\mathcal{K}$:

$$\min_{t\in\mathcal{T},k\in\mathcal{K}} a_{t,k} \;\leq\; \mathrm{MVPF}(\mathcal{P}) \;\leq\; \max_{t\in\mathcal{T},k\in\mathcal{K}} a_{t,k}. \tag{A.27}$$

Let $\mathrm{MVPF}_{j,0}$ denote the cost of the least distortionary instrument available for immediate revenue-neutral financing. If $\max_{t,k} a_{t,k} < \mathrm{MVPF}_{j,0}$, deferral is unambiguously superior regardless of the repayment rule. If $\min_{t,k} a_{t,k} > \mathrm{MVPF}_{j,0}$, revenue-neutral financing dominates. Otherwise, the conclusion depends on the actual repayment path, and the width of the bounds $[\min a_{t,k}, \max a_{t,k}]$ measures how much that uncertainty matters.

K Classic Optimal Tax Results as Projections

This appendix demonstrates that the projection framework nests several canonical results from optimal tax theory. While the main text focuses on policy evaluation, these derivations show the framework is consistent with standard normative prescriptions under various structural assumptions. The first two subsections restate the Ramsey and Barro conditions in the notation and geometry of this paper. The final subsection shows how the zero-capital-tax result can be interpreted as a special configuration of the welfare and revenue gradients under its usual assumptions. The Atkinson–Stiglitz theorem admits a similar interpretation: under their assumptions, the welfare and revenue gradients are collinear within the commodity-tax subspace, so revenue-neutral commodity-tax reforms yield zero first-order welfare gains.

K.1 The Ramsey Inverse Elasticity Rule

The Ramsey problem seeks the tax structure that minimizes welfare costs subject to raising a fixed revenue requirement. At the optimum, the first-order conditions imply that marginal costs of public funds are equalized across instruments. The projection framework recovers this result and clarifies its geometric interpretation.

Proposition 3 (Ramsey as Perfect Alignment)**.** *At a Ramsey optimum $\bar{\boldsymbol{\tau}}$, the welfare and revenue gradients are perfectly aligned:*

$$g(\bar{\boldsymbol{\tau}}) = \mu r(\bar{\boldsymbol{\tau}})$$

for some scalar $\mu > 0$. Equivalently, $\cos\theta = 1$ and the alignment is perfect.

Proof. The Ramsey problem is:

$$\min_{\boldsymbol{\tau}} W(\boldsymbol{\tau}) \quad \text{subject to} \quad R(\boldsymbol{\tau}) = \bar{R}.$$

The Lagrangian is $\mathcal{L} = W(\boldsymbol{\tau}) - \mu[R(\boldsymbol{\tau}) - \bar{R}]$. The first-order condition is:

$$\frac{\partial W}{\partial \tau_i} = \mu \frac{\partial R}{\partial \tau_i} \quad \text{for all } i,$$

which in vector form is $g(\bar{\boldsymbol{\tau}}) = \mu r(\bar{\boldsymbol{\tau}})$. Since g and r point in the same direction, the angle between them is zero: $\cos\theta = 1$. The sign $\mu > 0$ follows from the requirement that both $g_i > 0$ (positive tax bases) and $r_i > 0$ (the economy is below the Laffer peak) at the optimum. □

When gradients are perfectly aligned, the projection $P_{\perp r} g = 0$ is the zero vector: there is no component of the welfare gradient perpendicular to revenue. All welfare improvements require changing total revenue. This is the geometric interpretation of "no further revenue-neutral gains are available."

The Ramsey optimum also implies MVPF equalization. From $g_i = \mu r_i$, we have:

$$\text{MVPF}_i = \frac{g_i}{r_i} = \mu \quad \text{for all } i.$$

All MVPFs equal the Lagrange multiplier μ, the shadow cost of the revenue constraint.

The inverse elasticity rule. Under additional structure, the Ramsey formula yields the classic inverse elasticity result. Suppose:

1. Utilitarian welfare: $g_i = B_i$ (the tax base).

2. Ad valorem taxation with instrument-specific rates τ_i.

3. No cross-base spillovers: $\Lambda_i = 0$ (partial equilibrium).

Then from equation (3), the revenue gradient is:

$$r_i = B_i \left[1 - \frac{\tau_i}{1 - \tau_i} \varepsilon_{ii} \right].$$

MVPF equalization $g_i / r_i = \mu$ implies:

$$\frac{B_i}{B_i \left[1 - \frac{\tau_i}{1-\tau_i} \varepsilon_{ii} \right]} = \mu \quad \Rightarrow \quad \frac{\tau_i}{1 - \tau_i} \varepsilon_{ii} = 1 - \frac{1}{\mu}.$$

Since the right-hand side is constant across i, we have $\frac{\tau_i}{1-\tau_i} \varepsilon_{ii} =$ constant. For small tax rates, this implies:

$$\tau_i \propto \frac{1}{\varepsilon_{ii}}.$$

Taxes should be inversely proportional to elasticities. Highly elastic bases receive low taxes; inelastic bases receive high taxes. This is Ramsey's (1927) classic result, recovered as the shadow of perfect alignment under no cross-base spillovers.

K.2 Barro Tax Smoothing

Barro (1979) shows that when the government's borrowing rate equals the social discount rate, optimal tax policy smooths distortions over time. This result emerges directly from the intertemporal projection framework in Section 2.

Proposition 4 (Barro as Temporal MVPF Equalization)**.** *Consider the dynamic Ramsey problem from equation* (A.21)*. Suppose (1) the government's borrowing rate equals the social discount rate: $r_{debt} = \rho$, so $q = \beta$; and (2) tax bases and behavioral responses are stationary: $B_{i,t} = B_i$ and $\varepsilon_{ik,t} = \varepsilon_{ik}$ for all t.*

Then at the Ramsey optimum, MVPFs are constant over time:

$$MVPF_{i,t} = MVPF_{i,s} \quad \text{for all } t, s.$$

Proof. From equation (A.22), the first-order condition for the dynamic Ramsey problem is:

$$\beta^t \frac{\partial W_t}{\partial \tau_{i,t}} = \lambda q^t \frac{\partial R_t}{\partial \tau_{i,t}}.$$

When $q = \beta$, this simplifies to:

$$\frac{\partial W_t}{\partial \tau_{i,t}} = \lambda \frac{\partial R_t}{\partial \tau_{i,t}}.$$

The discount factors cancel. Under stationarity, $\partial W_t/\partial\tau_{i,t} = g_i$ and $\partial R_t/\partial\tau_{i,t} = r_i$ are constant across t. Therefore:

$$\text{MVPF}_{i,t} = \frac{g_i}{r_i} = \lambda \quad \text{for all } t.$$

MVPFs are equalized not only across instruments (the static Ramsey condition) but also across time. $\square$

This is Barro's tax smoothing result: optimal policy equalizes distortions intertemporally. When $r_{\text{debt}} = \rho$, there is no wedge between fiscal and social accounting, so the government should smooth MVPFs over time just as it smooths them across instruments. Temporary shocks to revenue needs should be financed primarily by debt, with small permanent tax adjustments to service the debt, maintaining equalized MVPFs across time. Permanent shocks should be financed by permanent tax increases.

When $r_{\text{debt}} \neq \rho$, the result no longer holds. From equation (A.23):

$$\text{MVPF}_{i,t} = \lambda \left(\frac{1 + r_{\text{debt}}}{1 + \rho}\right)^t .$$

If $r_{\text{debt}} < \rho$ (as in recent decades), the government borrows cheaply relative to its social discount rate, creating an incentive to back-load taxation. MVPFs should fall over time at rate $(1+r_{\text{debt}})/(1+\rho) < 1$, reflecting lower marginal costs of future financing. Conversely, if $r_{\text{debt}} > \rho$, MVPFs should rise over time, favoring front-loaded taxation.

K.3 Zero Capital Taxation as a Limiting Case

Chamley (1986) and Chari, Nicolini, and Teles (2020) show that under standard macroeconomic assumptions (infinitely lived agents, perfect capital markets, homothetic preferences, additively separable preferences over time), the optimal long-run tax on capital is zero. In those environments, a permanent change in the capital tax rate induces such a large long-run response of the capital stock that the present-value revenue raised by the tax tends to zero. The projection framework makes this logic transparent by expressing it in terms of the present-value revenue gradient.

In the notation of Section 2, consider a permanent increase in the capital tax $\tau_{K,0}$. Its present-value revenue effect is

$$r_{K,0}^{PV} = \sum_{t=0}^{\infty} \beta^t \frac{\partial R_t}{\partial \tau_{K,0}},$$

and the present-value marginal value of public funds for capital taxation is

$$\text{MVPF}_K^{PV} = \frac{g_{K,0}^{PV}}{r_{K,0}^{PV}}.$$

In a Chamley-type environment, long-run capital supply is effectively infinitely elastic: a permanent increase in τ_K causes the capital tax base to shrink so much over time that

$$r_{K,0}^{PV} \to 0.$$

Given a finite welfare gradient $g_{K,0}^{PV}$, this implies $\text{MVPF}_K^{PV} \to \infty$. In the projection formula, instruments with very high MVPFs are pushed toward their lower bounds. When the present-value revenue gradient for capital taxation vanishes, the optimal reform direction drives τ_K toward zero. The familiar zero-capital-tax result thus appears as a limiting case of the general projection framework, corresponding to an extreme configuration of the present-value revenue gradient.

This limiting case is not generic. If the present-value elasticity of capital supply is large but finite—because of adjustment costs, firm-specific capital, borrowing constraints, or heterogeneous discount rates—then $r_{K,0}^{PV}$ remains strictly positive and so does MVPF_K^{PV}. Straub and Werning (2020) show that relaxing the standard assumptions can overturn the zero-capital-tax result. In the projection framework, this fragility appears as sensitivity of the optimal capital tax to the shape of r_K^{PV}: whenever $r_{K,0}^{PV}$ does not collapse to zero, the optimal reform sets a finite capital tax rate determined by the same MVPF equalization logic as for any other instrument.